SHANGHAI JIAO TONG UNIVERSITY

学士学位论文

THESIS OF BACHELOR

论文题目： Some Topics on Theta Correspondence and Hecke Modules

学生姓名:	袁江帆
学生学号:	516071910049
专　　业:	数学与应用数学
指导教师:	马家骏教授
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Abstract

This thesis contains some topics on representation theory, especially on theta correspondences and representation theory of finite reductive groups.

The study of theta correspondence (or Howe correspondence) is basically on the representations of pairs of dual reductive groups (G, G'), where $G \times G'$ is embedded in a larger symplectic group $\mathrm{Sp}(\mathbb{W})$, and by 'dual' we mean that they are mutually centralized. We endow a representation $\omega_{\mathrm{Sp}_{2n}}$ to the surrounding symplectic group, called Weil representation. By theta correspondence we mean $\pi \otimes \pi'$ occurs in $\omega_{G\times G'}$, where π (resp. π') is the irreducible representation of G (resp. G') and $\omega_{G\times G'}$ is the restriction of Sp_{2n} on $G \times G'$.

In Chapter 1, we give some preliminaries. Since the change of base field, we may lose advantages of analytical properties of Lie groups, then the notion of algebraic groups is introduced to study the structure theory of reductive over various fields, under the help of algebraic geometry. We will see explicit construction of linear algebraic groups of classical types and several important but somehow abstract notions along with structure theory, eg. tori, Borel subgroups, Parabolic subgroups and Weyl groups. As a conclusion of first half part of preliminaries, we introduce finite group of Lie type (defined over finite field $\mathbb{F}_q$).
In Chapter 2, we provide a first glimpse into theta correspondence over non-Archimedean local field, including the construction of Weil representation of local case and the Howe duality principle.

Chapter 3 and Chapter 4 are the main parts of this thesis.
In Chapter 3, we give two approaches of constructing representations of finite reductive groups. The first one is Harish-Chandra induction, it gives an so called parabolic induction from $\mathbf{L}^F$ module to $\mathbf{G}^F$ module, with regard to the Levi decomposition $\mathbf{P} = \mathbf{LU}$. As an analog to the induced representation of finite group, we have the Mackey formula to compare two parabolic inductions. Also the Harish-Chandra series give the disjoint partition of irreducible characters of $\mathbf{G}^F$.
And the second one is due to Deligne and Lusztig, using the cohomological methods. It is a monumental work in the study of representation theory. After introducing some properties of $l-$adic cohomology, we introduce the construction of Deligne-Lusztig's virtual characters $R^{\mathbf{G}}_{\mathbf{T},\theta}$ and their properties with concrete computations. The main references we follow are [Car85] and [DM91], we make some simplification of the computing process, however, some steps associating to complex structure theory is omitted in order to give a clearer description, one may refer to these two texts for more details. As a end, we introduce Lusztig series. As an analog of Harish-Chandra series, it gives a partition of irreducible characters associating to geometric conjugacy classes of semisimple elements in the dual groups.
In Chapter 4, we turn to the study of finite theta correspondence by following its development (the dual pair we would like to focus on is of the type symplectic-even orthogonal) In Geradin's paper [Gér77] , he gave the explicit description of finite Weil representation and how elements acting. Inspired by this, Srinivasan gave the formula of uniform projection of Weil representation restricted on dual pairs (of any type). Srinivasan's work laid a foundation of [AM93] and [AMR96] . In [AM93] , Adams and Moy proved that unipotent representation is theta correspondent to unipotent representation, and for those groups of specific dimensions, the theta correspondence of unipotent cuspidal representations is illustrated. In

[AMR96] , Aubert-Michel-Rouquier classified all theta correspondences of unipotent characters of type A by considering the bipartition of Weyl groups. They also conjectured the case of symplectic-even orthogonal. We shall give a review of the above developments.

In recent years, Shuyen Pan did a series of works on theta correspondence, including give a complete description of the symplectic-even orthogonal case, which solved AMR's conjecture, by using the method of 'symbols'. We are curious if Pan's work can be realized by considering bi-Hecke algebra modules. Thus we introduce the basic knowledge of Hecke algebras in Chapter 1 and dig further on the structure of Hecke rings induced by cuspidal representations by following [HL80]. At last, we verify the case of the lowest dimension. Unfortunately, due to the author's lack of ability, there still exists a long way off a complete description.

The idea mentioned at last gives credit to my advisor Prof.Ma Jiajun and all faults made in the thesis are mine.

有限域上的theta对应和一些Hecke模

摘要

本论文主要包含了表示论的相关内容，包括theta对应和有限约化群的表示。
在第一章中，我们总结了一些预备知识。一部分是在代数几何的基础上介绍代数群，并且给出不同型线性代数群的具体构造，在此基础上梳理一些代数群的抽象结构理论，这部分最后我们 会介绍有限李型群的相关定义。另一部分是有关Iwahori-Hecke代数的定义，Hecke代数模的基本性质。
在第二章中，我们从局部域的情形入手，初步探究在该情形下theta对应的相关内容。包括Weil表示的构造，Howe对偶猜想等，由此为我们研究有限域上的theta对应做了铺垫和理论基础。
第三、四章是本论文的主要部分。在第三章中，我们探讨有限约化群表示的研究，其中主要涉及两方面，一是Harish-Chandra的诱导表示，我们在这里给出Mackey Formula的形式 并给出principal series的定义；其二是Deligne-Lusztig利用l-adic上同调来构造虚拟表示的方法，在这一部分我们会对该方法中的重要结论进行计算，并且知悉幂单表示的定义。最后通过Lusztig序列我们可以得到 用对偶群中半单元分类不可约表示的方法。
在第四章中，我们进入有限theta对应的研究。这一章中我们会介绍有限Weil表示的构造和基于此工作一系列的重要发展。我们将介绍Srinivasan关于Weil表示在不同型下的uniform space投影的计算， 并在此基础上证明Adams-Moy的幂单表示theta对应到幂单表示的结果，以及Aubert-Michel-Rouquier的一系列工作。 在此之后，我们给出Howlett-Lehrer文章中对更一般的尖表示诱导的Hecke环的结构的论述，并且用原创的方法计算$(\mathrm{Sp}_2, \mathrm{O}_2^+)$ 情形下的特殊Hecke模， 验证其和Shuyen Pan于19年最新文章中的解决AMR猜想的结果的一致性。

关键词：表示论，theta对应，有限theta对应，约化群，Weil表示，Hecke代数，Hecke代数模

Contents

Chapter 1

Preliminaries

1.1 Algebraic Groups

1.1.1 Algebraic Sets and Algebraic Groups

Let k be any field, infinite or finite, k^n the affine n-space over k, $k[X_1, ..., X_n]$ be the polynomial ring, A be a commutative ring.

Theorem 1.1.1 (Hilbert's basis theorem)**.** *If A is Noetherian, then $A[X]$ is also Noetherian. In particular, if k is a field, then $k[X_1, ..., X_n]$ is Noetherian.*

Definition 1.1.2. For any subset $S \subseteq k[X_1, ..., X_n]$, we call $\mathbf{V}(S) := \{x \in k^n | f(x) = 0, \forall f \in S\}$ the **algebraic set** realized by S. Conversely, for any subset $V \subseteq k^n$, we call $\mathbf{I}(V) := \{f \in k[X_1, ..., X_n] | f(x) = 0, \forall x \in V\}$ the **(vanishing) ideal** of V in $k[X_1, ..., X_n]$. And the quotient $A[V] := k[X_1, ..., X_n]/\mathbf{I}(V)$ is called the **affine algebra** or **affine coordinate ring** of V.

Proposition 1.1.3. *The union of two algebraic sets are algebraic, the intersection of any family of algebraic sets are algebraic. Moreover, the $\varnothing$ and the whole affine space k^n are algebraic sets.*

Definition 1.1.4. From the last proposition, we know that algebraic sets could define closed sets in k^n. We define the **Zarisiki Topology** on k^n by taking open sets to be the complements of algebraic sets in k^n. A topological space is said to be **irreducible** if it could not be expressed as the union of two proper sets, each one is closed in V, **Noetherian** if every descending chain of its closed subsets is stationary.

Proposition 1.1.5. *In a Noetherian topological space X, every non-empty closed subset Y could be expressed as a finite union $Y_1 \cup ... \cup Y_r$ of irreducible closed subset Y_i, without lose of generality, we require $Y_i \nsubseteq Y_j, i \neq j$, then Y_i are uniquely determined, called **irreducible components** of Y.*
Any non-empty algebraic set $V \in k^n$ is Noetherian w.r.t Zarisiki topology.

Proposition 1.1.6. *(a) If $S_1 \subseteq S_2$ are subsets of $k[X_1, ..., X_n]$, then $\mathbf{V}(S_1) \supseteq \mathbf{V}(S_2)$.*
(b) If $V_1 \subseteq V_2$ are subsets of k^n, then $\mathbf{I}(V_1) \supseteq \mathbf{I}(V_2)$.
(c) If V_1, V_2 are subsets of k^n, then $\mathbf{I}(V_1 \cup V_2) = \mathbf{I}(V_1) \cap \mathbf{I}(V_2)$.
(d) If S_1, S_2 are subsets in $k[X_1, ..., X_n]$, then $\mathbf{V}(\{fg | f \in S_1, g \in S_2\}) = \mathbf{V}(S_1) \cup \mathbf{V}(S_2)$. Moreover, if S_1, S_2 are ideals, then $\mathbf{V}(S_1 \cap S_2) = \mathbf{V}(S_1) \cup \mathbf{V}(S_2)$.
(e) For any subset $V \subseteq k^n$, define the closure $\overline{V}$ of V under the Zarisiki topology, then $\mathbf{V}(\mathbf{I}(V)) = \overline{V}$.

Proof. Only prove (e), first notice that $V \subseteq \mathbf{V}(\mathbf{I}(V))$, thus $\overline{V} \subseteq \mathbf{V}(\mathbf{I}(V))$. On the other hand, let W be any closed algebraic set containing V, then W is realized by some $S \in k[X_1, ..., X_n]$. We have $S \subseteq \mathbf{I}(\mathbf{V}(S)) = \mathbf{I}(W) \supseteq \mathbf{I}(V)$, then $\mathbf{V}(\mathbf{I}(V)) \subseteq \mathbf{V}(S) = W$. Thus $\mathbf{V}(\mathbf{I}(V)) = \overline{V}$. $\square$

Theorem 1.1.7 (Hilbert's Nullstellensatz)**.** *Assume that k is an algebraically closed field, let I be any ideal in $k[X_1, ..., X_n]$, for any $f \in k[X_1, ..., X_n]$ that vanishes at all points of $\mathbf{V}(I)$, there is some integer $r > 0$ such that $f^r \in I$.*

Corollary 1.1.8. *Assume that k is an algebraically closed field, then there is a one-to-one correspondence between algebraic sets in k^n and radical ideals(i.e, ideal $I = \sqrt{I} := \{f \in k[X_1, ..., X_n] | f^r \in I, \text{for some } r > 0\}$). Moreover, An algebraic set is irreducible if and only if its corresponding ideal is prime.*

Corollary 1.1.9 (Hilbert's Nullstellensatz for hypersurfaces)**.** *Let $f \in k[X_1, ..., X_n]$ be non-constant, define $H_f := \mathbf{V}(\{f\}) \subseteq k^n$. If $n = 2$, then it is called a plane curve, if $n = 3$, it is called a plane, if $n > 3$, it is called a hypersurface. Assume k is algebraically closed, and $H_f \neq \varnothing$. Write $f = f_1^{s_1} ... f_r^{s_r}, f_i irreducible$. Then $H_f = H_{f_1} \cup ... \cup H_{f_r}$, $\mathbf{I}(H_f) = (f_1, ..., f_r)$.*

Definition 1.1.10. Let $V \subseteq k^n, W \subseteq k^m$, realized by $S \subseteq k[X_1, ..., X_n]$ and $T \subseteq k[Y_1, ..., Y_m]$, respectively. Notice that $k^n \times k^m$ is identified with k^{n+m}, and $k[X_1, ..., X_n] \times k[Y_1, ..., Y_m]$ as the subring of $k[X_1, ..., X_n, Y_1, ..., Y_m]$.We define the **direct product of algebraic sets** to be

$$V \times W = \{(v, w) \in k^{n+m} | f(v) = 0 \text{ for all } f \in S, g(w) = 0 \text{ for all } g \in T\} = \mathbf{V}(S \cup T) \subseteq k^{n+m}$$

The direct product $V \times W$ is an algebraic set in k^{n+m}.

Remark. If V, W are irreducible respectively, then $V \times W$ is irreducible in k^{n+m}. To prove this, only need to prove that $\mathbf{I}(V \times W)$ is a prime ideal. Set $f_1, f_2 \in k[X_1, ..., X_n, Y_1, ..., Y_m]$ such that $f_1 f_2 \in \mathbf{I}(V \times W)$. Fix $w \in W$, let $V_i(w) := \{v \in V | f_i(v, w) = 0\}, i = 1, 2$. $\forall v \in V$, $f_1 f_2(v, w) = 0$ since $f_1 f_2 \in \mathbf{I}(V \times W)$, then $V = V_1(w) \cup V_2(w)$. $V_i(w)$ are clearly algebraic sets, and V is irreducible by assumption, then either $V = V_1(w)$ or $V = V_2(w)$. Similarly, $W = W_1 \cup W_2$, where $W_i := \{w \in W | V_i(w) = V\} = \{w \in W | f_i(v, w) = 0, \forall v \in V\}$. Then either $W = W_1$ or $W = W_2$. In the first case we have $f_1(v, w) = 0, \forall v \in V, \forall w \in W\}$, it means $f_1 \in \mathbf{I}(V \times W)$. The second case is the same, then $\mathbf{I}(V \times W)$ is prime, as desired.

Definition 1.1.11. Consider if $M_n(k) = k^{n^2}$ is an algebraic set. let $\mu : M_n(k) \times M_n(k) \to M_n(k)$ be the usual matrix multiplication such that $\mu : ((a_{ij}), (b_{kl})) \mapsto (c_{ij} := \sum_{k=1}^{n} a_{ik} b_{kj})$, easy to check that μ is regular map, $M_n(k)$ together with μ is called **the general linear algebraic monoid**. $A[M_n(k)] = k[X_{ij}]/\mathbf{I}(M_n(k))$, and μ induces $\mu^* : A[M_n(k)] \to A[M_n(k)] \otimes_k A[M_n(k)], \mu^*(\overline{X}_{ij}) = \sum_{k=1}^{n} \overline{X}_{ik} \otimes \overline{X}_{kj}$.
Further, if $\forall A \in G$ has an inverse and the map $\iota : G \to G, A \mapsto A^{-1}$ is regular, G is called the **linear algebraic group**.

Remark. We here actually consider algebraic groups in the language of variety for a clearer demonstration and a intuitive construction of examples in later. The notion of algebraic groups of course can be generalized into scheme, that is, (G, μ) scheme over k with $\mu : G \times G \to G$ a regular map and exists $k : \operatorname{Spec} k \to G, \operatorname{inv} : G \to G$ satisfying some commutative diagram, see [Mil17]. And μ induces $\mu^* : \mathcal{O}(G) \to \mathcal{O}(G) \otimes \mathcal{O}(G)$.
Under the setting of above , we see group varieties are actually geometrically reduced group schemes, and **affine** actually means $G = \operatorname{Spec}(\mathcal{O}(G))$. By all means should we acknowledge this notion, and we may us it in 1.1.18 without more declaration. For details , [Mil17] is a good reference.

Example 1.1.12. (i) Let $\mathrm{SL}_n(k) = \{A \in M_n(k) | det(A) = 1\}$, it is closed under multiplication, and $\mathrm{SL}_n(k) = \mathbf{V}(det - 1)$, thus an algebraic monoid. And $\forall A \in \mathrm{SL}_n(k)$, let $A^{-1} = det(A)^{-1}(A^*)^T$, where A^* is the cofactor matrix of A. Then $A^{-1} \in \mathrm{SL}_n(k)$, and $\iota : A \mapsto A^{-1}$ is regular. Hence $\mathrm{SL}_n(k)$ is a linear algebraic group, call it special linear algebraic group.
(ii) Let $e \geq 1$, set $\mathrm{SL}_n^{(e)}(k) = \{A \in M_n(k) | det(A)^e = 1\}$. It is a linear algebraic monoid, and $det(A)^{-1} = det(A)^{e-1}$, then A^{-1} could be expressed by the polynomial in the entries of A, a regular map clearly. Thus $\mathrm{SL}_n^{(e)}(k)$ is also a linear algebraic group.
(iii) Define $U_n(k) = (u_{ij}) \subseteq M_n(k)$, where $u_{ii} = 1, i = 1, ..., n$; $u_{ij} = 0, i > j$. It contains upper triangular matrices with diagonal 1. It is a linear algebraic group of dimension $\frac{n(n-1)}{2}$. Moreover, $U_2(k)$ is isomorphic to $\mathbb{G}_a(k)$, the

additive group of k.
(iv) For any finite group G, by Cayley's theorem, it could be expressed as a subgroup of $\mathfrak{S}_n$. And $\mathfrak{S}_n$ has generators $\{(1),(1,2),(2,3),...,(n,n-1)\}$, which could be realized by matrices in $M_n(k)$. It gives an embedding of G into $\mathrm{SL}_n^{(2)}(k)$, and tells that any finite group G is an algebraic group.

Proposition 1.1.13. *Let G be a linear algebraic group, G° be its irreducible component that containing 1, then*
(i) *G° is a closed normal subgroup of G of finite index, and the cosets of G° are precisely irreducible components of G°, they are disjoint and thus G° is uniquely determined.*
(ii) *Every closed subgroup of G of finite index contains G°.*
(iii) *G is irreducible if and only if G is connected.*

Proof. (i) First choose any irreducible $X \subseteq G^\circ$ with $1 \in X$, consider $\overline{X \cdot G^\circ} = \overline{\mu(X \times G^\circ)} \subseteq G^\circ$. By the remark of direct product of irreducible algebraic set, we konw that $\overline{X \cdot G^\circ}$ is irreducible closed subset of G. As $X, G^\circ \subseteq \overline{X \cdot G^\circ}$ as maximal closed irreducible closed subsets, then $X = \overline{X \cdot G^\circ}$, and $G^\circ = \overline{X \cdot G^\circ}$. Take $X = G^\circ$, we know $G^\circ \cdot G^\circ \subseteq \overline{G^\circ \cdot G^\circ} = G^\circ$, thus G° is closed under multiplication. Further, $\iota(G^\circ)$ is also an irreducible component with 1, thus closed under multiplication, it must be G°. G° is a closed subgroup of G.
For some $x \in G$, define the right translation map $R_x : G \to G, g \mapsto R_x(g) := \mu(g,x)$, it is regular since it is the composition of two regular maps $\delta_x : g \mapsto (g,x)$ and μ, with $R_x \circ R_y = R_{yx}, \forall x,y \in G$. For any x, R_x is an isomorphism with inverse R_{-x}. Then the cosets $G^\circ x = R_x(G^\circ)$ is an irreducible component of G.
Conversely, for any irreducible component X, choose $x \in G$ such that $x^{-1} \in X$, then $R_x X$ is irreducible component of G with 1, it must be G°, $X = G^\circ x^{-1}$ is a coset of G°. By finiteness of irreducible components, G° has finite index in G. The normalness is obvious. Hence we proved (i).
Let $H \subseteq G$ be closed subgroup of finite index. Then $G = \coprod_{i=1}^r H_{g_i}$, $G^\circ = G^\circ \cap G = G^\circ \cap (\coprod_{i=1}^r H_{g_i}) = \bigcup_{i=1}^r (G^\circ \cap Hg_i)$. Each Hg_i is closed subset of G, since G° is irreducible, $G^\circ = G^\circ \cap H_{g_i}$, $G^\circ \subseteq H_{g_i}$. Because $1 \in G^\circ$, $g_i = 1$, hence $G^\circ \subseteq H$, as desired.
(iii) is clear by (i). $\square$

Corollary 1.1.14. $\dim G = \dim G^\circ$.

Proof. Since any irreducible component of G is the isomorphic image under regular map R_x, for some $x \in G$, then dimensions of all irreducible components of G are as same as G°.
Use the fact that $\dim G = max \dim(G_i)$, G_i irreducible component ([Har77], Chap 1, Ex1.10), we proved the corollary. $\square$

1.1.2 Algebraic Groups of Classical Type

In the previous examples, we have seen $\mathrm{SL}_n(k)$ and $\mathrm{SL}_n^{(e)}(k)$, in this section, we would like to give constructions of other algebraic groups of Classical type.

Definition 1.1.15. Let $Q \in M_n(k)$ be an invertible matrix, define

$$\Gamma_n(Q,k) := \{A \in M_n(k) | A^T Q A = Q\}$$

From the above relationship, one immediately get that $\det(A) = \pm 1$, and if $A, B \in \Gamma_n(Q,k)$, then AB, A^{-1} are in $\Gamma_n(Q,k)$, and $\Gamma_n(Q,k)$ is closed if one writes down expression of entries of $A^T Q A = Q$. Thus $\Gamma_n(Q,k) \subseteq \mathrm{SL}_n^{(2)}(k)$ is a linear algebraic group, call it **classical group**.

Remark. Suppose $Q' \in M_n(k)$ is another invertible matrix, if there is some invertible $R \in M_n(k)$ such that $Q' = R^T Q R$, then $\phi_R : \Gamma_n(Q',k) \to \Gamma_n(Q,k), A \mapsto RAR^{-1}$ is an isomorphism of algebraic groups, we say Q' and Q are equivalent.
Q could define a bilinear form $\beta_Q : k^n \times k^n \to k, (v,w) \mapsto \beta_Q(v,w) := v^t Q w$, this bilinear form does not depend

on the choice of representation element in equivalent class of Q. There are two possible circumstances for this bilinear form.

(i) Suppose $\mathrm{char}(k) \neq 2$. We say it is **symmetric** if $\beta_Q(v,w) = \beta_Q(w,v), \forall v, w \in k^n$, this case is the same as $Q = Q^T$.

(ii) We say it is **alternating** if $\beta_Q(v,w) = -\beta_Q(w,v), \forall v, w \in k^n$. This condition implies that $Q = -Q^T$.

Example 1.1.16. **(i)** $\mathrm{char} k \neq 2$. Take $Q_n = \begin{bmatrix} 0 & \cdots & 0 & 1 \\ \vdots & \cdots & \cdots & 0 \\ 0 & 1 & \cdots & \vdots \\ 1 & 0 & \cdots & 0 \end{bmatrix} \in M_n(k)$. Then $\Gamma_n(Q_n, k)$ is called the **orthogonal group**, denote by $O_n(k)$. However, $O_n(k)$ could be checked not being irreducible, it is the union of two cosets of the closed normal subgroup $SO_n(k) := O_n(k) \cap \mathrm{SL}_n(k)$, the **special orthogonal group**.

When $n = 1$, $SO_1(k) = \{1\}$ and $O_1(k) = \{\pm 1\}$. When $n = 2$, $SO_2(k) = \left\{ \begin{bmatrix} a & 0 \\ 0 & a^{-1} \end{bmatrix} | 0 \neq a \in k \right\}$.

(ii) Let k have any characteristic. In this case $n = 2m$ must be even. Define

$$Sp_{2m}(k) := \Gamma_{2m}\left(Q_{2m}^-, k\right), \text{where } Q_{2m}^- := \left[\begin{array}{c|c} 0 & Q_m \\ \hline -Q_m & 0 \end{array}\right] \in M_{2m}(k), \text{and } Q_m \text{ is the same as in Eg(i).}$$

It is called the **symplectic group**. When $n = 2, Sp_2(k) = \mathrm{SL}_n(k)$.

Example 1.1.17. Let Q_{2m} be the same as the above example, let $P \in M_{2m}(k)$ such that $Q_{2m} = P + P^T$. Define the polynomial of $k[X_1, ..., X_{2m}]$.

$$f_P := \sum_{i,j=1}^{2m} p_{ij} X_i X_j \in k\left[X_1, \ldots, X_n\right]$$

And define the corresponding subset of $M_{2m}(k)$

$$\Gamma_{2m}\left(f_P, k\right) := \left\{A \in M_{2m}(k) | f_P(Ax) = f_P(x) \text{ for all } x \in k^{2m}\right\}$$

$\Gamma_{2m}(f_P, k)$ just defined is in fact a linear algebraic group. First, it is closed in $M_{2m}(k)$ since the condition implies a polynomial expression of elements of $\Gamma_{2m}(f_P, k)$. Second, it is closed under multiplication since $f_P(ABx) = f_P(A(Bx)) = f_P(Bx) = f_P(x)$ if $A, B \in \Gamma_{2m}(f_P, k)$. Finally, $\forall A \in \Gamma_{2m}(f_P, k)$, it has its inverse in $\Gamma_{2m}(f_P, k)$, since $f_P(x+y) - f_P(x) - f_P(y) = \sum_{i,j=1}^{2m} \left(p_{ij} + p_{ji}\right) x_i y_j = \beta_{Q_{2m}}(x,y)$. Thus $A^T Q_{2m} A = Q_{2m}, \det(A) = \pm 1$, and $\Gamma_{2m}(f_P, k) \subseteq \Gamma_{2m}(Q_{2m}, k)$ as a closed subset, thus the existence of inverse is proved.

Notice that the condition in the definition of $\Gamma_{2m}(f_P, k)$ implies $f_P(Ae_i) = f_P(e_i)$, and those $2m$ equations also shows that $(A^T PA)_{ii} = P_i i$, then $\Gamma_{2m}(f_P, k)$ can also be defined as

$$\Gamma_{2m}\left(f_P, k\right) = \left\{A \in \Gamma_{2m}\left(Q_{2m}, k\right) | \left(A^T PA\right)_{ii} = P_{ii} \text{ for } 1 \leqslant i \leqslant 2m\right\}$$

The equivalence of $\Gamma_{2m}(f_{P'}, k)$ and $\Gamma_{2m}(f_P, k)$ is provided if there is some invertible $R \in M_{2m}(k)$ such that $R^T Q_{2m} R = Q_{2m}$ and $f_P(Rx) = f_{P'}(x), \forall x \in k^{2m}$. Now let

$$f_{2m} := \sum_{i=1}^{m} X_i X_{2m+1-i}, \text{the determining matrix is } P_{2m} := \left[\begin{array}{c|c} 0 & Q_m \\ \hline 0 & 0 \end{array}\right], P_{2m}^T = \left[\begin{array}{c|c} 0 & 0 \\ \hline Q_m & 0 \end{array}\right]$$

Denote the corresponding $\Gamma_{2m}(f_P, k)$ by $O_{2m}^+(k)$.

(a) If $\mathrm{char}\, k \neq 2$, $O_{2m}^+(k) = O_{2m}(k)$, and $SO_{2m}^+(k) := O_{2m}^+(k) \cap SL_{2m}(k) = SO_{2m}(k)$.

(b) If $\mathrm{char}\, k = 2$, $\Gamma_{2m}(Q_{2m}, k)$. And $O_{2m}^+(k)$ is not connected, take $SO_{2m}^+(k) := O_{2m}^+(k)^\circ$.

The table of algebraic groups of classical types is given following.

1.1.3 Structure Theory and Finite Group of Lie Type

In this section, we shall introduce more structure theory of algebraic groups over infinite algebraically closed fields k , and their finite subgroup fixed by variety morphisms. Thus, for clarity, we use the **bold font** to denote algebraic groups

	G	Condition
Type A_{m-1}	$SL_m(k)$	any chara
Type B_m	$SO_{2m+1}(k)$	char$(k) \neq 2$
Type C_m	$Sp_{2m}(k)$	any chara
Type D_m	$SO^+_{2m}(k)$	any chara

over infinite algebraically closed fields. We may use notions in Remark 1.1.11.
Without special notations, we do not demand the connectedness of algebraic groups occur in this section.

Definition 1.1.18. (i)Define the **multiplicative group** $\mathbb{G}_m$ to be $\mathcal{O}(\mathbb{G}_m) = k[X, X^{-1}]$ with multiplication $\mu^* : X \mapsto X \otimes X$. Define the **additive group** $\mathbb{G}_a$ to be $\mathcal{O}(\mathbb{G}_a) = k[X]$ with multiplication $\mu^* : X \mapsto X \otimes 1 + 1 \otimes X$.
(ii) A **torus** of an algebraic group is a finite product of multiplicative groups.
(iii)Define the **rational character** of $\mathbf{G}$ to be one of $\mathrm{Hom}(\mathbf{G}, \mathbb{G}_m)$. The **character group** $X(\mathbf{G})$ of $\mathbf{G}$ is defined to be group of its rational characters.

Definition 1.1.19. Consider the commutator group $[\mathbf{H}_1, \mathbf{H}_2]$ that is generated by $g^{-1}h^{-1}gh, g \in \mathbf{H}_1, h \in \mathbf{H}_2$. We can construct a series $\mathbf{G} \supseteq \mathbf{G}^{(1)} \supseteq ... \supseteq \mathbf{G}^{(i)} \supseteq ...$, where $\mathbf{G} = \mathbf{G}^{(0)}$ and $\mathbf{G}^{(i)} = [\mathbf{G}^{(i-1)}, \mathbf{G}^{(i-1)}]$, notice each $\mathbf{G}^{(i)}$ is normal and $\mathbf{G}^{(i)}/\mathbf{G}^{(i+1}$ is commutative. We call $\mathbf{G}$ a **solvable group** if the series descends to 1 in finite length.
The similar series is constructed as $\mathbf{G} \supseteq \mathbf{G}^1 \supseteq ... \supseteq \mathbf{G}^i \supseteq ...$ where $\mathbf{G} = \mathbf{G}^0$ and $\mathbf{G}^i = [\mathbf{G}, \mathbf{G}^{i-1}]$, We call $\mathbf{G}$ a **nilpotent group** if the series descends to 1 in finite length.
It is clear from the construction of the series, nilpotent group is solvable.

Definition 1.1.20. Define the **unipotent radical** of a connected $\mathbf{G}$ to be $R_u(\mathbf{G}) := \bigcap_{\chi \in X(\mathbf{G})} \mathrm{Ker}\, \chi$. $\mathbf{G}$ is called **reductive** if its unipotent radical is trivial.
Define the **radical** of a connected linear algebraic group $\mathbf{G}$ to be $R(\mathbf{G}) := \langle \mathbf{H} \subseteq \mathbf{G} | \mathbf{H}$ connected, normal, solvable subgroup $\rangle$. $\mathbf{G}$ is called **semisimple** if its radical is trivial.

Remark. In fact, we have $R_u(\mathbf{G}) = (R(\mathbf{G}))_u$, then it shows a semisimple group implies a reductive group. Also, if a connected $\mathbf{G}$ is reductive, then its commutator group is semisimple.

Example 1.1.21. We have $\mathbf{GL}_n$ being reductive, and $\mathbf{SL}_n$ being semisimple.

Proposition 1.1.22. *(i) (Rigidity of tori) If $\boldsymbol{T}$ is a torus in $\boldsymbol{G}$, then $N_{\boldsymbol{G}}(\boldsymbol{T})^\circ = C_{\boldsymbol{G}}(\boldsymbol{T})^\circ$, where $N_{\boldsymbol{G}}(\boldsymbol{T})$ is the normalizer of $\boldsymbol{T}$ in $\boldsymbol{G}$ and $C_{\boldsymbol{G}}(\boldsymbol{T})$ the centralizer.*
(ii) If $\boldsymbol{G}$ is connected, all maximal tori in $\boldsymbol{G}$ are conjugate.
(iii) If connected $\boldsymbol{G}$ is solvable, it is nilpotent if and only if one (hence for all) maximal torus in contained in $Z(\boldsymbol{G})$.

Definition 1.1.23. Call the maximal closed connected solvable subgroup of some algebraic group the **Borel subgroup**.

Theorem 1.1.24. *For a connected algebraic group $\boldsymbol{G}$, denote by $\boldsymbol{B}$ its Borel subgroup, we have*
(i) All Borel subgroup are conjugate in $\boldsymbol{G}$.
(ii) Every element in $\boldsymbol{G}$ is contained in some Borel subgroup.
(iii) $N_{\boldsymbol{G}}(\boldsymbol{B}) = \boldsymbol{B}$.

Definition 1.1.25. For a group G (do not need to be algebraic here), a $(B, N)-$ **pair** or a **Tits system** means two subgroups B, N of G and a set S satisfies the following
(i) G is generated by B and N.
(ii) $H = B \cap N$ is normal in N, $W := N/B \cap N$ is generated by S, and it is a Coxeter group of order 2, called the **Weyl group**.
(iii) For $s \in S$, n_s the representative element of s in N, then $n_s B n_s \neq B$.
(iv) For all $s \in S, n \in N$, $n_s B n \subseteq B n_s n B \cup B n B$.
(v) $\bigcap_{n \in N} nBn^{-1} = H$.

Remark. Call $C_w = Bn_wB$ the **Bruhat cells** and define $l(w)$ to be the minimal length of the expression of w in its generators in S. Notice that for $s \in S$, $l(sw) = l(w) \pm 1$, then actually the condition (iv) could be refined by the following proposition.

Proposition 1.1.26 (Bruhat Decomposition)**.** *The group G could be decomposed into*

$$G = \coprod_{w \in W} Bn_wB$$

Moreover, we will get a refined condition instead of condition (iv) in Definition 1.1.25 .

$$Bn_sB \cdot Bn_wB = \begin{cases} Bn_sn_wB & \text{if } l(sw) = l(w) + 1 \\ Bn_sn_wB \cup Bn_wB & \text{if } l(sw) = l(w) - 1 \end{cases}$$

Proof. Take $S_0 \subseteq S$ be a subset of S, let W_{S_0} denote the subgroup of W generated by S_0, and N_{S_0} denote the preimage of W_{S_0} in N. Set $P_{S_0} := BN_{S_0}B$, we claim it is a subgroup of G.
Want to show $P_{S_0} \supseteq P_{S_0}P_{S_0} = BN_{S_0}B \cdot BN_{S_0}B$. Since $BP_{S_0} \subseteq P_{S_0}$, only need to show $N_{S_0}P_{S_0} \subseteq P_{S_0}$. And it is enough to show $n_sP_{S_0} \subseteq P_{S_0}$, which is clear by condition (iv). Take $S_0 = S$, then $G = BN_SB = BNB$, it proves the decomposition. Proof the resting formula is based on combinatorics of Coxeter group, which is omitted here. □

We would like to fit the above settings into (connected) reductive group **G** and its Borel subgroups. For any maximal torus, use the solvability of $C_{\mathbf{G}}(\mathbf{T})$, we have $\mathbf{T} \subseteq C_{\mathbf{G}}(\mathbf{T})$ is contained in some Borel. Since any two of Borels are conjugate, then any Borel subgroup contains a maximal torus **T**. Notice also

Lemma 1.1.27. *Nontrivial minimal closed unipotent subgroups that are normalized by maximal torus $\boldsymbol{T}$ are isomorphic to $\mathbb{G}_a$. Since the action of $\boldsymbol{T}$ on $\mathbb{G}_a$ is given as $x \mapsto \alpha(t)x$, $\alpha \in X(\boldsymbol{T})$, these minimal unipotent subgroups $\boldsymbol{U}_\alpha$ are parametrized by $\alpha \in X(\boldsymbol{T})$. Moreover, $\boldsymbol{G}$ is generated by $\boldsymbol{T}$ and $\{\boldsymbol{U}_\alpha\}$.*
Let $\Sigma := X(T) \otimes \mathbb{R}$, then $N_{\boldsymbol{G}}(\boldsymbol{T})/\boldsymbol{T}$ isomorphic to the Weyl group of Φ. There is a bijection between Borel subgroups containing $\boldsymbol{T}$ and basis of Σ.

Proof. See [DM91]. □

Theorem 1.1.28. *Suppose $\boldsymbol{G}$ is a connected reductive group, then if we take B as some Borel subgroup $\boldsymbol{B}$, and N as $N_{\boldsymbol{G}}(\boldsymbol{T})$, for some maximal torus contained in $\boldsymbol{B}$, they satisfies the axiom of $(B, N)-$pair with the Weyl group $N_{\boldsymbol{G}}(\boldsymbol{T})/\boldsymbol{T}$, and its generator set $S = \{s_\alpha\}$, where α running over the basis of Σ as in Lemma 1.1.27.*

We hence get a first glimpse at the root system and Weyl group of connected reductive group **G**. The Bruhat decomposition gives **G** a partition $\mathbf{G} = \coprod_{w \in W(\mathbf{T})} \mathbf{B}w\mathbf{B}$, the disjoint union of so called **Bruhat cells**.

Remark. We shall remark here that there are quite many geometries lying in the above notion. For example, $\mathbf{G}/\mathbf{B}$ is a homogeneous space, and hence a projective variety, called (full) flag variety. We can read off from the Bruhat decomposition that orbit of **B** action on $\mathbf{G}/\mathbf{B}$ is in bijection with $W(\mathbf{T})$, in fact, it is also isomorphic to $\mathbf{G}-$diagonal orbit on $\mathbf{G}/\mathbf{B} \times \mathbf{G}/\mathbf{B}$. This is kind of origin of geometric representation theory, one can refer to [Nei10] for details, which discuss mostly on $\mathbb{C}$.

Definition 1.1.29. Define the **Parabolic subgroup** of **G** to be the closed subgroup that containing some Borel subgroup.

Proposition 1.1.30. *As a corollary of Theorem 1.1.28, for a connected reductive group $\boldsymbol{G}$, if we fix a maximal torus $\boldsymbol{T}$ and a Borel $\boldsymbol{B}$ containing $\boldsymbol{T}$, then every Parabolic subgroup is conjugate to some $\boldsymbol{P}_J = \boldsymbol{B}W_J\boldsymbol{B}$, where W_J is subgroup of $W(\boldsymbol{T})$ that generated by subset $J \subseteq S$. We call P_J the **standard Parabolic subgroup**.*

Definition 1.1.31. By **Levi decomposition** for a subgroup $\mathbf{P} \subseteq \mathbf{G}$, we mean the partition $\mathbf{P} = R_u(\mathbf{P}) \rtimes \mathbf{L}$, for **L** some closed subgroup of **P**. Call **L** the **Levi subgroup** of **P**.

1.1.4 Rationality and Finite Group of Lie Type

Now we fix the algebraically closed field to be $k = \bar{\mathbb{F}}_q$, of odd characteristic p.

Definition 1.1.32. Consider the affine variety $\mathbf{X}$ over k with its algebra A. $F : \mathbf{X} \to \mathbf{X}$. is a morphism which induces $F^* : A \to A$. We say $(\mathbf{X}, A)$ has a $\mathbb{F}_q$ **structure** (or **defined over** $\mathbb{F}_q$) if F^* is injective , $F^*(A) = A^q$ and for $\forall f \in A$, $\exists n \geq 1 \in \mathbb{N}$ such that $(F^*)^n(f) = f^{q^n}$ and F here is called the **Frobenius map**. Denote by $\mathbf{X}^F$ the set of $F-$fixed points of $\mathbf{X}$, called $\mathbb{F}_q$ **rational points** of $\mathbf{X}$.
For the affine group variety $\mathbf{G}$, it is said to be **defined over** $\mathbb{F}_q$ if $\mu : \mathbf{G} \to \mathbf{G}$, $\iota : \mathbf{G} \to \mathbf{G}$ are defined over $\mathbb{F}_q$, equivalently, there exists a Frobenius map $F : \mathbf{G} \to \mathbf{G}$ that commutes with μ and ι.

Remark. We sometimes see the conceptions of **geometric** or **arithmetic Frobenius maps**. By geometric Frobenius map we mean a variety $\mathbf{X}_0$ defined over $\mathbb{F}_q$ with a morphism $F = F_0 \otimes \text{id}$ on $\mathbf{X} = \mathbf{X}_0 \otimes_{\mathbb{F}_q} k$ such that F_0 induces the map $A_0 \to A_0, f \mapsto f^q$. It is equivalent to the above definition 1.1.32, and thus we neglect the word geometric. By arithmetic Frobenius map we mean a morphism Φ on $\mathbf{X}$ induced by $\text{Gal}(k/\mathbb{F}_q) \ni \sigma : x \mapsto x^q$. Notice $F \circ \Phi$ on $A = A_0 \otimes_{\mathbb{F}_q} k$ is $f \otimes \lambda \mapsto f^q \otimes \lambda^q$. Their relation with the Definition 1.1.32 is shown in the following lemma. And by **generalized Frobenius map** we mean $F : \mathbf{G} \to \mathbf{G}$ such that $\exists d \geq 1$, F^d is the Frobenius morphism.

Lemma 1.1.33. *Let $A_0 := \{f \in A | F^*(f) = f^q\}$, where F is the Frobenius map over $(\mathbf{X}, A)$ define in 1.1.32. Then A_0 is a finitely generated $\mathbb{F}_q$ subalgebra of A, and there is an isomorphism $A_0 \otimes_{\mathbb{F}_q} k \cong A$. As is said in the Remark of 1.1.32, $F \circ \sigma$ is the q power action on A, we may also write the arithmetic Frobenius map as $\sigma : A \to A, f \mapsto (F^*)^{-1}(f^q)$. Then, we have $A_0 := \{f | f = \sigma(f)\} = \{f \in A | F^*(f) = f^q\}$.*

Proof. Fix $f \in A$, let S_f denote the set $\langle \sigma^i(f) | i = 0, 1, 2... \rangle_k$, by definition of F^*, we have $\sigma^n(f) = f$ for some $n \geq 1$. Consider $\mathbb{F}_{q^n} = \mathbb{F}_q[\zeta_1, ..., \zeta_{n-1}]$, define $\tilde{f}_i := \sum_{j=0}^{n-1} \sigma^j(\zeta_i f) = \sum_{j=0}^{n-1} \zeta_i^{q^j} \sigma^j(f)$, we have

$$\sigma(\tilde{f}_i) = \sum_{j=0}^{n-1} \sigma^{j+1}(\zeta_i f) = \tilde{f}_i$$

Since the matrix $\left(\zeta_i^{q^j}\right)_{0 \leq i,j \leq n-1}$ is Vandermonde, thus invertible, we have $S_f = \left\langle \tilde{f}_i | 0 \leq i \leq n-1 \right\rangle$. Choose f^j to run over the finite generators of A, we have finite k basis by taking generators of S_{f^j}. □

Proposition 1.1.34. *Suppose affine variety $(\boldsymbol{X}, A)$ has a $\mathbb{F}_q$ structure with Frobenius map F, then there exists an integer $n \geq 1$ and a closed embedding $\phi_n : \boldsymbol{X} \to k^n$ such that the following diagram commutative. Here F_q is the standard Frobenius map that induces F_q^* mapping an element to its $q-$power. Moreover, $\boldsymbol{X}^F$ is a finite set, call its elements **rational points**.*

$$\begin{array}{ccc} \boldsymbol{X} & \xrightarrow{\phi_n} & k^n \\ {\scriptstyle F}\downarrow & & \downarrow{\scriptstyle F_q} \\ \boldsymbol{X} & \xrightarrow{\phi_n} & k^n \end{array}$$

Proof. Notice $A = k[X_1, ..., X_n]/I$ for some n and as is proved in 1.1.33, we have $A_0 = \mathbb{F}_q[X_1, ..., X_n]/I_0$. Take ϕ_n to be the morphism that induces the projection $p : k[X_1, ..., X_n] \to A$, it is a closed embedding, then $p(F_q^*(f)) = p(f)^q = F^*(\pi(f))$, hence $(\phi_n \circ F)^* = (F_q \circ \phi_n)^*$, the commutativity is hence shown.
Notice $\phi_n(\mathbf{X}^F)$ is fixed by F_q, and ϕ_n is injective. Thus $\phi_n(\mathbf{X}^F)$ is a subset of $\mathbb{F}_q^n$, and $\left|\mathbf{X}^F\right|$ is finite. □

Remark. For a affine group variety $\mathbf{G}$ that has a $\mathbb{F}_q$ structure, we thus have $\mathbf{G}^F$ a finite group, called **finite algebraic group**, If $\mathbf{G}(k)$ is a reductive algebraic group, then $\mathbf{G}^F$ is finite group of classical (Lie) type. Take the Frobenius map to be the standard one, then we get $\mathbf{G}^F = \mathbf{G}(\mathbb{F}_q)$. For example, if $\mathbf{G} = \mathbf{GL}$, $\mathbf{G}^F = GL(\mathbb{F}_q)$ and $|GL(\mathbb{F}_q)| = q^{\frac{n(n-1)}{2}} \prod_{i=1}^n (q^i - 1)$.

Theorem 1.1.35 (Lang-Steinberg)**.** *For connected reductive $\mathbf{G}$ over k and generalized Frobenius map $F : \mathbf{G} \to \mathbf{G}$, define **the Lang map** to be $L : \mathbf{G} \to \mathbf{G}, g \mapsto g^{-1}F(g)$. Then L is dominant and moreover surjective.*

Proof. We first prove the dominance of L, that is $\overline{L(\mathbf{G})} = \mathbf{G}$. If $L(x) = L(y)$, then $xy^{-1} = F(xy^{-1})$, $xy^{-1} \in \mathbf{G}^F$. Thus fibres of L are finite, which implies $\dim \mathbf{G} = \dim \overline{L(\mathbf{G})}$. Notice $\mathbf{G}$ is connected, and $\overline{L(\mathbf{G})}$ is irreducible, thus $\mathbf{G} = \overline{L(\mathbf{G})}$, L is dominant.
Moreover, since finite and dominant morphism implies surjective morphism, only need to show L is finite. Denote by $N_d : g \mapsto gF(g)F^2(g)...F^{d-1}(g)$ and $L_d : g \mapsto g^{-1}F^d(g)$, then $L_d = N_d \circ L$ by straightforward computation. We prove L_d is a finite morphism, since F^d is a Frobenius map, thus we can assume $d = 1$ without losing generality. We can write $A = k[e_1, ..., e_n]$ where $e_i \in A_0, i = 1, ..., n$, similar to the diagram in Proposition 1.1.34, we can replace k^n by $\mathbf{GL}_n(k)$, hence $\mu^*(e_j) = \sum_{j=1}^n e_i \otimes a_{ij}$, where $a_{ij} \in A$. Consider

$$F^*(e_i)(x) = e_i^q(x) = e_i(F(x)) = e_i(\mu(x, L(x))) = \mu^*(e_i)(x, L(x)) = \sum_{j=1}^n e_j(x) \otimes a_{ij}(L(x)) = (\sum_{j=1}^n e_i L^*(a_{ij}))(x)$$

we have $e_i^q = e_j \sum_{j=1}^n L^*(a_{ij})$, then A is a finite generated $L^*(A)$ module, hence L is a finite morphism. □

We have some immediate corollaries

Corollary 1.1.36. *If a connected algebraic group $\mathbf{G}$ defined over $\mathbb{F}_q$ acting on $\mathbf{X}$ defined over $\mathbb{F}_q$ where the action is also defined over $\mathbb{F}_q$ (stable under F). Then any $F-$stable $\mathbf{G}-$orbit has a rational point.*

Proof. Suppose $x \in \mathbf{X}$ is of $F-$stable orbit, $\exists g \in \mathbf{G}$ with $F(x) = g^{-1}.x$, since $g = h^{-1}F(h)$ for some $h \in H$ by Theorem 1.1.35, then $h.x$ is fixed by F. □

Corollary 1.1.37. *For a connected algebraic group $\mathbf{G}$ defined over $\mathbb{F}_q$ with Frobenius morphism F, if $\mathbf{H} \subseteq \mathbf{G}$ is closed, connected and rational (stable under F) subgroup. Then $(\mathbf{G}/\mathbf{H})^F = \mathbf{G}^F/\mathbf{H}^F$.*

Proof. By Corollary 1.1.36, we have $\mathbf{G}^F/\mathbf{H}^F \to (\mathbf{G}/\mathbf{H})^F$ is surjective. If $g_1, g_2 \in \mathbf{G}^F$ are in the same $\mathbf{H}$ coset, then $xy^{-1} \in \mathbf{H}^F$. Thus the map is also injective. □

Corollary 1.1.38. *We can verify for $\mathbf{G}$ with BN-pair $\mathbf{B}, \mathbf{N}, \mathbf{H}$. The axiom of BN-pair holds also for $\mathbf{G}^F = \langle \mathbf{B}^F, \mathbf{N}^F \rangle$. $W^F = N_{\mathbf{G}}(\mathbf{T})^F/\mathbf{T}^F$ by Corollary 1.1.37 and $\mathbf{G}^F = \coprod_{w \in W^F} \mathbf{U}^F\mathbf{H}^F w \mathbf{U}_w^{-F}$.*

Proof. We shall see this statement later in 1.2.13. □

1.2 Iwahori-Hecke Algebras

1.2.1 Hecke Algebra Over Local Fields

Let G be a locally compact Hausdorff group, and $Z(G)$ be its center, F be a nonarchimedean local field. (π, V) representation of G, where V is the vector space over F.

Definition 1.2.1. Call the representation π **smooth** if $\{k \in G | \pi(k)v = v\}$ is open. For any open subgroup $K \subseteq G$, denote $V^K = \{v \in V | \pi(k)v = v, \forall k \in K\}$. To say π is smooth is also equivalently to have any $v \in V$ located in some V^K. Further, π is called **admissible** if V^K is additionally finite-dimensional.

A representation is admissible means we can translate it into finite dimension case since each V^K is finite. Before doing so, we first give some properties of smooth and admissible representations.
If (π, V) is a representation of G over F, then $(\pi, V \otimes_F E)$ is an extended representation, for E/F any field extension.

Proposition 1.2.2. *$(\pi, V \otimes_F E)$ is smooth(resp. admissible, resp. finitely generated) if and only if (π, V) is smooth(resp. admissible, resp. finitely generated).*

Proof. It is immediate by $(V\otimes_F E)^K = V^K\otimes_F E$. We call (π, V) **completely irreducible** if $(\pi, V\otimes_F E)$ is irreducible for evert extension E/F. $\square$

Assign G a rational Haar measure such that for any compact open subgroup K, $|K|\in\mathbb{Q}$. Then for a smooth representation (π, V) of G and a open subgroup K, define $\mathcal{P}_K(v) = \frac{\int_K \pi(k)vdk}{|K|}$. The operator $\mathcal{P}_K$ gives a projection from V to V^K. Denote by V_K the kernel of $\mathcal{P}_K$, then it is isomorphic to the space $\{\pi(k)v - v|k\in K\}$, and $V = V^K\oplus V_K$. The subspace V_K is importatnt to define the Jacquet module later.

Proposition 1.2.3. *If (π, V) is a smooth representation, then it is also admissible if and only if the restriction of π to K is a direct sum of irreducible finite-dimensional representations, K is any compact open subgroup.*

Proof. Assume π is admissible, for any K_1 normal open subgroup of K, $V = V^{K_1}\oplus V_{K_1}$, and the action of K on each summand is stable. Then V^{K_1} could be considered as a representation of K/K_1, is a direct sum of finite irreducible representations of K. Then one using Zorn's Lemma on the rest part V_{K_1} can show that V decomposes into direct sum of irreducible finite dimensional smooth representations of K. $\square$

Lemma 1.2.4. *The category of smooth and admissible representations are abelian categories*

Proposition 1.2.5. *Let (π_i, V_i) be smooth G representations, and K a compact, open subgroup, then $V_1\to V_2\to V_3$ exact sequence implies exactness $V_1^K\to V_2^K\to V_3^K$*

Proposition 1.2.6. *Let (π_i, V_i) be smooth G representations, and consider the exact sequence $0\to V_1\to V_2\to V_3\to 0$. The mid one is admissible if and only if the side two are.*

Definition 1.2.7. For each K compact open subgroup, define the Hecke algebra $\mathcal{H}_F(G, K)$, or abbreviate it as $\mathcal{H}_K$, to be the space of all functions $\phi: G\to F$ such that $\phi(k_1gk_2) = \phi(g)\forall k_1, k_2\in K$. The multiplication of ring structure is the convolution product $\phi_1 * \phi_2(g) = \int_G \phi_1(gx^{-1})\phi_2(x)dx$. Define $\mathcal{H}_F(G) := \bigcup_K \mathcal{H}_F(G, K)$.

If (π, V) is a smooth representation, then it could also be a $\mathcal{H}_F(G)-$ module by the action

$$\pi(\phi)(v) := \int_G \phi(g)\pi(g)vdg$$

Proposition 1.2.8. *If $(\pi_i, V_i), i = 1, 2$ are two smooth representations of G, then there is an isomorphism between morphisms in G smooth representation category and in $\mathcal{H}_G$ module category, induced by the natural map. That is,* $\mathrm{Hom}_G(V_1, V_2) = \mathrm{Hom}_{\mathcal{H}_G}(V_1, V_2)$.

Proof. Assume $f\in \mathrm{Hom}_G(V_1, V_2)$, then it is automatically a $\mathcal{H}_G$ morphism from above discussion. Conversely, assume $f\in \mathrm{Hom}_{\mathcal{H}_G}(V_1, V_2)$, then for any $v_1\in V_1$ and $g\in G$, $v_1\in V^{K_0}$ for some compact open K_0 from smoothness, if $f(v_1)\in V_2^{K_0'}$, one can find K compact, open such that $v_1\in V_1^K, \pi_1(g)v_1\in V_1^K, f(v_1)\in V_2^K, \pi_2(g)(f(v_1))\in V_2^K$, where $\pi_1(g)(v)$ can be understand as $\frac{\pi_1(KgK)v}{|KgK|}$, then we can apply $\mathcal{H}(G, K)$ module morphism f

$$f(\pi_1(g)v) = \frac{f(\pi_1(KgK)v)}{|KgK|} = \frac{\pi_2(KgK)f(v)}{|KgK|} = \pi_2(g)f(v).$$

Thus f is also a G smooth representation morphism. $\square$

Theorem 1.2.9. *(i) If (π, V) is an irreducible, admissible representation and $V^K\neq 0$, then V^K is a simple $\mathcal{H}_K-$module. (ii) If (π_i, V_i) are two irreducible, admissible representations, and $V_1^K\simeq V_2^K$ as $\mathcal{H}_K-$module, then $\pi_1\simeq\pi_2$.*

Proof. [Bum10], P20. $\square$

1.2.2 Hecke Algebras Of Finite Groups

G is a finite group, and (π, V) its representation, $(\hat{\pi}, \hat{V})$ its contragredient. Here $\hat{V} \subseteq V^*$ the dual space of V.
Consider $\mathcal{H}$ containing the functions $G \to \mathbb{C}$. Similar to the local case, it has a ring structure with convolution. In this case, the convolution will become $(\phi_1 * \phi_2)(g) = \frac{1}{|G|}\sum_{x\in G}\phi_1(x)\phi_2(x^{-1}g) = \frac{1}{|G|}\sum_{x\in G}\phi_1(gx)\phi_2(x^{-1})$.
Then there is a ring isomorphism from $\mathcal{H}$ to group algebra $\mathbb{C}[G]$ given by the map

$$\phi \in \mathcal{H} \mapsto \phi' = \frac{1}{|G|}\sum_{g\in G}\phi(g)g \in \mathbb{C}[G]$$

If (π, V) a complex representation of G, then it is also a $\mathcal{H}-$module, $\pi(\phi)(v) := \frac{1}{|G|}\sum_{g\in G}\phi(g)\pi(g)v$, and it is easy to check $\pi(\phi_1 * \phi_2) = \pi(\phi_1)\circ\pi(\phi_2)$.

Definition 1.2.10. Similarly, we define the **Hecke algebra** $\mathcal{H}_K$ to the $K-$biinvariant subspace of $\mathcal{H}$, where K is a subgroup of G.

Also, we consider the $K-$fixed subspace V^K, it is a $\mathcal{H}_K-$module, as an analog of the above theorem, we have

Theorem 1.2.11. *(i) If (π, V) is an irreducible representation and $V^K \neq 0$, then V^K is an irreducible $\mathcal{H}_K-$module.*
(ii) If (π_i, V_i) are two irreducible, admissible representations, and $V_1^K \simeq V_2^K$ as $\mathcal{H}_K-$module, then $\pi_1 \simeq \pi_2$.

Now for any subgroup $H \subseteq G$, with $\sigma \in \mathrm{Irr}(H)$ a linear character, define the Hecke algebra associates to σ by $\mathcal{H}_{G,\sigma} := \{\Delta : G \to \mathbb{C} | \Delta(h_1 g h_2) = \sigma(h)\Delta(g)\sigma(h_2)\}$. The ring multiplication is given by the convolution of map, and we get an isomorphism $\mathrm{End}_G(\mathrm{Ind}_H^G\,\sigma) \simeq \mathcal{H}_{G,\sigma}$. An immediate consequence of this isomorphism is that $\mathcal{H}_{G,\sigma}$ is commutative if and only if $\mathrm{Ind}_H^G\,\sigma$ is multiplicity free (i.e each irreducible component occurs only once).

Now consider a Coxeter group (W, S) where S is the set of simple reflection generators, we can associate W with Iwahori-Hecke algebra $\mathcal{H}_q(W)$ of generators $T_i, s_i \in S$ satisfying the braid relation $T_iT_jT_i... = T_jT_iT_j...$, the length of the braid depends on the order of s_is_j. And T_i satisfies the quadratic relation $T_i^2 = (q-1)T_i + q\,\mathrm{id}$ (Assume the field contains q).

Proposition 1.2.12. *We can choose W to be the Weyl group of some root system and S is the set of simple reflections of Weyl chambers For each w, we can associate it with $T_w \in \mathcal{H}_q(W)$ and $T_{s_i} = T_i$.*
If $l(w) + l(v) = l(wv)$, then $T_{wv} = T_wT_v$. If $s \in S$ is a simple reflection

$$T_sT_w = \begin{cases} T_{sw} & \text{if } l(sw) > l(w) \\ (q-1)T_w + qT_{sw} & \text{if } l(sw) < l(w) \end{cases}$$

Proof. Consider a reduced decomposition of w, $w = s_{i_1}, ..., s_{i_n}, n = l(w)$, define $T_w := T_{s_1}, ..., T_{s_{i_n}}$ clearly it is in $\mathcal{H}_q(W)$. We next need to prove T_w is well defined, that is, for any two reduced decomposition $w = s_{i_1}, ..., s_{i_{l(w)}} = t_{j_1}, ..., t_{j_{l(w)}}$, $T_{s_{i_1}}...T_{s_{i_{l(w)}}} = T_{t_{j_1}}...T_{j_{l(w)}}$. By the finiteness of order of simple reflections, the Braid relation of T_i is thus in finite length, according to the word property of Coxeter groups (see [And05]) , any two reduced expression of the same element w can achieve to each other by braid moves (hence finite step moves in our case), then apply the homomorphism $s_i \mapsto T_i$ we have the equality.
Moreover, if $l(wv) = l(w) + l(v)$, we can divide the reduced expression of wv into the composition of reduced expressions of w and v separately (this is plausible, otherwise we would get a reduced form of length less than $l(wv)$), then the equality $T_{wv} = T_wT_v$. If $s \in S$ and $l(sw) > l(w)$, then $l(sw) = l(w) + 1 = l(w) + l(s)$, we have $T_sT_w = T_{sw}$. If $l(sw) < l(w)$, then $w = sv$ by word property, and thus $T_sT_w = T_s^2T_v = ((q-1)T_s + q\,\mathrm{id})T_v = (q-1)T_sT_v + qT_v = (q-1)T_w + qT_{sw}$. □

Let $G = \coprod_{w\in W} BwB$ be the Bruhat decomposition of group G defined over $\mathbb{F}_q$, with (W, S) the Weyl group, we can define a convolution ring of $B-$biinvariant functions on G, denote by $\mathcal{H}$ (over $\mathbb{C}$), then $\mathcal{H}$ has canonical basis $\phi_w, s \in S$

the characteristic function on Bruhat cell $C_w = BwB$. The ring convolution is $(f_1 * f_2)(g) = \frac{1}{|B|}\sum_{x\in G} f_1(x)f_2(x^{-1}g)$ and $\phi_s * \phi_s = (q-1)\phi_s + q\phi_1$, $\phi_w\phi_v = \phi_{wv}$, if $l(w)+l(v) = l(wv)$.

Proposition 1.2.13. *Let $G = \coprod_{w\in W} BwB$ be the finite group with $BN-$pair, $U = R_u(B)$ the unipotent group, W the Weyl group, Σ^+ (resp. Σ^-) the positive (resp. negative) roots. Denote $U_w^- = U \cap wUw^{-1}, U_w^+ = U \cap (w_0w)U(w_0w)^{-1}$, then (i) There is a bijection $U_w^- \times U_w^+ \xrightarrow{\simeq} U$. (ii) $|U_w^-| = q^{l(w)} = [U : U_w^+]$.*

Remark. If $\mathbf{G}$ is a connected, reductive algebraic group, then $\mathbf{U}_w^- = \coprod_{\alpha\in\Sigma^+, w\alpha\in\Sigma^-} \mathbf{U}_\alpha, \mathbf{U}_w^+ = \coprod_{\alpha\in\Sigma^+, w\alpha\in\Sigma^+} \mathbf{U}_\alpha$, where $\mathbf{U}_\alpha$ is stated as in Lemma 1.1.27. Associating with Corollary 1.1.37, we have $\left|\mathbf{U}_w^{-F}\right| = q^{l(w)}$ and Corollary 1.1.38 holds.

The $\mathbb{C}-$algebra homomorphism $\mathcal{H} \to \mathbb{C}$ is given by $\varphi : f \mapsto \frac{1}{|B|}\sum_{x\in G} f(x)$, notice $BwB = U_w^- wB$, and $|U_w^-| = q^{l(w)}$, then $\varphi(\phi_w) = q^{l(w)}$. The ring homomorphism $\mathcal{H}_q(W) \to \mathcal{H}, T_{s_i} \mapsto \phi_{s_i}$ gives an isomorphism.

Chapter 2

On Local Theta Correspondences

2.1 Construction of Weil Representations

In this section, we shall introduce the construction of Weil representations (over local fields), which are fundamental in the study of local theta correspondences. There are some classical references, we mainly follow Wenwei Li's master thesis [Li08], which makes a good summary.

2.1.1 Some Settings

The field F in this passage is defaulted to be a non-Archimedean local field F and $charF \neq 2$. We fix a non-trivial character $\psi : F \longrightarrow \mathbb{C}$.
Let V be a finite dimensional vector space over F.

Definition 2.1.1. Define the α-densities on V

$$\Omega_\alpha^{\mathbb{R}}(V) := \left\{ \nu : \bigwedge^{\max} V \to \mathbb{R} : \forall x \in \bigwedge V, t \in F^\times, \quad \nu(tx) = |t|^\alpha \nu(x) \right\}$$

Remark. (i) Whenever $\alpha \in \mathbb{R}^\times$, elements in $\Omega_\alpha^{\mathbb{R}}(V)$ are unique up to constant, and when $\alpha = 1$, they could be viewed as the self-dual Haar measure on V(real).
(ii) There is an operation $\otimes : \Omega_\alpha^{\mathbb{R}}(V) \otimes \Omega_\beta^{\mathbb{R}}(V) \to \Omega_{\alpha+\beta}^{\mathbb{R}}(V)$ defined by $(\nu \otimes \omega)(x) := \nu(x)\omega(x)$.
(iii) Remark (ii) implies the duality between $\Omega_\alpha^{\mathbb{R}}(V) = \Omega_\alpha^{\mathbb{R}}(V^*)^*$ and $\Omega_{-\alpha}^{\mathbb{R}}(V)$.
(iv) Let $\Omega_\alpha(V) := \Omega_\alpha^{\mathbb{R}}(V) \otimes_{\mathbb{R}} \mathbb{C}$, then $\Omega_\alpha(V)$ contains complex measures on V.

Definition 2.1.2. Let $\mathcal{S}(V)$ denote the **Schwarz-Bruhat** functions on V, and let $\mathcal{D}(V)$ be the dual of $\mathcal{S}(V) \otimes \Omega_1(V)$, call it the space of **distributions** on V.

Definition 2.1.3. Define the Fourier transform on functions to be the map: $\mathcal{S}(V^*) \otimes \Omega_1^{\mathbb{R}}(V^*) \to \mathcal{S}(V), f \mapsto \hat{f}$, where

$$\hat{f}(v) := \int_{v^* \in V^*} f(v^*)\psi(\langle v^*, v \rangle)$$

And the Fourier transform on distributions: $\mathcal{D}(V) \otimes \Omega_1^{\mathbb{R}}(V) \to \mathcal{D}(V^*), f\nu \mapsto \widehat{f\nu}$, where

$$\left\langle \widehat{f\nu}, f_0 \right\rangle = \left\langle f, \hat{f}_0 \nu \right\rangle$$

Remark. Recall that double Fourier transform will produce a minus in the classical case. In this case is similar

$$\widehat{\widehat{f\nu}} = c_\psi \cdot \tau^*(f\nu), \quad \tau : x \mapsto -x, x \in V$$

A symplectic space (V, B) is a F-vector space V with a non-degenerate skew-symmetric bilinear form $B : V \times V \longrightarrow F$.

Definition 2.1.4. Let (V, B) be a symplectic space, the **lagrangian** l is a maximal isotropic space. Let $\Lambda(W)$ be the set of lagrangians of W.

Proposition 2.1.5. *Symplectic space V is even-dimensional and any lagrangian $l \subset V$ is of half-dimension of V.*

Proof. Equip B with a non-degenerate matrix $B_{i,j} = B(e_i, e_j)$, then $B^T = -B$. Then $det(B^T) = det(-B) = (-1)^{dim(W)}det(B)$ implies V is even dimensional. Moreover, any subspace $V_0 \subset V$, let $V_0^\perp := \{v \in V | B(v,u) = 0, \forall u \in V\}$. One can prove that $dim(V) = dim(V_0) + dim(V_0^\perp)$, and isotropic means $l^\perp = l$. Then $dim(l) = \frac{1}{2}dim(V) = n$. $\square$

Proposition 2.1.6. *For any lagrangians $l_1, l_2 \in W$, $dim(l_1 \cap l_2) = s$, there is a basis of W, $\{p_1, ..., p_s, p_{s+1}, ..., p_n, q_1, ..., q_s, q_{s+1}, ..., q_n\}$ such that (i)$\langle p_i, p_j\rangle = \langle q_i, q_j\rangle = 0$, $\langle p_i, q_j\rangle = \delta_{ij}$ (ii) $l_1 = \bigoplus_{i=1}^n p_i$ (iii) $l_1 \cap l_2 = \bigoplus_{i=1}^s p_i$ (iv) $l_2 = \bigoplus_{i=1}^s p_i \bigoplus_{j=s+1}^n q_j$.*

Definition 2.1.7. Fix a symplectic vector space W with n-lagrangians, $\{l_1, ..., l_n\}$. define $\tau(l_1, ..., l_n) \in \mathcal{W}(F)$ to be the **Maslov Index** satisfy the following properties ($n \geq 3$)
(i) $\tau(l_1, ..., l_n) = \tau(gl_1, ..., gl_n), \forall g \in Sp(W)$ (ii) $\tau(l_1, ..., l_n) = \tau(l_2, ..., l_n, l_1)$, $\tau(l_1, ..., l_n) = -\tau(l_n, ..., l_1)$
(iii) $\tau(l_1, ..., l_n) = \tau(l_1, ..., l_s) + \tau(l_1, l_s, ..., l_n), s \geq 3, n > 3$.

Two vector spaces $(V, q), (V', q')$ with quadratic form q, q' are called isometric if there is a map $\phi : V \longrightarrow V'$ such that $q'(\phi(v)) = q(v)$. And a hyperbolic plane H is the space isometric to $(F \times F, q), q : (v_1, v_2) \mapsto v_1^2 - v_2^2$.

Definition 2.1.8. Define the **Witt group** $\mathcal{W}(F) := \mathcal{G}(F)/\mathbb{Z}H$, where $\mathcal{G}(F)$ is the Grothendieck group of isometric classes of quadratic forms.

Fix $\psi : F \longrightarrow \mathbb{C}$, and (V, q) a quadratic vector space. Let $f_q(x) = \psi(\frac{q(x)}{2})$.

Theorem 2.1.9 (Weil)**.** *There exists a constant $\gamma(q) \in \mathbb{S}^1$ such that $\widehat{f_q dq} = \gamma(q) f_{-q^*}$. Thus $\gamma(-) : \mathcal{W}(F) \longrightarrow \mathbb{S}^1$ is a character, we call it **Weil character**. In particular, if we choose $f \in \mathcal{S}(V)$, then*

$$\int\int f(x-y)f_q(y)dxdy = \gamma(q)\int f(x)dx$$

Where the integral measure is taken to be the self dual measure dq.

Theorem 2.1.10. *$\gamma(q)$ is the eighth root of unity.*

Proof. For any quadratic form q, we could diagonalize it to the quadratic form $q : x \mapsto cx^2$, where $c \in F^\times$ is fixed, denote the Weil character of this form by $\gamma(c)$. After diagonlizing, we next only consider the Weil character $\gamma(c)$. Let $a, b \in F^\times$, we have $\frac{\gamma(ab)\gamma(1)}{\gamma(a)\gamma(b)} = (a, b)$, where $(-,-)$ is the quadratic Hilbert symbol. First $\gamma(H) = \gamma(1)\gamma(-1) = 1$, take $a = b = -1$, $\gamma(1)^4 = (-1,-1) = \pm 1$. Thus $\gamma(1)^8 = 1$. For any $\gamma(a)$, $\gamma(a)^2 = \gamma(1)^2(a, a)$. It implies $\gamma(a)^4 = \gamma(1)^4$ and $\gamma(a)^8 = \gamma(1)^8 = 1$. $\square$

Let W be symplectic space, with n lagrangians $l_1, ..., l_n$. We construct the following diagram. Defining

$$\tilde{\partial} : \bigoplus_{i \in \mathbb{Z}/n\mathbb{Z}} W \longrightarrow \bigoplus_{i \in \mathbb{Z}/n\mathbb{Z}} W, \quad (w_i) \mapsto (w_i - w_{i-1})$$

$$\tilde{\sigma} : \bigoplus_{i \in \mathbb{Z}/n\mathbb{Z}} W \longrightarrow W, \quad (w_i) \mapsto \sum_i w_i$$

where $\tilde{\partial}, \tilde{\sigma}$ has natural restriction ∂, σ on l_i and $l_i \cap l_{i+1}$. Moreover consider $\bigoplus_{i \in \mathbb{Z}/n\mathbb{Z}} l_i \cap l_{i+1} \xrightarrow{\partial} \bigoplus_{i \in \mathbb{Z}/n\mathbb{Z}} l_i \xrightarrow{\sigma} W$, we have $Ker(\partial) = Im(\sigma) = \bigcap_{i \in \mathbb{Z}/n\mathbb{Z}} l_i$. Defining a quadratic space (T, q) where $T = Ker(\sigma)/Im(\partial)$ with quadratic form q, $q(v, w) := \sum_{i \in \mathbb{Z}/n\mathbb{Z}} \langle v_i, \hat{w}_i\rangle, \forall \hat{w}$ with $\tilde{\partial}(\hat{w}) = w$.

Remark. Value of q does not depend on the choice of $\hat{w}$. $q(v, w) = 0$ if $v \in Im(\partial)$ or $w \in Ker(\sigma)$.

Proposition 2.1.11. $\dim T = \frac{(n-2)\dim W}{2} - \sum_{i \in \mathbb{Z}/n\mathbb{Z}} \dim(l_i \cap l_{i+1}) + 2\dim \bigcap_{i \in \mathbb{Z}/n\mathbb{Z}} l_i$

For any vector space V, orientations on V, denoted by $\mathfrak{o}(V)$ makes up the set $\{\Lambda^{max}/\mathbb{F}^{\times 2}\}$. Defining $A_{l_1, l_2} := \langle e_1, \bar{e}_2\rangle$. Where $e_i \in \mathfrak{o}(l_i), i = 1, 2$, and $\bar{e}_i \in \mathfrak{o}(l_i/l_1 \cap l_2), i = 1, 2$ satisfying $e \wedge \bar{e}_i = e_i, \forall e \in \mathfrak{o}(l_1 \cap l_2)$, '$\wedge$' means exterior product pairing.

2.1.2 Weil Representations and Schrödinger Models

Let (W, B) be a symplectic space over F, we intensionally assume that $\psi \circ B : W \times W \longrightarrow \mathbb{S}^1 \subset \mathbb{C}$

Definition 2.1.12. A **Heisenberg group** $H(W)$ is $W \times F$ with a binary operation, abbreviated by H, where $(W, <, >)$ is a symplectic space. (Here we write in short $B(\cdot, \cdot)$ as $\langle \cdot, \cdot \rangle$The group operation in $H(W)$ is as follows:

$$(w_1, t_1) \cdot (w_2, t_2) = (w_1 + w_2, t_1 + t_2 + \frac{\langle w_1, w_2 \rangle}{2})$$

The action of $Sp(W)$ on H is trivially raised from its action on W, that is, $g \cdot (w, t) := (g \cdot w, t)$.
The center Z of $H(W)$ is $\{0\} \times F$, $H/Z = W$, and in the following we have well-known theorem by Stone-Von Neumann telling us something important about the irreducible representation of H.

Theorem 2.1.13 (Stone-Von Neumann)**.** *There is a unique (up to isomorphism) smooth irreducible representation ρ of H that has central character ψ (i.e. $\rho|_{Z=\{0\}\times F} = \psi$). We denote it by ρ_ψ.*

With Stone-Von Neumann theorem ensuring the uniqueness of ρ_ψ, we will have an intertwining operator $M(-)$ such that $\rho_\psi(gw, t) = M(g)\rho_\psi(w, t)M(g)^{-1}$. We need to further clarify the structure of $M(-) : Sp(W) \longrightarrow GL(S)$ by determining the representative vector space S. But at the first glance, we could roughly define a covering of $Sp(W)$ called **metaplectic group**.
As M is unique up to constant isomorphism by **Schur's Lemma**, we now could define the metaplectic group $\widetilde{\mathrm{Sp}}_\psi(W)$ with the projective representation $\bar{\omega}_\psi : \mathrm{Sp}(W) \to \mathrm{PGL}(S)$, that is

$$\begin{aligned} \widetilde{\mathrm{Sp}}_\psi(W) :&= Sp(W) \times_{PGL(S)} GL(S) \\ &= \left\{ (g, M) \in \mathrm{Sp}(W) \times \mathrm{GL}(S) | M \circ \rho_\psi = \rho_\psi^g \circ M \right\} \end{aligned}$$

Definition 2.1.14. The **Weil Representation** is the representation $\omega_\psi : \widetilde{\mathrm{Sp}}_\psi(W) \longrightarrow GL(S)$, lift by $\bar{\omega}_\psi$. We could view it through the following diagram.

$$\begin{tikzcd}
1 \arrow[r] & \mathbb{C}^\times \arrow[r] \arrow[d, "\simeq"] & \widetilde{\mathrm{Sp}}_\psi(W) \arrow[r, "\widetilde{p}"] \arrow[d, "\omega_\psi"] & Sp(W) \arrow[r] \arrow[d, "\bar{\omega}_\psi"] & 1 \\
1 \arrow[r] & \mathbb{C}^\times \arrow[r] & GL(S) \arrow[r, "id/\sim"] & PGL(S) \arrow[r] & 1
\end{tikzcd}$$

We use Schrödinger models to realize the concrete construction of Weil representation. In addition, what we care about in θ-correspondence is the two-fold cover of $Sp(W)$, whose existence is proved by Weil, and moreover, we could build it in explicit way, with the help of Maslov index.

Remark. The construction of Heisenberg group and Stone-Von Neumann theorem are compatible with the circumstance of finite field $\mathbb{F}_q$. Thus we could still construct a projective representation of $Sp(2n, \mathbb{F}_q)$ and lift it to representation of $\widehat{Sp}(2n, \mathbb{F}_q)$.

Theorem 2.1.15 (Weil)**.** *The surjective map $\widetilde{p}$ restricted on the derived group(commutator group)$\widehat{Sp}_\psi(W)$ of $\widetilde{Sp}_\psi(W)$ is a two-fold surjection on $Sp(W)$, and if we denote the non-trivial kernal by ε, then*

$$\widetilde{Sp}_\psi(W) = \widehat{Sp}_\psi(W) \times_{\{1,\varepsilon\}} \mathbb{C}^\times$$

Remark. Need to notice that $\widehat{Sp}(W)$ is not a trivial two cover on $Sp(W)$ when $F \neq \mathbb{C}$.

2.1.3 Schrödinger Models

Let $A \subset W$ be a closed subgroup, define $A^\perp := \{w \in W | \psi(B(w, a)) = 1, \forall a \in A\}$, which is again a closed subgroup. Fix A such that $A = A^\perp$(clearly lagrangian meets this condition). Define $H(A) := A \times F$, and let $\psi_A : H(A) \longrightarrow \mathbb{S}^1$, whose restriction on $\{0\} \times F$ is acquired to be ψ. Let (ρ, S_A) be the induced representation, where S_A is defined to be $Ind_{H(A)}^{H}\psi_A$.

$$Ind_{H(A)}^{H}\psi_A := \{f : H \longrightarrow \mathbb{C} | f(ah) = \psi_A(a)f(h), \forall a \in H(A), h \in H\}$$

Moreover any $f \in Ind_{H(A)}^{H}\psi_A$ is fixed by compact open subgroup $L \subset W$. The Heisenberg group $H(W)$ acts on S_A by right translation $\rho(g)f(h) = f(hg) \quad h, g \in H$

Lemma 2.1.16. *(i) $S_A = c - Ind_{H(A)}^{H}\psi_A$, which means all $f \in S_A$ has compact support.*
(ii) The representation (ρ, S_A) is smooth and irreducible with central character ψ. It implies $(\rho, S_A)$is one of the models of ρ_ψ.
(iii) (ρ, S_A) is admissible (i.e. any vector space of functions fixed by compact open subgroup is finite dimensional).

Proof. (i) Let $f \in S_A$ is fixed by a compact open subgroup $L \subset W$. To show f is compactly supported on $H(A)\backslash H(W) = A\backslash W$, we could just assume that any $(w, t) \in Supp(f)$ has 'coordinate' $\{0\}$ on F. For any $l \in L \cap A$

$$f((w, 0)) = f((w, 0)(l, 0)) = f(w + l, \frac{\langle w, l\rangle}{2}) = f((l, \langle w, l\rangle)(w, 0)) = \psi(\langle w, l\rangle)\psi_A((l, 0))f((w, 0)).$$

Thus f is supported on$A\backslash(L \cap A)^{\perp}$, where $(L \cap A)^{\perp} = L^{\perp} + A^{\perp} = L^{\perp} + A^{\perp}$. $L^{\perp}$ is additionally compact. Thus $f \in C_c^{\infty}(H; \mathbb{C})$.
(ii),(iii) c.f. $\square$

Now we take A to be l, the lagrangian, as well as a maximal isotropic subspace of W. The completion of l in W is again isotropic, thus lagrangian, denote it by l'. There is an isomorphism that $S_l \simeq C_c^{\infty}(l'), f \mapsto f'$, where

$$f'(x) = f((x, 0))$$

Moreover, consider the restriction of H on $W \times \{0\}$, it gives an isomorphism between S_l and the space

$$S_l^W\{f : W \longrightarrow \mathbb{C} | f(x + a) = \psi(\frac{1}{2}\langle x, a\rangle)f(x), \forall x \in W, a \in l\}$$

Also, f in above space is fixed by some compact, open subgroup $L \subset W$. H action on S_l^W is given by

$$(\rho_l(x, t)f)(x_0) = \psi(\frac{\langle x_0, x\rangle}{2} + t)f(x_0 + x), \forall(x, t) \in H, x_0 \in W$$

Proposition 2.1.17. *Similarly, H action on $C_c^{\infty}(l') = S_l$ is given by*

$$(\rho_l(x + y, t)f')(y_0) = \psi(\langle y_0, x\rangle + \frac{\langle y, x\rangle}{2} + t)f'(y_0 + y)$$

Proof. Use the isomorphic map between S_l and $C_c^{\infty}(l')$, then

$$\begin{aligned}(\rho_l(x + y, t)f')(y_0) &= f((y_0, 0)(x + y, t)) = f((x + y_0 + y, t + \frac{\langle y_0, y\rangle}{2})) \\ &= f((x, t + \langle y_0, x\rangle + \frac{\langle y, x\rangle}{2})(y_0 + y, 0)) = \psi(\langle y_0, x\rangle + \frac{\langle y, x\rangle}{2} + t)f'(y_0 + y)\end{aligned}$$

$\square$

The representation (ρ_l, S_l) is called Schrödinger representation. Fix a lagrangian l, assume $W = l \bigoplus l'$. Define $\mathcal{F}(l)$ to be the space $\{f : W \longrightarrow \Omega_{\frac{1}{2}}(l')\}$. Then ρ_l could be inductively regard as $\rho_l : \mathcal{S}(H) \otimes \Omega_1(W \times F) \longrightarrow End(\mathcal{F}(l))$,

$$\rho_l(h)s = \int_H h((w, t)) \cdot \rho_l((w, t)) \cdot sdg, \qquad h \in \mathcal{S}(H) \otimes \Omega_1(W \times F), s \in \mathcal{F}(l)$$

Given two different lagrangian l_i, l_j, define $\mathscr{F}_{l_j, l_i} : \mathcal{F}_{l_i} \longrightarrow \mathcal{F}_{l_j}$ by

$$(\mathscr{F}_{l_j, l_i}f)(y) = \int_{x \in \frac{l_j}{l_i \cap l_j}} f(x + y)\psi(\frac{\langle x, y\rangle}{2})\mu_{i,j}^{\frac{1}{2}}$$

Remark. (i) Here the measure $\mu_{i,j} \in \Omega_1(\frac{l_i + l_j}{l_i \cap l_j}) = \Omega_1(l_i) \otimes \Omega_1(l_j) \otimes \Omega_1(l_i \cap l_j)^* \otimes \Omega_1(l_i \cap l_j)^*$. And it is ordinary Haar measure in W if l_i, l_j are complementary lagrangians. In fact here we might as well regard $\mathcal{F}(l), L^2(l') \otimes \Omega_{\frac{1}{2}}(l'), \mathcal{S}(l)$, the Schwarz-Bruhat set of functions, as the same.
(ii) $\mathscr{F}_{l_j, l_i}$ is a unitray operator. $\mathscr{F}_{l_j, l_i} \circ \mathscr{F}_{l_i, l_j} = id$.

Lemma 2.1.18. *With lagrangians $l_1, ..., l_n$ ($n \geq 3$), define $\mathscr{F}_{1,...,n} := \mathscr{F}_{l_1,l_n} \circ \mathscr{F}_{l_n,l_{n-1}} \circ ... \circ \mathscr{F}_{l_2,l_1} : \mathcal{F}_{l_1} \longrightarrow \mathcal{F}_{l_1}$. Then*

$$\mathscr{F}_{1,...,n} = \gamma(-\tau(l_1, ..., l_n)) \cdot id$$

For some $g \in Sp(W)$, g will permute space W, thus it will induce a intertwine of function space on lagrangian(pull-back), i.e, $g_* : \mathcal{F}(l) \simeq \mathcal{F}(gl), \quad g_* f(-) \mapsto f(g^{-1}(-))$. Define $M_l^{Sch}[g] := \mathscr{F}_{l,gl} \circ g_* = g_* \circ \mathscr{F}_{g^{-1}l,l}$. $\mathscr{F}_{l,gl}$ as is shown in the above remark is actually a unitary, also g_* is also unitary, since

$$\langle g_* f_1, g_* f_2 \rangle = \int_{W/gl} f_1(gh)\overline{f_2}(gh) = \int_{W/l} f_1(h)\overline{f_2}(h) = \langle f_1, f_2 \rangle$$

Definition 2.1.19. Call $M_l^{Sch}[-] : Sp(W) \longrightarrow End(\mathcal{F}(l))$ associated with ρ_l the **Schrödinger Model**.

Remark. (i)The Schrödinger Model just defined fits the demands we want for a model of Weil representation. First, it is a unitary operator as we have shown that both $\mathscr{F}_{l,gl}$ and g_* are unitary. Second, $M_l^{Sch}[g] \circ \rho_l = \mathscr{F}_{l,gl} \circ g_* \circ \rho_l = \mathscr{F}_{l,gl} \circ \rho_{gl}^g \circ g_* = \rho_l^g \circ \mathscr{F}_{l,gl} \circ g_* = \rho_l^g \circ M_l^{Sch}[g]$
(ii) $M_l^{Sch}[g] \circ M_l^{Sch}[h] = \gamma(\tau(l, gl, ghl))M_l^{Sch}[gh]$, here the non-trivial coefficient is why we study on the projective representation $\bar{\omega}_\psi : Sp(W) \longrightarrow GL(S)$.

Definition 2.1.20. Fix a lagrangian l, define $c_{g,h}(l) := \gamma(\tau(l, gl, ghl))$ to be the **Maslov cocycle**. (It is clearly a cocycle according to the above remark).

Definition 2.1.21. Define the group $\widehat{Sp}(W)$ containing the pairs $(g, \phi), g \in Sp(W), \phi : \Lambda(W) \longrightarrow \mathbb{C}^\times$, whose multiplication is: $(g_1, \phi_1) \cdot (g_2, \phi_2) := (g_1 g_2, \phi_1 \phi_2 \cdot c_{g_1,g_2})$. Moreover ϕ is demanded to satisfy the following
(i)$\phi(l_2) = \gamma(\tau(l_1, gl_1, gl_2, l_2))\phi(l_1), \forall l_1, l_2 \in \Lambda(W)$.
(ii) $\phi(l)^2 = m_g(l)^2$, $m_g(l) := \gamma(1)^{\frac{\dim W}{2} - \dim gl \cap l - 1}\gamma(A_{gl,l})$.

Proposition 2.1.22. *The projective map $\widehat{Sp}(W) \xrightarrow{p} Sp(W), p((g, \phi)) := g$ is a two-fold covering map.*

Proposition 2.1.23. *Fix $l \in \Lambda(W)$, there is a group embedding of $\widehat{Sp}(W)$ into $\widetilde{Sp}(W)$ by taking $(g, \phi) \mapsto (g, \phi(l)M_l^{Sch}[g])$.*

2.2 Dual Reductive Pairs and Local Theta Correspondence

Definition 2.2.1. Let W be a symplectic vector space over F, associated with $Sp(W)$. Then the **dual reductive pair** means a pair (G, G') of reductive subgroups of $Sp(W)$, and satisfies $Cent_{Sp(W)}(G) = G', Cent_{Sp(W)}(G') = G$. If $W = W_1 \oplus W_2$, then dual pairs $(G_1, G_1^{'})$ in $Sp(W_1)$ (resp. $(G_2, G_2^{'})$ in $Sp(W_2)$) form a dual pair $(G_1 \times G_2, G_2^{'} \times G_2^{'})$ in $Sp(W)$. By **irreducible** we mean dual pairs do not raise in such way.

Let D be F-algebra with an involution ι, and (D, ι) satisfying one of these three cases:
(i) $(D, \iota) = (F, id)$. (ii)$(D, \iota) = (E, \sigma)$, where E/F is a quadratic extension and $\sigma \in Gal(E/F)$ nontrivial
(iii) $(D, \iota) = (B, \sigma)$, where B is a division quaternion algebra with center F, and ι is the main involution of B.
There are two types of dual reductive pairs classified by how they were constructed.
Type I: Fix $\varepsilon = \pm 1$, Let $(W, \langle, \rangle_W)$, $(V, \langle, \rangle_V)$ be two finite dimensional vector spaces over D withε-skew Hermitian forms. $\langle, \rangle_W$ satisfies $\langle ax, by \rangle_W = a\langle x, y \rangle b^\iota$, $\langle y, x \rangle = -\varepsilon \langle x, y \rangle^\iota$. $\langle, \rangle_V$ satisfies $\langle xa, yb \rangle_V = a^\iota \langle x, y \rangle b$, $\langle y, x \rangle = \varepsilon \langle x, y \rangle^\iota$
Take isometry group $G(W), G(V)$ of W, V.

$$G(W) := \{g \in GL(W, D) | \langle xg, yg \rangle_W = \langle x, y \rangle_W, \forall x, y \in W\}$$

$$G(V) := \{g \in GL(V, D) | \langle gx, gy \rangle_V = \langle x, y \rangle_V, \forall x, y \in V\}.$$

Construct the F-vector space $\mathbb{W} := V \otimes_D W$, and define a F-bilinear alternating form $\langle\langle, \rangle\rangle$ on $\mathbb{W}$ by

$$\langle\langle x_1 \otimes y_1, x_2 \otimes y_2 \rangle\rangle := Tr_{D/F}(\langle x_1, y_1 \rangle_V \langle x_2, y_2 \rangle_W^\iota)$$

Where the trace is either field trace or reduced trace depending on choice of D.
Further, there is a natural map $G(V) \times G(W) \longrightarrow Sp(\mathbb{W}), (h, g) \mapsto h \otimes g$. We get a irreducible reductive pair $(G(V), G(W))$ in $Sp(\mathbb{W})$ which is classified as Type I.
Type II: Let D be a division algebra with center F, A, B be finite dimensional vector spaces over D. Let $A^* = Hom_D(A, D)$ (resp. B^*) with action of D is left translation. It gives a natural pairing of $A^* \times A \longrightarrow D, (a^*, a) \mapsto a^*(a)$(resp. B), where $da^*(a) = d \cdot a^*(a)$. Let $X := A \otimes_D B^*$, $Y := B \otimes_D A^*$.
The pairing $X \times Y \longrightarrow F$ is given by $a \otimes b^*(b \otimes a^*) := Tr_{D/F}(a^*(a) \cdot b^*(b))$. Let $\mathbb{W} = X + Y$, thus a symplectic form could be defined on it

$$\langle\langle x_1 + y_1, x_2 + y_2 \rangle\rangle = x_1(y_2) - y_1(x_2)$$

Associating with a homomorphism

$$GL(A, D) \times GL(B, D) \longrightarrow Sp(\mathbb{W}), (g, h) \mapsto g \otimes h^* + h^{-1} \otimes (g^*)^{-1}$$

where $g^* \in GL(A^*, D)$ is given by $a^* g^*(a) = a^*(ga)$(resp. h^*), we get an irreducible reductive pair $(GL(A), GL(B))$ in $Sp(\mathbb{W})$ of Type II.
Take F to be $\mathbb{R}$ or $\mathbb{C}$, and D to be the division algebra, the table of irreducible dual pairs could be drawn.

	Division Algebra & Involution	G	G'	Sp(W)
Type I	($\mathbb{R}$,id)	Sp(2n,$\mathbb{R}$)	O(p,q,$\mathbb{R}$)	Sp(2n(p+q),$\mathbb{R}$)
	($\mathbb{C}$,id)	Sp(2n,$\mathbb{C}$)	O(m,$\mathbb{C}$)	Sp(4nm,$\mathbb{C}$)
	($\mathbb{C}$, -)	U(r,s)	U(p,q)	Sp(2(p+q)(r+s),$\mathbb{R}$)
	($\mathbb{H}$, -)	O*(2n)	Sp(p,q)	Sp(4n(p+q),$\mathbb{R}$)
Type II	$\mathbb{R}$	GL(n,$\mathbb{R}$)	GL(m,$\mathbb{R}$)	Sp(2nm,$\mathbb{R}$)
	$\mathbb{C}$	GL(n,$\mathbb{C}$)	GL(m,$\mathbb{C}$)	Sp(4nm,$\mathbb{C}$)
	$\mathbb{H}$	GL(n,$\mathbb{H}$)	GL(m,$\mathbb{H}$)	Sp(8nm,$\mathbb{H}$)

Table 2.1: Irreducible pairs of two types (over $\mathbb{R}$)

	Division Algebra	G	G'	Sp(W)
Type I	$\mathbb{C}$	Sp(2n,$\mathbb{C}$)	O(m,$\mathbb{C}$)	Sp(2nm,$\mathbb{C}$)
Type II	$\mathbb{C}$	GL(n,$\mathbb{C}$)	GL(m,$\mathbb{C}$)	Sp(4nm,$\mathbb{C}$)

Table 2.2: Irreducible pairs of two types (over $\mathbb{C}$)

2.2.1 Local Theta Correspondence

For E a reductive subgroup of $Sp(W)$, Let $\widetilde{E}$ denote its inverse image in $\widetilde{Sp}(W)$. Let $\mathscr{R}(\widetilde{E})$ denote the set of infinitesimal equivalent classes of admissible representations of $\widetilde{E}$ on locally convex spaces. For the Weil representation ω we construct, denote in general $\mathscr{Y}$ as its realization Hilbert space. Let ω^∞ be the smooth representation of $\widetilde{Sp}(W)$ associated with ω, and its realization $\mathscr{Y}^\infty$ is the subspace of $\mathscr{Y}$ of smooth vectors. Let $\mathscr{R}(\widetilde{E}, \mathscr{Y}^\infty)$ denote the subset of $\mathscr{R}(\widetilde{E})$ which could be realized as the quotient of $\mathscr{Y}^\infty$ by some $\omega^\infty(\widetilde{E})$-invariant subspace.
For the dual pair (G, G'), consider their inverse image $(\widetilde{G}, \widetilde{G'})$ in $\widetilde{Sp}$. As $\widetilde{G}, \widetilde{G'}$ commute, $\omega^\infty|_{\widetilde{G} \cdot \widetilde{G'}}$ could be regard as a representation of $\widetilde{G} \times \widetilde{G'}$. There is a identity map between $\mathscr{R}(\widetilde{G} \times \widetilde{G'})$ and $\mathscr{R}(\widetilde{G}) \times \mathscr{R}(\widetilde{G'})$, given by tensor product, that is $(\rho, \rho') \mapsto \rho \otimes \rho' : ((g, \widetilde{g})) \mapsto \rho(g) \otimes \rho'(\widetilde{g'})$, where $\rho \in \mathscr{R}(\widetilde{G}), \rho' \in \mathscr{R}(\widetilde{G'})$. Hence it gives the correspondence between $\mathscr{R}(\widetilde{G} \cdot \widetilde{G'}, \mathscr{Y}^\infty)$ and subsets of $\mathscr{R}(\widetilde{G}, \mathscr{Y}^\infty)$ and $\mathscr{R}(\widetilde{G'}, \mathscr{Y}^\infty)$, since if $\rho \otimes \rho'$ is also in $\mathscr{R}(\widetilde{G} \cdot \widetilde{G'}, \mathscr{Y}^\infty)$, then we have $\rho \in \mathscr{R}(\widetilde{G}, \mathscr{Y}^\infty)$, $\rho' \in \mathscr{R}(\widetilde{G'}, \mathscr{Y}^\infty)$.
What is exactly such correspondence, Howe in his paper made a precise description.

Theorem 2.2.2. *The set $\mathscr{R}(\widetilde{G} \cdot \widetilde{G'}, \mathscr{Y}^\infty)$ is the graph of a bijection between all of $\mathscr{R}(\widetilde{G}, \mathscr{Y}^\infty)$ and all of $\mathscr{R}(\widetilde{G'}, \mathscr{Y}^\infty)$. Moreover, an element $\rho \otimes \rho'$ of $\mathscr{R}(\widetilde{G} \cdot \widetilde{G'}, \mathscr{Y}^\infty)$ occurs as a quotient of $\mathscr{Y}^\infty$) in a unique way.*

To proceed the proof of the theorem and other conclusions, we might introduce the conception of $(\mathfrak{g}, K)$**-module**.

Definition 2.2.3. Let $\mathfrak{g}$ be the Lie algebra of some compact Lie group G, and K be the maximal compact subgroup of G, with Lie algebra $\mathfrak{k}$. By $(\mathfrak{g}, K)$**-module** we mean a vector space V which is simultaneously the module of $\mathfrak{g}$ and K and satisfies the following:
(i)$k \cdot (X \cdot v) = (\mathrm{Ad}(k)X) \cdot (k \cdot v), \forall v \in V, k \in K, X \in \mathfrak{g}$.
(ii) dim span$\{K \cdot v\} \leq \infty, \forall v \in V$. (iii) $\left(\frac{d}{dt} \exp(tY) \cdot v\right)\big|_{t=0} = Y \cdot v, \forall v \in V, Y \in \mathfrak{k}$.

Remark. For any $(\mathfrak{g}, K)$-module V, $V = \oplus_{\tau \in \widehat{K}} V(\tau)$, where $V(\tau)$ is τ-isotypic component. In addition, $\forall v \in V$ is a finite sum of $v_\tau \in V(\tau)$. The projection from V to $V(\tau)$ could be realized as integral against conjugation of χ_τ on K. And for each $v \in V$, the integration of $\overline{\chi_v} = \sum_{v_\tau \neq 0} \overline{\chi_\tau}$ will give a projection onto $\oplus_{v_\tau \neq 0} V(\tau)$.

Turning back on our discussion of dual reductive pairs. As the maximal compact subgroup of Sp_{2n} is U_n, one has the Cartan decomposition $\mathfrak{sp} = \mathfrak{u} \oplus \mathfrak{q}$. Furthermore, for a dual pair (G, G'), fix appropriate U_n such that the Cartan decomposition of $\mathfrak{sp}$ could induce the Cartan decompositions of $\mathfrak{g}, \mathfrak{g}'$, that is

$$\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}, \mathfrak{k} = \mathfrak{u} \cap \mathfrak{g}, \mathfrak{p} = \mathfrak{q} \cap \mathfrak{g}.\mathfrak{g}' = \mathfrak{k}' \oplus \mathfrak{p}', \mathfrak{k}' = \mathfrak{u} \cap \mathfrak{g}', \mathfrak{p}' = \mathfrak{q} \cap \mathfrak{g}'.$$

Let $\mathscr{P}$ be subspace of $\widetilde{U}$-finite of $\mathscr{Y}^\infty$, then it is a $(\mathfrak{g}, \widetilde{K})$-module and a $(\mathfrak{g}', \widetilde{K'})$-module. Let $\rho \in \mathscr{R}(\mathfrak{g}, \widetilde{K}, \mathscr{P})$, which denotes the set of irreducible $(\mathfrak{g}, \widetilde{K})$-modules which could be realized as the quotient of $\mathscr{P}$ by a $\mathfrak{g}$ and $\widetilde{K}$ invariant space. Define

$$\mathscr{N}(\rho) := \bigcap_{\lambda \in \mathrm{Hom}_{(\mathfrak{g}, \widetilde{K})}(\mathscr{Y}^\infty, \rho)} \mathrm{Ker}(\lambda)$$

Then $\mathscr{P}/\mathscr{N}(\rho)$ is a $\left(\mathfrak{g} \oplus \mathfrak{g}', \widetilde{K} \cdot \widetilde{K'}\right)$-module.

Lemma 2.2.4. *[MVW06] Let V_1 be an admissible, irreducible $(\mathfrak{g}, K)$-module, V_2 be a $(\mathfrak{g}', K')$-module, for any V a subspace of $V_1 \otimes V_2$, invariant under $(\mathfrak{g}, K) \times (\mathfrak{g}', K')$, there exists a subspace V_2' of V_2 such that $V = V_1 \otimes V_2'$.*

Proof. Let $V_2' := \{v_2 \in V_2 | V_1 \otimes \mathbb{C}v_2 \subset V\}$. Then $V_1 \otimes V_2' \subset V$, only need to prove that if $V_2' = 0$, then V must be 0. Suppose not, $\exists 0 \neq v \in V$, by the property of $(\mathfrak{g}, K)$-module, it could be decomposed into isotypic components. Write $v = \sum_{i=1}^n v_1^i \otimes v_2^i$, where $\{v_1^i\}$ are linearly independent and at least some $v_2^i \neq 0$, may assume $v_2^1 \neq 0$. Since V_1 is irreducible, there exists f in the sum of $\mathfrak{U}(\mathfrak{g})^K$ and K, such that $f \cdot \mathrm{v}_1^{\mathrm{i}} = \begin{cases} 0, & \mathrm{i} \neq 1 \\ \mathrm{v}_1^1, & \mathrm{i} = 1 \end{cases}$. Hence, $v_1^1 \otimes v_2^1 = f \cdot v \neq 0$, then $v_2^1 \neq 0$, contradicts to $V_2' = 0$. □

Lemma 2.2.5. *[MVW06] Let V_1 be an admissible, irreducible $(\mathfrak{g}, K)$-module, V be a $(\mathfrak{g}, K) \times (\mathfrak{g}', K')$-module. Suppose that $\bigcap_{\lambda \in \mathrm{Hom}_{(\mathfrak{g}, \widetilde{K})}(V, V_1)} \mathrm{Ker}(\lambda) = 0$. Then there exists unique(up to isomorphism) $(\mathfrak{g}', K')$-module V_2 such that V is isomorphic to $V_1 \otimes V_2$.*

Proof. For any G module U, We denote U_G to be the maximal quotient of U that G acts trivially. Similarly for the $(\mathfrak{g}, K)$ module case. Especially, for an irreducible $(\mathfrak{g}, K)$ module V_1 and its contragredient module $\widetilde{V_1}$ for $(\mathfrak{g}, K)$, $(\widetilde{V_1} \otimes V_1)_{\mathfrak{g}, K} \simeq \mathbb{C}$. Moreover, suppose V_2 exists $(\widetilde{V_1} \otimes V_1)_{\mathfrak{g}, K} \simeq (\widetilde{V_1} \otimes V_1 \otimes V_2)_{\mathfrak{g}, K} \simeq (\widetilde{V_1} \otimes V_1)_{\mathfrak{g}, K} \otimes V_2 \simeq V_2$. By the uniqueness of V_2, $(\widetilde{V_1} \otimes V_1)_{\mathfrak{g}, K} = V_2'$, Let $p : \widetilde{V_1} \otimes V_1 \longrightarrow V_2'$ be the projection map.
We could define the map $\phi : V \longrightarrow Hom_{\mathbb{C}}(\widetilde{V_1}, V_2'), v \mapsto (\widetilde{v_1} \mapsto p(\widetilde{v_1} \otimes v))$. As is showed in Remark, we could show that $\phi(v) \in Hom_{\mathbb{C}}(\oplus \widetilde{V_1}_\tau, V_2') \subset Hom_{\mathbb{C}}(\widetilde{V_1}, V_2')$. It shows $\phi(v)$ is K-finite. Thus ϕ could be translated to $\phi' : V \longrightarrow V_1 \otimes V_2'$.
In addition, ϕ is injective. For $v \in V$, $v \neq 0$. Gives $f \in Hom_{G_1}(V, V_1)$ with $f(v) \neq 0$, associated with $f' : (\widetilde{V_1} \otimes V)_{\mathfrak{g}, K} \longrightarrow (\widetilde{V_1} \otimes V_1)_{\mathfrak{g}, K}$. On a$f' \circ p(\widetilde{v_1} \otimes v) = \widetilde{v_1} \circ f(v) \neq 0$, we prove ϕ is injective, as well as ϕ'.
Hence V could be realized as $(\mathfrak{g}, K) \times (\mathfrak{g}', K')$ submodule of $V_1 \otimes V_2'$, using the last lemma, we prove this lemma. □

Corollary 2.2.6. *As an immediate consequence of the above Lemma, there is a smooth $(\mathfrak{g}', \widetilde{K'})$-module $\Theta(\rho)$ such that $\mathscr{N}(\rho) \simeq \rho \otimes \Theta(\rho)$, and $\Theta(\rho)$ is unique up to isomorphism.*

Moreover, Howe showed a more powerful statement.

Theorem 2.2.7 (Howe Duality Principle)**.** *The* $(\mathfrak{g}', \widetilde{K'})$*-module* $\Theta(\rho)$ *above is finitely generated, admissible and quasisimple. Further,* $\Theta(\rho)$ *has a unique quotient* $\theta(\rho)$ *that the correspondence between* ρ *and* $\theta(\rho)$ *gives a bijection between* $\mathscr{R}(\mathfrak{g}, \widetilde{K}, \mathscr{Y}^{\infty})$*and* $\mathscr{R}(\mathfrak{g}', \widetilde{K'}, \mathscr{Y}^{\infty})$.

Remark. (i) For the quasisimple representation ρ of a connected Lie group G we mean that ρ acts on $Z\mathfrak{U}(\mathfrak{g})$, the center of universal enveloping algebra, by scalar.
(ii) Theorem 2.2.1 is the consequence of the above theorem.

Proof of the Remark (ii). See [How89] □

Chapter 3

Representation Theory of Finite Reductive Groups

3.1 Harish-Chandra Inductions and Cuspidal Representations

Consider G is a finite group and H is its subgroup, M is a $G-$module thus is automatically a $H-$module, for E a $H-$module, then there is a functor that lifts $\mathbb{C}[H]-$module to $\mathbb{C}[G]-$module given by $E \mapsto M \otimes_{\mathbb{C}[H]} E$(Notice for group module, there is a natural group algebra action). This is so called induced representation.
We consider $\mathbf{G}$ being a reductive algebraic group over k, where $k = \bar{\mathbb{F}}_q$, for q is odd and the prime power, let $F : \mathbf{G} \to \mathbf{G}$ denote the generalized Frobenius map with respect to $\mathbb{F}_q$. For $\mathbf{P}$ the rational parabolic subgroup of $\mathbf{G}$, associated with the **Levi decomposition** $\mathbf{P} = \mathbf{LU}$, by **Harish-Chandra induction** we mean the functor $R_{\mathbf{L}}^{\mathbf{G}}$ lifting $\mathbf{L}^F-$module to $\mathbf{G}^F-$module. Similar to the first paragraph, choose M to be the $\mathbf{G}^F$-$\mathbf{L}^F$ module given by $\mathbb{C}[\mathbf{G}^F/\mathbf{U}^F]$, where $\mathbf{G}^F$ acting by left translation and $\mathbf{L}^F$ acting by right translation since $\mathbf{L}^F$ normalizes $\mathbf{U}^F$. To clarify this notation, we also write $R_{\mathbf{L}}^{\mathbf{G}}$ as $R_{\mathbf{L}\subseteq\mathbf{P}}^{\mathbf{G}}$ if there is a need. Denote by ${}^*R_{\mathbf{L}}^{\mathbf{G}}$ the adjoint functor of $R_{\mathbf{L}}^{\mathbf{G}}$.

Proposition 3.1.1 (Transitivity of the functor)**.** *Let $\boldsymbol{P}, \boldsymbol{Q}$ be rational parabolic subgroups of $\mathbf{G}$, and suppose $\boldsymbol{Q} \subset \boldsymbol{P}$. Let $\boldsymbol{L}, \boldsymbol{M}$ be Levi subgroups of $\boldsymbol{P}$ and $\boldsymbol{Q}$ respectively, and $\boldsymbol{M}$ contained in $\boldsymbol{L}$, then*

$$R_{\boldsymbol{L}\subset\boldsymbol{P}}^{\boldsymbol{G}} \circ R_{\boldsymbol{M}\subset\boldsymbol{L}\cap\boldsymbol{Q}}^{\boldsymbol{L}} = R_{\boldsymbol{M}\subset\boldsymbol{Q}}^{\boldsymbol{G}}.$$

Proof. Let the rational Levi decomposition be $\mathbf{P} = \mathbf{LU}, \mathbf{Q} = \mathbf{MV}$. From the assumption, $\mathbf{Q} \subset \mathbf{P}$, then the unipotent radicals $\mathbf{U} \subset \mathbf{V}$, and $\mathbf{V}^F = \mathbf{U}^F(\mathbf{L}^F \cap \mathbf{V}^F)$. By definition, the above equation holds when there is a bijection:

$$\mathbb{C}[\mathbf{G}^F/\mathbf{U}^F] \otimes_{\mathbb{C}[\mathbf{L}^F]} (\mathbb{C}[\mathbf{L}^F/\mathbf{L}^F \cap \mathbf{V}^F] \otimes_{\mathbb{C}[\mathbf{M}^F]} \mathbb{C}[\mathbf{M}^F]) \overset{\cong}{\to} \mathbb{C}[\mathbf{G}^F/\mathbf{V}^F] \otimes_{\mathbb{C}[\mathbf{M}^F]} \mathbb{C}[\mathbf{M}^F]$$

Or equivalently, $\mathbb{C}[\mathbf{G}^F/\mathbf{U}^F] \otimes_{\mathbb{C}[\mathbf{L}^F]} \mathbb{C}[\mathbf{L}^F/\mathbf{L}^F \cap \mathbf{V}^F] \overset{\cong}{\to} \mathbb{C}[\mathbf{G}^F/\mathbf{V}^F]$, this will be an immediate conclusion if

$$\mathbf{G}^F/\mathbf{U}^F \times_{\mathbf{L}^F} \mathbf{L}^F/\mathbf{L}^F \cap \mathbf{V}^F \to \mathbf{G}^F/\mathbf{V}^F, \text{ given by } (g\mathbf{U}^F, l(\mathbf{L}^F \cap \mathbf{V}^F)) \mapsto gl\mathbf{V}^F = gl\mathbf{U}^F(\mathbf{L}^F \cap \mathbf{V}^F)$$

is a bijection. And this isomorphism is clear, thus the transitivity in the proposition holds. □

Recall that in the representation theory of finite groups, we could use the Mackey formula to partition the H induced G representation(where H is a subgroup of G) into K representations (some subgroup K), for finite group of Lie type, we have a similar Mackey formula.

Theorem 3.1.2 (The Mackey Formula for Harish-Chandra Inductions). *Let $\boldsymbol{P}, \boldsymbol{Q}$ be rational parabolic subgroups of $\boldsymbol{G}$, and $\boldsymbol{L}, \boldsymbol{M}$ be Levi subgroups of $\boldsymbol{P}$ and $\boldsymbol{Q}$ respectively, $\mathcal{S}(\boldsymbol{L}, \boldsymbol{M}) := \{g \in \boldsymbol{G} | \exists$ maximal torus$\boldsymbol{T} \subset \boldsymbol{L} \cap^g \boldsymbol{M}\}$*

$$^*R^{\boldsymbol{G}}_{\boldsymbol{L}\subset\boldsymbol{P}} \circ R^{\boldsymbol{G}}_{\boldsymbol{M}\subset\boldsymbol{Q}} = \sum_{\bar{g}} R^{\boldsymbol{L}}_{\boldsymbol{L}\cap^g\boldsymbol{M}\subset\boldsymbol{L}\cap^g\boldsymbol{Q}} \circ^* R^{^g\boldsymbol{M}}_{\boldsymbol{L}\cap^g\boldsymbol{M}\subset\boldsymbol{P}\cap^g\boldsymbol{M}} \circ \operatorname{ad} g$$

Where $g \in \mathcal{S}(\boldsymbol{L}, \boldsymbol{M})^F$, $\bar{g}$ is the representative element of g in the coset $\boldsymbol{L}^F \backslash \mathcal{S}(\boldsymbol{L}, \boldsymbol{M})^F / \boldsymbol{M}^F$.

Notice that we take g running over cosets of $\mathcal{S}(\mathbf{L}, \mathbf{M})^F$ rather than cosets of $\mathbf{G}^F$ to ensure the functors $R^{\mathbf{L}}_{\mathbf{L}\cap^g\mathbf{M}}$ and $R^{^g\mathbf{M}}_{\mathbf{L}\cap^g\mathbf{M}}$ are well-defined. Further, for the set $\mathcal{S}(\mathbf{L}, \mathbf{M})$, we have some important remarks.

Proposition 3.1.3. *There is an isomorphism $\boldsymbol{L}\backslash\mathcal{S}(\boldsymbol{L}, \boldsymbol{M})/M \xrightarrow{\sim} \boldsymbol{P}\backslash\boldsymbol{G}/\boldsymbol{Q}$. Moreover, $(\boldsymbol{P}\backslash\boldsymbol{G}/\boldsymbol{Q})^F \cong \boldsymbol{P}^F\backslash\boldsymbol{G}^F/\boldsymbol{Q}^F$, and $(\boldsymbol{L}\backslash\mathcal{S}(\boldsymbol{L}, \boldsymbol{M})/M)^F \cong \boldsymbol{L}^F\backslash\mathcal{S}(\boldsymbol{L}, \boldsymbol{M})^F/M^F$*

To prove the Proposition, we recall that Bruhat decomposition divides $\mathbf{G}$ into disjoint Bruhat cells, labelled by elements in Weyl group W. The generators of W is a set S, for a subset I of S, we have the subgroup W_I of W and the standard parabolic subgroups of $\mathbf{G}$ indexed by I is the union of Bruhat cells associate to W_I, denoted by $\mathbf{P}_I$. For $I \subseteq S$, by **I-reduced** (resp. **reduced-I**) elements we mean those $w \in W$ such that $l(v) + l(w) = l(vw), \forall v \in W_I$ (resp. $l(w) + l(v) = l(wv)$) , and we have

Proposition 3.1.4. *Fix $\boldsymbol{T}$ a maximal torus in $\boldsymbol{G}$, and $\boldsymbol{B}$ a Borel subgroup containing $\boldsymbol{T}$. Then for $\boldsymbol{P}_I$, $\boldsymbol{P}_J$ parabolic subgroups that containing $\boldsymbol{B}$ and indexed by I, J respectively, we have a bijection $W_I\backslash W/W_J \xrightarrow{\sim} \boldsymbol{P}_I\backslash\boldsymbol{G}/\boldsymbol{P}_J$.*

Proof. We consider $\mathbf{P}_I = \mathbf{B}W_I\mathbf{B}, \mathbf{P}_I = \mathbf{B}W_J\mathbf{B}$, and thus $\forall \mathbf{P}_I g \mathbf{P}_J \subseteq \mathbf{P}_I\backslash\mathbf{G}/\mathbf{P}_J = \mathbf{B}W_I\mathbf{B}W\mathbf{B}W_J\mathbf{B}$, it is the image of some $W_I w W_J$, then the map $W_I w W_J \to \mathbf{P}_I g \mathbf{P}_J \cong \mathbf{P}_I w \mathbf{P}_J$ is surjective. Without lose of generality, assume w and w' are in reduced form, if $\mathbf{P}_I w \mathbf{P}_J = \mathbf{P}_I w' \mathbf{P}_J$, then $\mathbf{B}W_I\mathbf{B}w\mathbf{B}W_J\mathbf{B} = \mathbf{B}W_I\mathbf{B}w'\mathbf{B}W_J\mathbf{B}$. consider $\forall x \in W_I, y \in W_J$, $\mathbf{B}x\mathbf{B}w\mathbf{B}y\mathbf{B}$ is the union of those $\mathbf{B}x'\mathbf{B}w\mathbf{B}y'\mathbf{B}$ such that $x' \in W_I, y' \in W_J$ with $l(x'wy') = l(x') + l(w) + l(y')$. According the uniqueness of expression of elements in coset $W_I w W_J$ in $I-$reduced$-J$ form, we deduce the injectivity. □

Remark. By $I-$reduced$-J$ element we mean $w \in W$ such that $\forall x \in W_I, y \in W_J$ and $l(xwy) = l(x) + l(w) + l(y)$. Notice $W_I\backslash W/W_J$ is bijective to set of $I-$reduced$-J$ elements.

Proof of Proposition 3.1.3. The first half is a corollary of Proposition 3.1.4. Fix $\mathbf{P}$, we make a conjugate on $\mathbf{Q}$ such that $\mathbf{L} \cap {}^g\mathbf{M}$ contains a maximal torus $\mathbf{T}$. Take the Weyl group W associated with $\mathbf{T}$, then $\mathbf{P}$ and ${}^g\mathbf{Q}$ are standard parabolics $\mathbf{P}_I$ and $\mathbf{P}_J$. Now since $\mathbf{P}g'\mathbf{Q} = \mathbf{P}g'g^{-1g}\mathbf{Q}g$ and $\mathcal{S}(\mathbf{L}, \mathbf{M}) \cong \mathcal{S}(\mathbf{L}, {}^g\mathbf{M})$, we are able to take ${}^g\mathbf{Q}$ instead of $\mathbf{Q}$ and consider the standard case.
The remaining is to show that $\mathbf{L}\backslash\mathcal{S}(\mathbf{L}, \mathbf{M})/\mathbf{M}$ is in bijection with $W_I\backslash W/W_J$. Consider $\mathbf{L} \cap {}^x\mathbf{M}$ contains a unique maximal torus means $\mathbf{P}_I \cap {}^x\mathbf{P}_J$ contains some Borel ${}^w\mathbf{B}$ (but unique), that is $\mathbf{B}w\mathbf{B}w^{-1}\mathbf{B} \subseteq \mathbf{B}W_I\mathbf{B} \cap \mathbf{B}x\mathbf{B}W_J\mathbf{B}x^{-1}\mathbf{B}$, which implies a unique correspondence $\mathbf{B}W_I\mathbf{B}w\mathbf{B}W_J\mathbf{B} \leftrightarrow \mathbf{P}_I x \mathbf{P}_J$, and thus $W_I w W_J \leftrightarrow W_I x W_J$ by Proposition 3.1.4. Notice also $\mathbf{L}$ and $\mathbf{M}$ are also uniquely determined by W_I, W_J (in the setting of standard case), then we conclude the first half of the Proposition 3.1.3. □

Definition 3.1.5. For pairs $(\mathbf{L}, \delta) \in \operatorname{Irr}(\mathbf{L}^F))$, we define a partial order on their set, say $(\mathbf{L}, \delta) \leq (\mathbf{L}', \delta')$ if $\mathbf{L} \subseteq \mathbf{L}'$ and $\langle \delta', R^{\mathbf{L}'}_{\mathbf{L}} \delta \rangle_{\mathbf{L}'^F} \neq 0$. The transitivity of this partial order is given by Proposition 3.1.1. Then we say $\delta \in \operatorname{Irr}(\mathbf{L}^F)$ is **cuspidal** of $(\mathbf{L}, \delta)$ is minimal under this partial order, or equivalently, $^*R^{\mathbf{L}}_{\mathbf{L}'}(\delta) = 0$, for all $\mathbf{L}' \subset \mathbf{L}$. By **Harish-Chandra series** we mean $\mathcal{E}(\mathbf{G}, (\mathbf{L}, \delta)) = \{\chi \in \operatorname{Irr}(\mathbf{G}^F) | \langle \chi, R^{\mathbf{G}}_{\mathbf{L}}(\delta) \rangle \neq 0\}$.

Theorem 3.1.6 (Harish-Chandra). *There is a partition of irreducible characters given by Harish-Chandra series,* $\operatorname{Irr}(\boldsymbol{G}^F) = \coprod_{(\boldsymbol{L},\delta)/\sim} \mathcal{E}(\boldsymbol{G}, (\boldsymbol{L}, \delta))$.

Theorem 3.1.7 (Howlett-Lehrer). *Assume $\boldsymbol{G}$ is connected. For $\delta \in \operatorname{Irr}(\boldsymbol{L}^F)$, set $W_{\boldsymbol{G}}(\boldsymbol{L}, \delta) := \{w \in N_{\boldsymbol{G}^F}(\boldsymbol{L})/\boldsymbol{L}^F | {}^w\delta = \delta\}$ being the stabilizer of δ, then there is an isomorphism $\operatorname{End}_{\boldsymbol{G}^F}(R^{\boldsymbol{G}}_{\boldsymbol{L}}(\delta)) \cong \mathbb{C}[W_{\boldsymbol{G}}(\boldsymbol{L}, \delta)]$. It is equivalent to say there is a naturally isomorphic functor: $\operatorname{Irr}(W_{\boldsymbol{G}}(\boldsymbol{L}, \delta)) \to \mathcal{E}(\boldsymbol{G}, (\boldsymbol{L}, \delta))$.*

Definition 3.1.8. Call the series that associated to a rational maximal torus which is included in some rational Borel subgroup **principle series**. (Notice all such tori are $\mathbf{G}^F$ conjugate, thus we could choose any rational torus).

3.2 Deligne-Lusztig Inductions and Unipotent Representations

Fix $k = \bar{\mathbb{F}}_q$, where q is odd and the prime power. Let $\mathbf{G}$ be a connected reductive algebraic group over k, and $F : \mathbf{G} \to \mathbf{G}$ be the generalized Frobenius map with respect to $\mathbb{F}_q$. Denote by $G = \mathbf{G}^F$ the fixed points of F, which is called **finite group of Lie type**. To study the unipotent representation, we must acknowledge Deligne-Lusztig's definition of **virtual character**, by following [DL76].

Definition 3.2.1. $\mathbf{G}$ reductive algebraic group admits with a $BN-$ pair, and $\mathbf{G} = \coprod_{w\in W} \mathbf{B}w\mathbf{B}$ is the Bruhat decomposition. Let $\mathcal{B}$ denote the set of all Borels of $\mathbf{G}$, then there is a bijection between $\mathbf{G}$-orbits on $\mathcal{B}\times\mathcal{B}$ and W. We write $\mathbf{B}' \xrightarrow{w} \mathbf{B}''$ if $(\mathbf{B}', \mathbf{B}'')$ is in the $\mathbf{G}$-orbit contains the pair $(\mathbf{B}, w\mathbf{B}w^{-1})$, or equivalently, there exists $\mathbf{B} \in \mathcal{B}, g \in \mathbf{G}$, such that $\mathbf{B}' = g\mathbf{B}g^{-1}, \mathbf{B}'' = gw\mathbf{B}w^{-1}g^{-1}$. For $w \in W$, define the **Deligne-Lusztig variety** associated with w to be $X_w := \{\mathbf{B} \in \mathcal{B} | \mathbf{B} \xrightarrow{w} F(\mathbf{B})\}$. Notice $\mathcal{B} \cong \mathbf{G}/\mathbf{B}$ is a homogeneous space, thus a projective variety, for more detailed discussion, see [Nei10].

Choose $w = 1$, then $X_1 = \{\mathbf{B} \in \mathcal{B} | F(\mathbf{B}) = \mathbf{B}\}$, and there is a $G-$equivariant bijection between X_1 and $G/B = \mathbf{G}^F/\mathbf{B}^F$. For the longest element $w_0 \in W$, $X_{w_0} = \{\mathbf{B} \in \mathcal{B} | \mathbf{B} \cap F(\mathbf{B})$ is a maximal torus $\}$.

Proposition 3.2.2. *Recall that* $\boldsymbol{B}$=$\boldsymbol{UH}$, *where* $\boldsymbol{U}$ *is the unipotent radical and* $\boldsymbol{H}$=$\boldsymbol{B} \cap \boldsymbol{N}$ *is a torus. For* $w \in W$, *set* $\boldsymbol{H}^w := \{t \in \boldsymbol{H} | F(t) = w^{-1}tw\}$ *and* $\boldsymbol{U}_w^- := \boldsymbol{U} \cap w\boldsymbol{U}w^{-1}$ *(the same as 1.2.13), then*
(i) $\boldsymbol{U}_w^-\boldsymbol{H}^w$ *acts on* $L^{-1}(w\boldsymbol{U})$ *by right multiplication.*
(ii) $g \mapsto g\boldsymbol{B}g^{-1}$ *induces a surjective map* $L^{-1}(w\boldsymbol{U}) \to X_w$, *whose fibres are* $\boldsymbol{U}_w^-\boldsymbol{H}^w$*-orbits on* $L^{-1}(w\boldsymbol{U})$. *Set the variety* Y_w *to be the quotient* $Y_w := L^{-1}(w\boldsymbol{U})/\boldsymbol{H}^w$.

Theorem 3.2.3. *For an affine varieties* X, Y *over* k, *fix a prime* $l \neq p$, $\bar{\mathbb{Q}}_l$ *the* $l-$*adic number field, then the finite morphism* $\phi : X \to Y$ *will induce* $\phi^* : H_c^i(Y, \bar{\mathbb{Q}}_l) \to H_c^i(X, \bar{\mathbb{Q}}_l)$, *which is a linear map between finite dimensional* $\bar{\mathbb{Q}}_l$ *vector spaces. Now in our case, we take* X *defined over* k *and* F *the corresponding Frobenius map as the morphism on itself. The* ***Grothendieck's trace formula*** *tells* $|X^F| = \sum_i (-1)^i Tr(F^*, H_c^i(X, \bar{\mathbb{Q}}_l))$.

For the any automorphism $g : X \to X$ of finite order, define the **Lefschetz Number** of g to be

$$\mathcal{L}(g, X) := \sum_i (-1)^i Tr(g^*, H_c^i(X, \bar{\mathbb{Q}}_l)).$$

We have the following properties 3.2.4 to 3.2.6 for the cohomology groups $H_c^i(X, \bar{\mathbb{Q}}_l)$ and Lefschetz numbers.

Theorem 3.2.4. *For the above* $g : X \to X$, *there is a Frobenius map* $F : X \to X$ *with* $gF = Fg$, *and* $P, Q \in \mathbb{Q}[t]$ *such that*

$$-\sum_{n=1}^{\infty} |X^{gF^n}| t^n = P/Q, \qquad \mathcal{L}(g, X) = \lim_{t\to\infty} P(t)/Q(t)$$

And $\mathcal{L}(g, X) = \mathcal{L}(g^{-1}, X) \in \mathbb{Z}$ *is independent of the choice of* l.

Theorem 3.2.5. *(i)*$H_c^i(X, \bar{\mathbb{Q}}_l) \neq 0$ *only possibly when* $0 \leq i \leq 2\dim X$.
(ii) If X *is a dimension* n *affine space, then* $\dim H_c^i(X, \bar{\mathbb{Q}}_l) = 1$, *if* $i = 2n$, *and* $\dim H_c^i(X, \bar{\mathbb{Q}}_l) = 0$ *for other cases.*
(iii) For two varieties X_1 *and* X_2, *with corresponding automorphisms* g_1, g_2 *of finite order, then* $g_1 \times g_2$ *is an automorphism on* $X_1 \times X_2$ *of finite order, and*

$$H_c^i(X_1 \times X_2, \bar{\mathbb{Q}}_l) \cong \bigoplus_{j+k=i} H_c^i(X_1, \bar{\mathbb{Q}}_l) \otimes H_c^i(X_2, \bar{\mathbb{Q}}_l).$$

As a result, the Lefschetz number will satisfy $\mathcal{L}(g_1 \times g_2, X_1 \times X_2) = \mathcal{L}(g_1, X_1)\mathcal{L}(g_2, X_2)$.
(iv) If $X = \bigcup_j X_j$ *is the finite disjoint union of closed subsets* X_j *and automorphism* g *leaves each* X_j *invariant, then* $\mathcal{L}(g, X) = \sum_j \mathcal{L}(g, X_j)$.

Theorem 3.2.6. *(i)(Suppose there is a finite group G of automophisms on X and assume the quotient space X/G exists, then $H_c^i(X/G,\bar{\mathbb{Q}}_l)\cong H_c^i(X,\bar{\mathbb{Q}}_l)^G$. Let $\varphi: X\to X/G$ be the quotient map, if there exists $\bar{g}$ a automorphism on X/G such that $\bar{g}\circ\varphi=\varphi\circ g$, where g is an automorphism on X commutes with all of G, we have $\mathcal{L}(\bar{g},X/G)=\frac{1}{|G|}\sum_{x\in G}\mathcal{L}(gx,X)$.*
(ii) If X is finite, then $\mathcal{L}(g,X)=|X^g|$.

Remark. We will repeatedly use these properties above in the following definitions and computations, and they lead to a branch of beautiful results.

The map $R_w:\mathbf{G}^F\to\mathbb{Z},g\mapsto\mathcal{L}(g,Y_w)$ then defines a virtual character of $\mathbf{G}^F$. An irreducible representation of $\mathbf{G}^F$ is called **unipotent** if its character χ has non-zero scalar product with R_w, that is $\langle\chi,R_w\rangle_{\mathbf{G}^F}\neq 0$.
The above construction of virtual characters can be generalized for each $F-$stable maximal torus $\mathbf{T}$ of $\mathbf{G}$ and irreducible character θ of $\mathbf{T}^F$. Since $\mathbf{T}$ is conjugate to $\mathbf{H}$, there is another Borel $\tilde{\mathbf{B}}$, such that $\tilde{\mathbf{B}}=\tilde{\mathbf{U}}\mathbf{T}$. Then $\mathbf{G}^F$ and $\mathbf{T}^F$ act on $L^{-1}(\tilde{\mathbf{U}})$ by respectively left and right multiplication, and the action leaves $L^{-1}(\tilde{\mathbf{U}})$ stable since

$$L(gxt)=(gxt)^{-1}F(gxt)=t^{-1}x^{-1}g^{-1}F(g)F(x)F(t)=t^{-1}x^{-1}F(x)t\in\tilde{\mathbf{U}},\quad x\in L^{-1}(\tilde{\mathbf{U}}),g\in\mathbf{G}^F,t\in\mathbf{T}^F.$$

Then define a complex function on $\mathbf{G}^F$ by $R^{\mathbf{G}}_{\mathbf{T},\tilde{\mathbf{U}}}(\theta)(g):=\frac{1}{|\mathbf{T}^F|}\sum_{t\in\mathbf{T}^F}\theta(t^{-1})\mathcal{L}((g,t),L^{-1}(\tilde{\mathbf{U}})),\forall g\in\mathbf{G}^F,\forall\theta\in\mathrm{Irr}(\mathbf{T}^F)$.

Proposition 3.2.7. *$R^{\mathbf{G}}_{\mathbf{T},\tilde{\mathbf{U}}}(\theta)$ just defined has values in algebraic integers, moreover, $R^{\mathbf{G}}_{\mathbf{T},\tilde{\mathbf{U}}}(\theta)$ defines a virtual character of $\mathbf{G}^F$.*

Theorem 3.2.8 (Deligne-Lusztig)**.** *For two $F-$stable maximal tori $\mathbf{T}$ and $\mathbf{T}'$, $\tilde{\mathbf{U}},\tilde{\mathbf{U}}'$ unipotent radicals, $\theta\in\mathrm{Irr}(\mathbf{T})$, $\theta'\in\mathrm{Irr}(\mathbf{T}')$, let $N_{\mathbf{G}}(\mathbf{T},\mathbf{T}'):=\{n\in\mathbf{G}|^n\mathbf{T}=n^{-1}\mathbf{T}n=\mathbf{T}'\}$. Then $\langle R^{\mathbf{G}}_{\mathbf{T},\tilde{\mathbf{U}}}(\theta),R^{\mathbf{G}}_{\mathbf{T}',\tilde{\mathbf{U}}'}(\theta')\rangle_{\mathbf{G}^F}=\left|\{w\in\mathbf{W}(\mathbf{T},\mathbf{T}')^F;\theta={}^w\theta'\}\right|$.*

Proof. By definition of the character and property of $l-$adic cohomology, we have

$$\begin{aligned}\left\langle R^{\mathbf{G}}_{\mathbf{T},\tilde{\mathbf{U}}}(\theta),R^{\mathbf{G}}_{\mathbf{T}',\tilde{\mathbf{U}}'}(\theta)\right\rangle&=\frac{1}{\left|\mathbf{G}^F\right|}\frac{1}{\left|\mathbf{T}^F\right|}\frac{1}{\left|\mathbf{T}'^F\right|}\sum_{g\in\mathbf{G}^F}\sum_{t\in\mathbf{T}^F}\sum_{t'\in\mathbf{T}'^F}\theta(t^{-1})\overline{\theta(t'^{-1})}\mathcal{L}((g,t),L^{-1}(\tilde{U}))\mathcal{L}((g,t'),L^{-1}(\tilde{U}'))\\
(\text{by Theorem 3.2.5})&=\frac{1}{\left|\mathbf{G}^F\right|}\frac{1}{\left|\mathbf{T}^F\right|}\frac{1}{\left|\mathbf{T}'^F\right|}\sum_{g\in\mathbf{G}^F}\sum_{t\in\mathbf{T}^F}\sum_{t'\in\mathbf{T}'^F}\theta(t^{-1})\theta(t)\mathcal{L}((g,t)\times(g,t'),L^{-1}(\tilde{U})\times L^{-1}(\tilde{U}'))\\
(\text{by Theorem 3.2.6})&=\frac{1}{\left|\mathbf{T}^F\right|}\frac{1}{\left|\mathbf{T}'^F\right|}\sum_{t\in\mathbf{T}^F}\sum_{t'\in\mathbf{T}'^F}\theta(t^{-1})\theta(t)\mathcal{L}((t,t'),L^{-1}(\tilde{U})\times L^{-1}(\tilde{U}')/\mathbf{G}^F)\end{aligned}$$

We consider the space $S=L^{-1}(\tilde{U})\times L^{-1}(\tilde{U}')/\mathbf{G}^F$ with $\mathbf{T}^F\times\mathbf{T}'^F$ acting on it, and the space

$$S':=\left\{(x,x',y)\in\tilde{\mathbf{U}}\times\tilde{\mathbf{U}}'\times\mathbf{G}|xF(y)=yx'\right\}$$

The map $L^{-1}(\tilde{U})\times L^{-1}(\tilde{U}')\to S'$ is given by $(g,g')\mapsto(L(g),L(g'),g^{-1}g')$. Easy to check $L(g)F(g^{-1}g')=g^{-1}g'L(g')$, and for $\forall x\in\mathbf{G}^F$, (xg,xg') is mapped to the same image in S', hence we get a map from S to S'. This map is a bijection, see [Car85]. Due to the Bruhat decomposition of $\mathbf{G}$, S' facts into disjoint union $S'=\bigcup_w S'_w:=\{(x,x',y)\in S'|y\in\mathbf{B}w\mathbf{B}\}$, and the $\mathbf{T}^F\times\mathbf{T}'^F$ action leaves each S'_w invariant. By 3.2.5, we have

$$\left\langle R^{\mathbf{G}}_{\mathbf{T},\tilde{\mathbf{U}}}(\theta),R^{\mathbf{G}}_{\mathbf{T}',\tilde{\mathbf{U}}'}(\theta)\right\rangle=\frac{1}{\left|\mathbf{T}^F\right|}\frac{1}{\left|\mathbf{T}'^F\right|}\sum_{w\in\mathbf{W}}\sum_{t\in\mathbf{T}^F}\sum_{t'\in\mathbf{T}'^F}\theta(t^{-1})\theta(t)\mathcal{L}((t,t'),S'_w)$$

the $\mathbf{T}^F\times\mathbf{T}'^F$ extend to $H_w=\left\{(t,t')\in\mathbf{T}\times\mathbf{T}'|L(t')=F(w)^{-1}L(t)F(w)\right\}$. Take H_w^0 the connected component of H_w, we have $\mathcal{L}((t,t'),S'_w)=\mathcal{L}((t,t'),S_w'^{H_w^0})$. By 3.2.6, we know the later one is $\left|S_w'^{H_w^0}\right|$ if $S_w'^{H_w^0}$ is finite. In [Car85], its structure is well illustrated. And as a conclusion, we gives

$$\left\langle R^{\mathbf{G}}_{\mathbf{T},\tilde{\mathbf{U}}}(\theta), R^{\mathbf{G}}_{\mathbf{T}',\tilde{\mathbf{U}}'}(\theta)\right\rangle = \frac{1}{\left|\mathbf{T}^F\right|}\sum_{w\in\mathbf{W}(\mathbf{T},\mathbf{T}')^F}\left|\mathbf{T}^F\right|\left\langle \theta, {}^w\theta'\right\rangle = \left|\{w\in\mathbf{W}(T,T')^F;\theta={}^w\theta'\}\right|$$

and that ends the proof. □

Corollary 3.2.9. *From Theorem 3.2.8, the virtual character $R^{\mathbf{G}}_{\mathbf{T},\tilde{\mathbf{U}}}(\theta)$ does not depend on $\tilde{\mathbf{U}}$, thus we can abbreviate it as $R^{\mathbf{G}}_{\mathbf{T},\theta}$.*
(i) For two $F-$stable maximal tori $\mathbf{T}$ and $\mathbf{T}'$, $R^{\mathbf{G}}_{\mathbf{T},\theta}$ and $R^{\mathbf{G}}_{\mathbf{T}',\theta'}$ are either equal or orthogonal.
(ii) Choose $\mathbf{T}$ obtained from $\mathbf{H}$ by twisting with $w\in W$, then $R_w = R^{\mathbf{G}}_{\mathbf{T},1}$.

Remark. Hence we see that $R^{\mathbf{G}}_{\mathbf{T},\theta}$ is indeed the generalization of the virtual character R_w, and one irreducible character is **unipotent** if it has non-zero scalar product with $R^{\mathbf{G}}_{\mathbf{T},1}$. We introduce R_w here is to obtain a geometric understanding of the construction, i.e. the relationship with the Deligne-Lusztig varieties, we shall see later the explicit relation between unipotent characters and Weyl groups.
For disconnected reductive algebraic group $\mathbf{G}$, consider the connected component $\mathbf{G}^\circ$ of $\mathbf{G}$ which contains the identity. Then $\mathbf{T}\subset\mathbf{G}^\circ$, define $R^{\mathbf{G}}_{\mathbf{T},\theta} := \mathrm{Ind}^{\mathbf{G}^F}_{\mathbf{G}^{\circ F}}(R^{\mathbf{G}^\circ}_{\mathbf{T},\theta})$.

Remark. Notice the proof can be largely simplified if the Mackey formula 3.1.2 holds under our construction of the character. The computation is straight forward

$$\left\langle R^{\mathbf{G}}_{\mathbf{T},\theta}, R^{\mathbf{G}}_{\mathbf{T}',\theta'}\right\rangle = \left\langle\theta, {}^*R^{\mathbf{G}}_{\mathbf{T}}\circ R^{\mathbf{G}}_{\mathbf{T}'}(\theta')\right\rangle = \left\langle \theta, \sum_{w\in\mathbf{T}^F\backslash\mathcal{S}(\mathbf{T},\mathbf{T}')/\mathbf{T}'^F} R^{\mathbf{T}}_{\mathbf{T}\cap{}^w\mathbf{T}'}\circ{}^*R^{{}^w\mathbf{T}'}_{\mathbf{T}\cap{}^w\mathbf{T}'}\circ\mathrm{ad}\, w(\theta')\right\rangle \text{ (by Mackey formula)}$$

Since $\mathbf{T}$ and $\mathbf{T}'$ are rational maximal tori, then $w\in\mathbf{W}(\mathbf{T},\mathbf{T}')^F$, and $\mathbf{T}={}^w\mathbf{T}'$. The rest of proof can be omitted since θ and θ' are irreducible. However, the reason why we do not use Mackey formula is that the proof of it holds overlaps our straightforward proof in the main text.

We explore more interesting properties about $R^{\mathbf{G}}_{\mathbf{T},\theta}$, without declaring, we consider only connected reductive groups.

Proposition 3.2.10. *(i)*

$$\left\langle R^{\mathbf{G}}_{\mathbf{T},\theta}, 1_{\mathbf{G}^F}\right\rangle = \begin{cases} 1 & \text{if } \theta = 1\\ 0 & \text{if } \theta\neq 1\end{cases}$$

(ii) Moreover, denote by $(\mathbf{T})$ the conjugacy class of fixed F-stable, maximal tori $\mathbf{T}$, that is, those $\mathbf{T}_w$ obtained by twisting $\mathbf{T}$ with $w\in\mathbf{W}(T)$, then we have $1_{\mathbf{G}^F} = \frac{1}{|\mathbf{W}(\mathbf{T})^F|}\sum_{(\mathbf{T})} R^{\mathbf{G}}_{\mathbf{T},1}$.

Proof. (i) Bland computations

$$\begin{aligned}
\left\langle R^{\mathbf{G}}_{\mathbf{T},\theta}, 1_{\mathbf{G}^F}\right\rangle &= \frac{1}{\left|\mathbf{G}^F\right|}\sum_{g\in\mathbf{G}^F}\frac{1}{\left|\mathbf{T}^F\right|}\sum_{t\in\mathbf{T}^F}\theta(t^{-1})\mathcal{L}((g,t),L^{-1}(\tilde{\mathbf{U}})).\\
&= \frac{1}{\left|\mathbf{T}^F\right|}\sum_{t\in\mathbf{T}^F}\theta(t^{-1})\mathcal{L}(t,L^{-1}(\tilde{\mathbf{U}})). \quad \left(\frac{1}{\left|\mathbf{G}^F\right|}\sum_{g\in\mathbf{G}^F}\mathcal{L}((g,t),L^{-1}(\tilde{\mathbf{U}})) = \mathcal{L}(t,L^{-1}(\tilde{\mathbf{U}})/\mathbf{G}^F),\ \text{by } 3.2.6\right)\\
&= \frac{1}{\left|\mathbf{T}^F\right|}\sum_{t\in\mathbf{T}^F}\theta(t^{-1}) = \begin{cases} 1 & \text{if } \theta=1\\ 0 & \text{if }\theta\neq 1\end{cases}
\end{aligned}$$

(ii) Let $\phi = \frac{1}{|\mathbf{G}^F|}\sum_{\mathbf{T}\text{ F-stable, maximal torus}}\left|\mathbf{T}^F\right| R^{\mathbf{G}}_{\mathbf{T},1}$, we prove that $\phi = 1_{\mathbf{G}^F}$, notice also $\langle\phi, 1_{\mathbf{G}^F}\rangle = \frac{1}{|\mathbf{G}^F|}\sum_{\mathbf{T}}\left|\mathbf{T}^F\right|$, then

$$\frac{1}{\left|\mathbf{G}^F\right|}\sum_{\mathbf{T}}\left|\mathbf{T}^F\right| = \frac{1}{\left|\mathbf{G}^F\right|}\sum_{(\mathbf{T})}\frac{\left|\mathbf{G}^F\right|}{\left|N_{\mathbf{G}}(\mathbf{T})^F\right|}\left|\mathbf{T}^F\right| = \sum_{(\mathbf{T})}\frac{1}{\left|\mathbf{W}(\mathbf{T})^F\right|} = 1$$

Compute $\langle \phi, \phi \rangle = \frac{1}{\left|\mathbf{G}^F\right|^2} \left\langle \sum_{\mathbf{T}} \left|\mathbf{T}^F\right| R^{\mathbf{G}}_{\mathbf{T},1}, \sum_{\mathbf{T}'} \left|\mathbf{T}'^F\right| R^{\mathbf{G}'}_{\mathbf{T}',1} \right\rangle = \frac{1}{\left|\mathbf{G}^F\right|^2} \sum_{\mathbf{T}} \sum_{\mathbf{T}'} \left|\mathbf{T}^F\right| \left|\mathbf{T}'^F\right| \left|\mathbf{W}(\mathbf{T}, \mathbf{T}')^F\right| = 1$,
where the second equality above is from the Theorem 3.2.8, and the last equality comes from the fact that $\left|\mathbf{W}(\mathbf{T}, \mathbf{T}')^F\right| \neq 0$ if and only if $\mathbf{T}, \mathbf{T}'$ are conjugate, and hence $\left|\mathbf{T}^F\right| = \left|\mathbf{T}'^F\right|$ under such case. Thus ϕ must be 1, and ϕ is exactly the right hand side of our formula given in the proposition. □

We consider value of $R^{\mathbf{G}}_{\mathbf{T},\theta}$ regarding to the Jordan decomposition. For any $g \in \mathbf{G}^F$, take its Jordan decomposition $g = su$, where s is semisimple and u is unipotent. Denote by $C(s)$ the centralizer of s in $\mathbf{G}$, and $C^0(s)$ its connected component.

Proposition 3.2.11. *For $u \in \boldsymbol{G}^F$ a unipotent, $R^{\boldsymbol{G}}_{\boldsymbol{T},\theta}(u)$ is independent on θ, we then denote it as $Q^{\boldsymbol{G}}_{\boldsymbol{T}}(u)$, called the **Green function**. And the value of $R^{\boldsymbol{G}}_{\boldsymbol{T},\theta}(g)$ can be expressed as $R^{\boldsymbol{G}}_{\boldsymbol{T},\theta}(g) = \frac{1}{|C^0(s)^F|} \sum_{x \in \boldsymbol{G}^F, {}^x\boldsymbol{T} \subseteq C^0(s)} {}^x\theta(s) Q^{C^0(s)}_{{}^x\boldsymbol{T}}(u)$*

Proof. Compute $R^{\mathbf{G}}_{\mathbf{T},\theta}(u) = \frac{1}{\left|\mathbf{T}^F\right|} \sum_{t \in \mathbf{T}^F} \theta(t^{-1}) \mathcal{L}((u,t), L^{-1}(\tilde{\mathbf{U}})) = \frac{1}{\left|\mathbf{T}^F\right|} \sum_{t \in \mathbf{T}^F} \theta(t^{-1}) \mathcal{L}((u,1), L^{-1}(\tilde{\mathbf{U}})^t)$, since $L^{-1}(\tilde{\mathbf{U}})$ is only fixed by $t = 1$, we have $R^{\mathbf{G}}_{\mathbf{T},\theta}(u) = \frac{1}{\left|\mathbf{T}^F\right|} \mathcal{L}((u,1), L^{-1}(\tilde{\mathbf{U}}))$ is independent of θ, and hence $\sum_{\theta \in \hat{\mathbf{T}}^F} R^{\mathbf{G}}_{\mathbf{T},\theta}(u) = \left|\mathbf{T}^F\right| Q^{\mathbf{G}}_{\mathbf{T}}(u)$.
We then give the proof of a restricted version of the proposition, we consider $R^{\mathbf{G}}_{\mathbf{T},\theta}(s)$ where s is a regular semisimple element, i.e. $C(s)$ is a maximal torus. By 3.2.6, $\mathcal{L}((s,t), L^{-1}(\tilde{\mathbf{U}})) = \mathcal{L}(1, L^{-1}(\tilde{\mathbf{U}})^{(s,t)})$.
If $g \in L^{-1}(\tilde{\mathbf{U}})$ is fixed by (s,t), then $g = sgt^{-1}$, and $L(g) = g^{-1}F(g) \in C_{\mathbf{G}}(\mathbf{T}) \cap \tilde{\mathbf{U}} = \{1\}$, then $g \in \mathbf{G}^F$, and $s = {}^tg$. $\mathcal{L}((s,t), L^{-1}(\tilde{\mathbf{U}})) = \left|L^{-1}(\tilde{\mathbf{U}})^{(s,t)}\right| = \left|\left\{g \in \mathbf{G}^F | s = {}^tg\right\}\right| = \left|C(s)^F\right| = \left|\mathbf{T}^F\right|$. Hence under above assumption, $R^{\mathbf{G}}_{\mathbf{T},\theta}(s) = \sum_{w \in \mathbf{W}(\mathbf{T})^F} {}^w\theta(s)$. Not hard to see it is the special case of the formula in the proposition. For a complete and general proof, please see [Car85]. □

Corollary 3.2.12. *As a corollary, we get the orthogonality of Green functions, $\frac{1}{\left|\boldsymbol{G}^F\right|} \sum_{u \text{ unipotent}} Q^{\boldsymbol{G}}_{\boldsymbol{T}'}(u) Q^{\boldsymbol{G}}_{\boldsymbol{T}}(u) = \frac{\left|N(\boldsymbol{T},\boldsymbol{T}')^F\right|}{\left|\boldsymbol{T}^F\right|\left|\boldsymbol{T}'^F\right|}$.*

Proposition 3.2.13. *We assume $\boldsymbol{T}_w$ is obtained by twisting with $w \in W$, then we have $R^{\boldsymbol{G}}_{\boldsymbol{T}_w,\theta}(1) = (-1)^{l(w)} |\boldsymbol{G}^F|_{p'} |\boldsymbol{T}^F|^{-1}$, where $l(w)$ is the length of w, and for a positive integer z, $z_{p'} := z/z_p$, where z_p is the maximal power of p that divides z.*

Before proving the proposition, we would like to study on the dimension of $R^{\mathbf{G}}_{\mathbf{T},\theta}$, for a better illustration(and actually as a must for the proof), we introduce the **Steinberg character**.

Definition 3.2.14. Recall in the section 3.1, we introduce the functor $R^{\mathbf{G}}_{\mathbf{L}}$ and its adjoint ${}^*R^{\mathbf{G}}_{\mathbf{L}}$. By $\varepsilon_{\mathbf{G}}$ we mean $(-1)^{\text{rank}\,\mathbf{G}}$, here rank is the dimension of maximal $\mathbb{F}_q$-split torus of $\mathbf{G}$. Define the duality functor on the class function space $\mathcal{V}(\mathbf{G}^F)$ to be $D_{\mathbf{G}} := \sum_{\mathbf{P} \supseteq \mathbf{B}} (-1)^{r(\mathbf{P})} R^{\mathbf{G}}_{\mathbf{L}} \circ^* R^{\mathbf{G}}_{\mathbf{L}}$ (by $r(\mathbf{G})$ we mean a semisimple rank of $\mathbf{G}$).
Define the **Steinberg character** on $\mathbf{G}^F$ to be $\mathbf{St} := D_{\mathbf{G}}(1_{\mathbf{G}^F})$.

Remark. For a general finite group G with a split $BN-$pair, U is a finite $p-$group, H is an abelian group of order prime to p. Consider the group algebra $K[G]$ of G over field K, the Steinberg character is constructed as the following. Set $e := \sum_{b \in B} \sum_{w \in W} (-1)^{l(w)} bw \in K[G]$, and $I = (e) \subseteq K[G]$ be the right ideal. We obtained a representation $\rho : G \to \text{End}_K(I)$ by setting $G-$action by right multiplication. When $K = \mathbb{C}$, then ρ is irreducible, and has multiplicity 1 in the permutation representation of G on $B\backslash G/B$. The character of ρ is denoted by St_G, is the generalization of definition of Steinberg character, it satisfies $\text{St}_G(1) = |U|, \text{St}_G(u) = 0, \forall$ non-unit $u \in U$, for details one can refer to [Hum].

We list some properties of the duality functor $D_{\mathbf{G}}$, for detailed proof, refer to [Dig].

Proposition 3.2.15. *(i) The definition of $D_{\boldsymbol{G}}$ does not depend on the choice of $\boldsymbol{B}$.*
(ii) $D_{\boldsymbol{G}}$ is self adjoint, and $D_{\boldsymbol{G}} \circ D_{\boldsymbol{G}}$ is the identity functor.
(iii) $\varepsilon_{\boldsymbol{G}} D_{\boldsymbol{G}} \circ R^{\boldsymbol{G}}_{\boldsymbol{L}} = \varepsilon_{\boldsymbol{L}} R^{\boldsymbol{G}}_{\boldsymbol{L}} \circ D_{\boldsymbol{L}}$
(iv) For $\chi \in \text{Irr}(\boldsymbol{G}^F)$, and Jordan decomposition$x = su \in \boldsymbol{G}^F$, $\varepsilon_{\boldsymbol{G}} D_{\boldsymbol{G}}(\chi) = \varepsilon_{C^0(s)} D_{C^0(s)}(\chi|_{C^0(s)})$.

Corollary 3.2.16. *As a corollary of the above, we see* $\langle \boldsymbol{St}_{\mathbf{G}^F}, \boldsymbol{St}_{\mathbf{G}^F}\rangle = \langle D_{\mathbf{G}}(1), D_{\mathbf{G}}(1)\rangle = \langle 1, D_{\mathbf{G}}^2(1)\rangle = 1$. *And*

$$\boldsymbol{St}_{\mathbf{G}^F}(x) = \begin{cases} \varepsilon_{\mathbf{G}}\varepsilon_{(C^0(x))} \left|C^0(x)^F\right|_p, & \textit{if } x \textit{ is semisimple} \\ 0, & \textit{other} \end{cases} \qquad \textit{Hence we have } \boldsymbol{St}(1) = \left|\boldsymbol{G}^F\right|_p.$$

Proof of Proposition 3.2.13. Bland computation:

$$\left\langle R^{\mathbf{G}}_{\mathbf{T},\theta}, \mathbf{St}_{\mathbf{G}^F}\right\rangle = \left\langle R^{\mathbf{G}}_{\mathbf{T},\theta}, D_{\mathbf{G}}(1_{\mathbf{G}^F})\right\rangle = \left\langle D_{\mathbf{G}} \circ R^{\mathbf{G}}_{\mathbf{T}}(\theta), 1_{\mathbf{G}^F}\right\rangle \quad \text{(by adjointness in proposition 3.2.15)}$$
$$= \varepsilon_{\mathbf{G}}\varepsilon_{\mathbf{T}}\langle \theta, 1_{\mathbf{T}^F}\rangle_{\mathbf{T}^F} \quad \text{(by (ii) of proposition 3.2.15)} \quad = \left\langle \sum_{\theta\in \mathrm{Irr}(\mathbf{T}^F)} R^{\mathbf{G}}_{\mathbf{T},\theta}, \mathbf{St}_{\mathbf{G}^F}\right\rangle$$

Hence $\frac{1}{|\mathbf{G}^F|}\left(\sum_\theta R^{\mathbf{G}}_{\mathbf{T},\theta}(1)\right)\mathbf{St}_{\mathbf{G}^F}(1) = \frac{1}{|\mathbf{G}^F|}\left|\mathbf{T}^F\right| R^{\mathbf{G}}_{\mathbf{T},\theta}(1)\,\mathbf{St}_{\mathbf{G}^F}(1) = \varepsilon_{\mathbf{G}}\varepsilon_{\mathbf{T}}$. Use 3.2.16, easy to deduce that $\varepsilon_{\mathbf{G}}\varepsilon_{\mathbf{T}}R^{\mathbf{G}}_{\mathbf{T},\theta}(1) = |\mathbf{G}^F|_{p'}|\mathbf{T}^F|^{-1}$. We see $\varepsilon_{\mathbf{G}}\varepsilon_{\mathbf{T}}R^{\mathbf{G}}_{\mathbf{T},\theta}$ then has a positive dimension.
For $\mathbf{T}_w$ get from twisting maximal $\mathbf{T}$ with w, we have $\varepsilon_{\mathbf{G}}\varepsilon_{\mathbf{T}_w} = (-1)^{l(w)}$, see [Dig] for a proof. □

Now we denote by $\mathcal{V}(\mathbf{G}^F)$ the set of class functions on $\mathbf{G}^F$ and by **uniform functions** we mean those class functions as linear combinations of Deligne-Lusztig characters. Denote the set of the uniform functions by $\mathcal{V}(\mathbf{G}^F)^\sharp$.
As the end of this section, for any class function f, we would like to give its projection $f^\sharp$ onto uniform function space.

Definition 3.2.17. For any $\mathbf{T} \in \mathbf{G}$ and $\forall f \in \mathcal{V}(\mathbf{G}^F)$, define the **principal part** of f on $\mathcal{V}(\mathbf{T}^F)$ by

$$f_{\mathbf{T}^F}(x) := \frac{|\mathbf{T}^F|}{|C^0(x)^F|}\sum_{u\in C^0(x), u \text{ unipotent}} f(xu)Q^{C^0(x)}_{\mathbf{T}}(u), \quad \forall x\in T$$

We shall see this definition is actually adjoint to Deligne-Lusztig induction.

Lemma 3.2.18. *The formula of adjoint functor* ${}^*R^{\boldsymbol{G}}_{\boldsymbol{T}}$ *is given as* $\frac{1}{|\boldsymbol{G}^F|}\sum_{g\in \boldsymbol{G}^F}\mathcal{L}((g,t), L^{-1}(U))$, *and* ${}^*R^{\boldsymbol{G}}_{\boldsymbol{T}}(f) = f_{\boldsymbol{T}^F}$.

Theorem 3.2.19. *The projection of f onto uniform function space is* $f^\sharp = \sum_{(\boldsymbol{T})}\frac{1}{|\boldsymbol{W}(\boldsymbol{T})^F|}R^{\boldsymbol{G}}_{\boldsymbol{T},f_{\boldsymbol{T}^F}} = \sum_{(\boldsymbol{T})}\frac{1}{|\boldsymbol{W}(\boldsymbol{T})^F|}R^{\boldsymbol{G}}_{\boldsymbol{T}}\circ {}^*R^{\boldsymbol{G}}_{\boldsymbol{T}}(f)$. *Notice here $f_{\boldsymbol{T}^F}$ may not be irreducible, but $R^{\boldsymbol{G}}_{\boldsymbol{T},f_{\boldsymbol{T}^F}}$ can be regard as the linear combination of induced irreducible component of $f_{\boldsymbol{T}^F}$. And the sum is running over the conjugacy classes of tori, $\boldsymbol{T}$ is the representative of each class.*

Proof. The right hand side is already of the form of Deligne-Lusztig character, thus only need to verify it coincides with the uniform constitute of f, that is, for any $\mathbf{T}$ and $\theta\in \mathrm{Irr}(\mathbf{T}^F)$, $\left\langle f, R^{\mathbf{G}}_{\mathbf{T},\theta}\right\rangle = \left\langle f^\sharp, R^{\mathbf{G}}_{\mathbf{T},\theta}\right\rangle$. In fact

$$\left\langle f^\sharp, R^{\mathbf{G}}_{\mathbf{T},\theta}\right\rangle = \left\langle \sum_{(\mathbf{T}')}\frac{1}{|\mathbf{W}(\mathbf{T}')^F|}R^{\mathbf{G}}_{\mathbf{T}'}\circ {}^*R^{\mathbf{G}}_{\mathbf{T}'}(f), R^{\mathbf{G}}_{\mathbf{T},\theta}\right\rangle_{\mathbf{G}^F} = \sum_{(\mathbf{T}')}\left\langle \frac{1}{|\mathbf{W}(\mathbf{T}')^F|}R^{\mathbf{G}}_{\mathbf{T}'}\circ {}^*R^{\mathbf{G}}_{\mathbf{T}'}(f), R^{\mathbf{G}}_{\mathbf{T},\theta}\right\rangle_{\mathbf{G}^F}$$
$$= \sum_{(\mathbf{T})'}\left\langle \frac{1}{|\mathbf{W}(\mathbf{T}')^F|}{}^*R^{\mathbf{G}}_{\mathbf{T}'}(f), {}^* R^{\mathbf{G}}_{\mathbf{T}'}\circ R^{\mathbf{G}}_{\mathbf{T}}(\theta)\right\rangle_{\mathbf{G}^F}$$

By Theorem 3.2.8, for $\forall\theta' \in \mathrm{Irr}(\mathbf{T}'^F)$, $\langle \theta', {}^*R^{\mathbf{G}}_{\mathbf{T}'}\circ R^{\mathbf{G}}_{\mathbf{T}}(\theta)\rangle = \left\langle R^{\mathbf{G}}_{\mathbf{T}',\theta'}, R^{\mathbf{G}}_{\mathbf{T},\theta}\right\rangle \neq 0$ only if $(\mathbf{T}') = (\mathbf{T})$, and under the circumstance that $\mathbf{T} = \mathbf{T}'$, ${}^*R^{\mathbf{G}}_{\mathbf{T}}\circ R^{\mathbf{G}}_{\mathbf{T}}(\theta) = \sum_{w\in \mathbf{W}(\mathbf{T})^F} {}^w\theta$ (by Mackey formula 3.1.2, we state it for $R^{\mathbf{G}}_{\mathbf{T},\theta}$ in the remark of Theorem 3.2.8), then

$$\left\langle f^\sharp, R^{\mathbf{G}}_{\mathbf{T},\theta}\right\rangle = \frac{1}{|\mathbf{W}(\mathbf{T})^F|}\left\langle {}^*R^{\mathbf{G}}_{\mathbf{T}}(f), \sum_{w\in\mathbf{W}(\mathbf{T})^F} {}^w\theta\right\rangle = \left\langle f, R^{\mathbf{G}}_{\mathbf{T},{}^w\theta}\right\rangle = \left\langle f, R^{\mathbf{G}}_{\mathbf{T},\theta}\right\rangle.$$

Hence the theorem is proved. And we see Proposition 3.2.10 is actually one of the corollaries of this theorem. □

In the above theorem, we see that for a connected $\mathbf{G}$, the identity representation of $\mathbf{G}^F$ is uniform, we also want to show under the connectedness condition, the regular representation of $\mathbf{G}^F$ is uniform.

Proposition 3.2.20. *For a semisimple $s \in \boldsymbol{G}^F$, the function $f = \frac{\varepsilon_{C^0(s)}}{|C(s)^F||C^0(s)^F|_p} \sum_{\boldsymbol{T}, s\in \boldsymbol{T}} \varepsilon_{\boldsymbol{T}} \sum_{\theta \in \mathrm{Irr}(\boldsymbol{T}^F)} \theta(s)^{-1} R^{\boldsymbol{G}}_{\boldsymbol{T},\theta}$ is in fact the characteristic function of $\boldsymbol{G}^F$ conjugacy class of s.*

Proof. We briefly sketch the proof. Assume f' is the characteristic function of $\mathbf{G}^F$ conjugacy class of s. The goal is to prove $\langle f, f\rangle = \langle f, f'\rangle = \langle f', f'\rangle$, and since f, f' are class functions, $f - f' = 0$.
By computation, $\langle f', f'\rangle = \frac{1}{|\mathbf{G}^F|} \sum_{\mathbf{G}^F \text{conjugate to } s} 1 = \frac{1}{|\mathbf{G}^F|} \left|\mathbf{G}^F : C(s)^F\right| = \frac{1}{|C(s)^F|}$, and,

$$\begin{aligned}
\langle f, f'\rangle &= \frac{1}{\left|\mathbf{G}^F\right|} \left|\mathbf{G}^F : C(s)^F\right| f(s) = \frac{1}{|C(s)^F|} f(s) = \frac{\varepsilon_{C^0(s)}}{|C(s)^F|^2 \, |C^0(s)^F|_p} \sum_{\mathbf{T}, s\in \mathbf{T}} \varepsilon_{\mathbf{T}} \sum_{\theta \in \mathrm{Irr}(\mathbf{T}^F)} \theta(s)^{-1} R^{\mathbf{G}}_{\mathbf{T},\theta}(s) \\
&= \frac{\varepsilon_{C^0(s)}}{|C(s)^F|^2 \, |C^0(s)^F|_p} \sum_{\mathbf{T}, s\in \mathbf{T}} \varepsilon_{\mathbf{T}} \sum_{\theta \in \mathrm{Irr}(\mathbf{T}^F)} \theta(s)^{-1} \frac{\varepsilon_{\mathbf{T}} \varepsilon_{C^0(s)}}{\left|\mathbf{T}^F\right| |C^0(s)^F|_p} \sum_{g \in \mathbf{G}^F, {}^g s \in \mathbf{T}^F} {}^g\theta(s) \text{ (by Prop 3.2.11)} \\
&= \frac{1}{|C(s)^F|^2 \, |C^0(s)^F|_p^2} \sum \frac{1}{\left|\mathbf{T}^F\right|} \sum_{\theta} \sum_{g} \theta(s^{-1} g^{-1} s g) = \frac{1}{|C(s)^F|^2 \, |C^0(s)^F|_p^2} \sum_{\mathbf{T}} \sum_{g \in C(s)^F} 1 = \frac{1}{|C(s)|^F}.
\end{aligned}$$

While the second last equation holds since $\sum_\theta \theta(x) \neq 0$ only when $x = 1$, and that means $g \in C(s)^F$ and the sum takes value $\left|\mathbf{T}^F\right|$. Through the same spirit, one can also compute that $\langle f, f\rangle = \frac{1}{|C(s)^F|}$. Hence the proposition is proved. $\square$

Corollary 3.2.21. *The above derives to several corollaries. The proofs are omitted.*
(i) Since the regular character $\mathrm{reg}_{\boldsymbol{G}^F}$ of $\boldsymbol{G}^F$ takes value $\left|\boldsymbol{G}^F\right|$ at 1 and 0 elsewhere, then it can be expressed as $\mathrm{reg}_{\boldsymbol{G}^F} = \frac{1}{\left|\boldsymbol{G}^F\right|_p} \sum_{\boldsymbol{T}, F(\boldsymbol{T}=\boldsymbol{T})} \varepsilon_{\boldsymbol{G}} \varepsilon_{\boldsymbol{T}} \sum_{\theta \in \widehat{\boldsymbol{T}^F}} R^{\boldsymbol{G}}_{\boldsymbol{T},\theta}$. Thus, the regular character is a uniform function.
(ii) Those class functions that vanishes at $\boldsymbol{G}^F$ but some semisimple elements must be the linear combination of $R^{\boldsymbol{G}}_{\boldsymbol{T},\theta}$. And since every irreducible character of $\boldsymbol{G}^F$ has nonzero multiplicity in $\mathrm{reg}_{\boldsymbol{G}^F}$, thus there must be some $R^{\boldsymbol{G}}_{\boldsymbol{T},\theta}$ has nonzero inner product with it. And the dimension of a character is equal to its uniform projection since $f(1) = \langle f, \mathrm{reg}_{\boldsymbol{G}^F}\rangle = \left\langle f^\sharp, \mathrm{reg}_{\boldsymbol{G}^F}\right\rangle f^\sharp(1)$.
(iii) The number of rational maximal tori of $\boldsymbol{G}$ is $\left|\boldsymbol{G}^F\right|_p^2$.

Proposition 3.2.22. *(i) If ρ_1, ρ_2 are irreducible unipotent characters of Sp_{2n}, then $\rho_1^\sharp = \rho_1^\sharp$ implies $\rho_1 = \rho_2$.*
(i) If ρ_1, ρ_2 are irreducible unipotent characters of O^ϵ_{2m}, then $\rho_1^\sharp = \rho_1^\sharp$ implies $\rho_1 = \rho_2$ or $\rho_1 = \rho_2 \cdot \mathrm{sgn}$.

Proof. See [Pan19]. $\square$

3.3 Geometry Conjugacy Classes and Lusztig Series

We have known that whether two Deligne-Lusztig characters are orthogonal depends on their conjugacy relation, however, since these are virtual character, they still may have irreducible constituents in common. We shall introduce the notion of geometry conjugacy classes and Lusztig's paramatrization of irreducible characters.
Fix $\mathbf{T}$ a maximal rational torus, and denote by $k = \bar{F}_p$, F the Frobenius map. Let $X(\mathbf{T}) = \mathrm{Hom}(\mathbf{T}, \mathbb{G}_m)$ be the **character group**(roots), containing algebraic group morphism from $\mathbf{T}$ to $\mathbb{G}_m$, where $\mathbb{G}_m$ is the muliplicative group scheme, and $Y(\mathbf{T}) = \mathrm{Hom}(\mathbb{G}_m, \mathbf{T})$ is its dual(coroots), the duality comes from the composition of morphisms $f \circ g \in \mathrm{Hom}(\mathbb{G}_m, \mathbb{G}_m) \cong \mathbb{Z}$.

Definition 3.3.1. We call two (connected) reductive algebraic groups $\mathbf{G}$ and $\mathbf{G}^*$ are **dual** if there exists pair of maximal tori $(\mathbf{T}, \mathbf{T}^*)$ in $(\mathbf{G}, \mathbf{G}^*)$ and a morphism $X(\mathbf{T}) \to Y(\mathbf{T}^*)$ such that it is an isomorphism.

Give the dual torus $\mathbf{T}^*$, the Frobenius structure F^* is obtained from F action on $Y(\mathbf{T}^*) = X(\mathbf{T})$ by identifying $(\mathbf{T}^F)^\vee$ and $\mathbf{T}^{*F^*}$ (denote by $(\mathbf{T}^F)^\vee := \mathrm{Hom}(\mathbf{T}^F, \bar{\mathbb{Q}}_l^\times) \cong \mathrm{Hom}(\mathbf{T}^F, \mathbb{Q}/\mathbb{Z})$).

There are some classical pairs of dual groups
(i) $\mathbf{G} = \mathbf{GL}_n, \mathbf{G}^* = \mathbf{GL}_n$; (ii) $\mathbf{G} = \mathbf{SL}_n, \mathbf{G}^* = \mathbf{PGL}_n$; (iii)$\mathbf{G} = \mathbf{U}_n, \mathbf{G}^* = \mathbf{U}_n$;
(iv)$\mathbf{G} = \mathbf{Sp}_{2n}, \mathbf{G}^* = \mathbf{SO}_{2n+1}$ (v) $\mathbf{G} = \mathbf{SO}^{\epsilon}_{2n}, \mathbf{G}^* = \mathbf{SO}^{\epsilon}_{2n}$.

We now consider the relationship between $Y(\mathbf{T})$ and $\mathbf{T}^F$. Let $(\mathbb{Q}/\mathbb{Z})_{p'}$ be the group of elements in $\mathbb{Q}/\mathbb{Z}$ that has order prime to p. Consider the map $\exp : x + \mathbb{Z} \mapsto e^{2\pi i x}$, where the later one is of the roots of unity, thus we obtain the order of elements $\mathbb{Q}/\mathbb{Z}$ from the order of their corresponding roots of unity. Notice also $k^\times \cong (\mathbb{Q}/\mathbb{Z})_{p'}$, $k^\times \hookrightarrow \bar{\mathbb{Q}}_l^\times$, then

$$\mathbf{T} = \mathbf{T}(k) \cong Y(\mathbf{T}) \otimes k^\times = Y(\mathbf{T}) \otimes (\mathbb{Q}/\mathbb{Z})_{p'}$$

since $\mathbf{T}^F$ is the fixed points of F acting on $\mathbf{T}$, thus $\mathbf{T}^F = \mathrm{Ker}(F-1)$ in $Y(\mathbf{T}) \otimes (\mathbb{Q}/\mathbb{Z})_{p'}$, since F is divisible by p, then it is also the kernel of $F-1$ in $Y(\mathbf{T}) \otimes (\mathbb{Q}/\mathbb{Z})$, and the exact sequence follows

$$0 \longrightarrow \mathbf{T}^F \longrightarrow Y(\mathbf{T}) \otimes (\mathbb{Q}/\mathbb{Z}) \xrightarrow{F-1} Y(\mathbf{T}) \otimes (\mathbb{Q}/\mathbb{Z}) \longrightarrow 0$$

By using the snake lemma on the diagram

$$\begin{tikzcd}
0 \arrow[r] & Y(\mathbf{T}) \arrow[r] \arrow[d, "F-1"] & Y(\mathbf{T}) \otimes \mathbb{Q} \arrow[r] \arrow[d, "F-1"] & Y(\mathbf{T}) \otimes \mathbb{Q}/\mathbb{Z} \arrow[r] \arrow[d, "F-1"] & 0 \\
0 \arrow[r] & Y(\mathbf{T}) \arrow[r] & Y(\mathbf{T}) \otimes \mathbb{Q} \arrow[r] & Y(\mathbf{T}) \otimes \mathbb{Q}/\mathbb{Z} \arrow[r] & 0
\end{tikzcd}$$

We obtain a map $\mathbf{T}^F = \mathrm{Ker}(F-1)$ to $\mathrm{Coker}(F-1) = Y(\mathbf{T})/(F-1)Y(\mathbf{T})$. And the exact sequence is as follows,

$$0 \longrightarrow Y(\mathbf{T}) \xrightarrow{F-1} Y(\mathbf{T}) \longrightarrow \mathbf{T}^F \longrightarrow 0$$

map the exact sequence into $\mathbb{Q}/\mathbb{Z}$ will produce another exact sequence

$$0 \longrightarrow X(\mathbf{T}) \xrightarrow{F-1} X(\mathbf{T}) \longrightarrow (\mathbf{T}^F)^\vee \longrightarrow 0$$

Going from $\mathbb{F}_q$ to $\mathbb{F}_{q^n}$ and replace F by F^n, consider $N = \frac{F^n-1}{F-1} = \sum_{i=0}^{n-1} F^i : \mathbf{T}^{F^n} \to \mathbf{T}^F$, the commutative diagram is as

$$\begin{tikzcd}
0 \arrow[r] & Y \arrow[r, "F^n-1"] \arrow[d, "N"] & Y \arrow[r] \arrow[d, equal] & \mathbf{T}^{F^n} \arrow[r] \arrow[d, "N"] & Y \otimes \mathbb{Q}/\mathbb{Z} \arrow[r] \arrow[d, "N"] & Y \otimes \mathbb{Q}/\mathbb{Z} \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & Y \arrow[r, "F-1"] & Y \arrow[r] & \mathbf{T}^F \arrow[r] & Y \otimes \mathbb{Q}/\mathbb{Z} \arrow[r] & Y \otimes \mathbb{Q}/\mathbb{Z} \arrow[r] & 0
\end{tikzcd}$$

$$\begin{tikzcd}
0 \arrow[r] & X \arrow[r, "F-1"] \arrow[d, equal] & X \arrow[r] \arrow[d] & (\mathbf{T}^F)^\vee \arrow[r] \arrow[d, "N^\vee"] & X \otimes \mathbb{Q}/\mathbb{Z} \arrow[r] \arrow[d, equal] & X \otimes \mathbb{Q}/\mathbb{Z} \arrow[r] \arrow[d, "N"] & 0 \\
0 \arrow[r] & X \arrow[r, "F^n-1"] & X \arrow[r] & (\mathbf{T}^{F^n})^\vee \arrow[r] & X \otimes \mathbb{Q}/\mathbb{Z} \arrow[r] & X \otimes \mathbb{Q}/\mathbb{Z} \arrow[r] & 0
\end{tikzcd}$$

Here we abbreviate $X(\mathbf{T})$, $Y(\mathbf{T})$ as X, Y, and $N^\vee : (\mathbf{T}^F)^\vee \to (\mathbf{T}^{F^n})^\vee$ is the induced map of N. For a character θ of $\mathbf{T}^F$, it is embedded in $\mathrm{Hom}(\mathbf{T}^F, \bar{\mathbb{Q}}_l)$, by the commutative diagram, we have θ and $\theta \circ N$ has the same image in $X \otimes \mathbb{Q}/\mathbb{Z}$.

Definition 3.3.2. For two $F-$stable maximal tori $\mathbf{T}$ and $\mathbf{T}'$, with $\theta \in \mathrm{Irr}(\mathbf{T}^F)$, $\theta' \in \mathrm{Irr}(\mathbf{T}^{F'})$, regard them as characters of $Y(\mathbf{T})$ and $Y(\mathbf{T}')$. Now we call these two pairs $(\mathbf{T}, \theta)$, $(\mathbf{T}', \theta')$ are **geometrically conjugate** if there exist $g \in \mathbf{G}$ such that $\mathbf{T}' = {}^g\mathbf{T}$ and θ, θ' are conjugate by $\mathrm{ad}\, g$ as characters of $Y(\mathbf{T})$ and $Y(\mathbf{T}')$.

Remark. (i) The above diagrams show the conditions in definition of geometrically conjugation is invariant under the replacement of F by F^n. Thus to say $(\mathbf{T}, \theta)$ and $(\mathbf{T}', \theta')$ are geometrically conjugate is equivalent to say $\theta \circ N$ and $\theta' \circ N$ are $\mathbf{G}^{F^n}$ conjugate for some n such that $g \in \mathbf{G}^{F^n}$, $\mathbf{T}' = {}^g\mathbf{T}$.

(ii) Since all maximal $F-$stable tori are $\mathbf{G}$ conjugate, thus fix a maximal torus $\mathbf{T}$, $\mathbf{T}' = {}^g\mathbf{T}$ implies $g \in \mathbf{W}(\mathbf{T})$, and since we have the identification $Y(\mathbf{T}) = X(\mathbf{T}) \otimes k^\times \cong X(\mathbf{T}) \otimes (\mathbb{Q}/\mathbb{Z})_{p'}$, then geometric conjugacy classes are one to one correspond to $\mathbf{W}(\mathbf{T})$ orbit in $X(\mathbf{T}) \otimes (\mathbb{Q}/\mathbb{Z})_{p'}$.

Proposition 3.3.3. *Consider dual groups $(\boldsymbol{G}, F)$ and $(\boldsymbol{G}^*, F^*)$ with corresponding dual torus $\boldsymbol{T}$ and $\boldsymbol{T}^*$. Then geometric conjugacy classes in $\boldsymbol{G}$ are one to one correspond to F^*-stable conjugacy classes of semisimple elements in $\boldsymbol{G}^*$.*

Proof. Use the Remark (ii) of Definition 3.3.2, the isomorphism $X(\mathbf{T}) \cong Y(\mathbf{T}^*)$ and the isomorphism $\mathbf{T}^* \cong Y(\mathbf{T}^*) \otimes k^\times$ stated before, we have the one to one correspondence between geometric conjugacy classes in $\mathbf{G}$ and F^*-stable $\mathbf{W}(\mathbf{T}^*)$ orbits in $\mathbf{T}^*$. Notice also every semisimple element in $\mathbf{G}^*$ is in some maximal torus $\mathbf{T}'^*$. Two elements in $\mathbf{T}^*$ are conjugate if and only if they are in the same $\mathbf{W}(\mathbf{T}^*)$ orbit, hence we get the conclusion. □

It is known in 3.2.8 that $R^{\mathbf{G}}_{\mathbf{T},\theta}$ are parametrized by $\mathbf{G}^F$ conjugacy classes of $(\mathbf{T}, \theta)$, we have (independent of Proposition 3.3.3)

Proposition 3.3.4. *$\boldsymbol{G}^F$ conjugacy classes of $(\boldsymbol{T}, \theta)$ are in one to one correspondence to the $\boldsymbol{G}^{*F^*}$ conjugacy classes of $(\boldsymbol{T}^*, s)$, where s is a semisimple element contained in maximal torus $\boldsymbol{T}^*$ of $\boldsymbol{G}^*$*

Definition 3.3.5. For any geometric conjugacy class (s) of semisimple element s in $\mathbf{G}^*$, we define the **Lusztig series** $\mathcal{E}(\mathbf{G}^F, (s))$ to be the subset of $\mathrm{Irr}(\mathbf{G}^F)$ that has nonzero product with $R^{\mathbf{G}}_{\mathbf{T},\theta}$, where the corresponding $(\mathbf{T}, \theta)$ is in geometric conjugacy class associated to (s) by Proposition 3.3.3.
Call elements of $\mathcal{E}(\mathbf{G}^F, (1))$ the **unipotent characters**.

With Proposition 3.3.4, we can write $R^{\mathbf{G}}_{\mathbf{T},\theta}$ as $R^{\mathbf{G}}_{\mathbf{T}^*,s}$, then $\mathcal{E}(\mathbf{G}^F, (s))$ can be regarded as irreducible constituents of $R^{\mathbf{G}}_{\mathbf{T}^*,s'}$, where s' and s are in the same geometric conjugacy class. And Lusztig series gives a partition of irreducible characters.

Proposition 3.3.6. $\mathrm{Irr}(\boldsymbol{G}^F) = \coprod_{(s)} \mathcal{E}(\boldsymbol{G}^F, (s))$. *It gives an answer to the question at the beginning of this section.*

Proof. Since every irreducible character has nonzero product with the regular character, it should be constituent of some Deligne-Lusztig characters thus it lies in some Lusztig series. Moreover, two Deligne-Lusztig characters corresponding to different series have no common constituent by above. Hence we show the partition. □

Lusztig gave the following correspondence

Theorem 3.3.7. *(Lusztig) For s semisimple element in $\boldsymbol{G}^{*F^*}$, there is a bijection between $\mathcal{E}(\boldsymbol{G}^F, (s))$ and $\mathcal{E}(C_{\boldsymbol{G}^*}(s)^{F^*}, (1))$. The bijection takes $\varepsilon_{\boldsymbol{G}} R^{\boldsymbol{G}}_{\boldsymbol{T}^*,s}$ to $\varepsilon_{C^0_{\boldsymbol{G}^*}(s)} R^{C_{\boldsymbol{G}^*}(s)}_{\boldsymbol{T}^*,1_{\boldsymbol{T}^*}}$. Here $\boldsymbol{G}$ is assumed to be connected, connectedness of $C_{\boldsymbol{G}^*}(s)$ is not required.*

Chapter 4

Finite Weil Representation and Finite Theta Correspondences

4.1 The Weil Representation over Finite Fields

4.1.1 The Weil Representations of General Linear Groups

Let k be a finite field with q elements, V the finite dimensional vector space over k, and V^* its dual space associated with a symplectic form $\langle y, x\rangle$ where $x \in V, y \in V^*$ satisfying $\langle y, x\rangle + \langle x, y\rangle = 0$.

Definition 4.1.1. Define the **Heisenberg group** associated with V by $H(V) := V \times V^* \times k$ with the group multiplication written into the matrix form as $\begin{pmatrix} 1 & y & z \\ & 1 & x \\ & & 1 \end{pmatrix}$, where $y \in V^*, x \in V, z \in k$, and the decomposition of $H(V)$ is given by the sequence

$$0 \longrightarrow k \xrightarrow{e} H(V) \xrightarrow{e'} V \times V^* \longrightarrow 0, \text{where}\, e : z \mapsto \begin{pmatrix} 1 & 0 & z \\ & 1 & 0 \\ & & 1 \end{pmatrix} \text{and}\, e' : \begin{pmatrix} 1 & y & 0 \\ & 1 & x \\ & & 1 \end{pmatrix} \mapsto (x, y).$$

One can easily verify that the Heisenberg group $H(V)$ is a two-step nilpotent group with center $Z = e(k)$, and the reverse of $V \times k$(resp. $V^* \times k$) in $H(V)$ is a maximal commutative subgroup of $H(V)$.

Proposition 4.1.2. *For any nontrivial character $\psi : k^+ \longrightarrow \mathbb{C}$, associate it with a nontrivial character ζ of Z by $\psi(z) = \zeta(e(z))$, then for each ζ, one has a unique(under class equivalence) irreducible representation η_ζ of $H(V)$.*

Proof. First notice that the non-degeneracy of the symplectic form on dual spaces implies that for any pair $(x, y) \in V \times V^*$, one could find another pair (x', y') such that $\psi(\langle y, x'\rangle + \langle x, y'\rangle) \neq 1$. Translating it into $H(V)$ means for any $h \in H(V)$, with resp. to (x, y), there is $h' \in H(V)$, with resp. to (x', y'), such that $\zeta((h', h)) \neq 1$, where (h', h) the commutator of two group elements, must in the center Z.
Then we consider the irreducible components of the induce representation of $H(V)$ by ζ, the Frobenius reciprocity, they gives all classes of irreducible representations of H that are given by ζ. We choose η some irreducible representation of H given by ζ and $h \in H, h \notin Z$, find $h' \in H$ such that $\zeta((h', h)) \neq 1$, then $\operatorname{Tr}\eta(h) = \operatorname{Tr}\eta(h'hh'^{-1}h^{-1}h) = \operatorname{Tr}\eta((h', h)h) = \zeta((h', h))\operatorname{Tr}\eta(h)$. By the assumption, $\operatorname{Tr}\eta(h) = 0$, thus the support of character of η is Z, and $\operatorname{Tr}\eta(z) = \zeta(z)\deg(\eta)$. By the irreduciblity of η, we have the character satisfying $\sum_{h\in H} |\operatorname{Tr}\eta(h)|^2 = |H|$, and $\sum_{z\in Z} |\zeta(z)|^2 = |Z|$, thus $\deg(\eta) = [H : Z]^{\frac{1}{2}}$. So η is irreducible if and only if its degree is the square root of $[H : Z]$, further, it shows that the class of irreducible representation given by ζ is uniquely determined, denote it by η_ζ. $\square$

Next we give a realization of irreducible representation class η_ζ of $H(V)$, associated with ζ(or, equivalently, ψ). It is given in the space of complex functions on V, for any $F \in \mathbb{C}[V]$, the action is given by

$$\eta((x_0)F)(x) = F(x+x_0), \quad x_0 \in V, (x_0) \text{ is the pull back of } x_0, (x_0) = \begin{pmatrix} 1 & 0 & 0 \\ & 1 & x_0 \\ & & 1 \end{pmatrix},$$

$$\eta((y_0)F)(x) = \psi(\langle x, y_0\rangle)F(x), \quad x_0 \in V, (y_0) \text{ is the pull back of } y_0.$$

The realization space clearly could also be complex functions on V^*, the intertwining of these two realization space is given by the Fourier transform. Let V and V^* admit the Haar measures which take the value $|V|^{\frac{1}{2}}$, the square root of the order of V, then the Fourier transform is given as

$$F^*(y) = \int_V F(x)\psi(\langle x, y\rangle)dx; \qquad F'(x) = \int_{V^*} F'^*\psi(\langle y, x\rangle)dy, \text{where } F \in \mathbb{C}[V], F'^* \in \mathbb{C}[V^*].$$

We study the representation of $G(V)$, a subgroup of $GL(V \times V^*)$ that makes the form $\langle y, x'\rangle$ invariant for $(x, y), (x', y')$ $\in V \times V^*$. It is actually the group of pair (g, g^*), where $g \in GL(V)$, and g^* its contragredient makes $\langle g^*y, gx\rangle = \langle y, x\rangle$. Thus the $G(V)$ action on $H(V)$ could also be expressed as conjugate multiplication by $\begin{pmatrix} 1 & & \\ & g & \\ & & 1 \end{pmatrix}$. $G(V)$ thus is the subgroup of automorphisms of $H(V)$, let $GH(V)$ be their semidirect product, by the **Weil representation**, we mean a unique class of representation ω_V of $G(V)$ with the following properties.

(i) For any irreducible representation η of $H(V)$ which is nontrivial on center Z, there is a representation ρ of $GH(V)$ in the same realization space of η such that the restriction of ρ on $H(V)$ is η, and to $G(V)$ is in the class ω_V.

(ii) The character of ω_V has positive values. We now proof the existence and uniqueness of Weil representation.

Proof. For the character ζ of Z, the representation $1 \cdot \zeta$ of $V^* \times Z$ could naturally extend to the representation $1 \cdot 1 \cdot \zeta$ of $G(V) \times V^* \times Z$ by unit representation of $G(V)$, then its induced representation ρ_ζ onto $GH(V)$ satisfies the condition (i), where the restriction on $H(V)$ is in η_ζ. In the same realization space $\mathbb{C}[V]$, the action of $g \in G(V)$ is given by $g.F(x) : F(x) \mapsto F(g^{-1}x), F \in \mathbb{C}[V]$, and it has positive character. By the irreduciblity of η, we know two representations of $GH(V)$ satisfying (i) will differ by a one-dimensional representation of $G(V)$, together with the positive value of the induced representation, the uniqueness of the representation class is also shown. $\square$

Moreover, in the realization space $\mathbb{C}[V]$ of $GH(V)$, we know the bilinear form of two $F, G \in \mathbb{C}[V]$ is $\int_V F(x)G(x)dx$, this form is invariant under the $G(V)$ action. For $g \in G(V)$, the character of ω_V is $\chi_{\omega_V}(g) = q^{\frac{1}{2}\dim \mathrm{Ker}(g-1)}$, this is followed by the character of natural representation of $GL(V)$ on $\mathbb{C}[V]$.

For the direct sum decomposition of the vector space $V = \oplus_{i=1}^n V_i$ over k, the restriction $\omega_V|_{\prod_{i=1}^n G(V_i)} = \omega_{V_1} \otimes ... \otimes \omega_{V_n}$.

4.1.2 The Weil Representations of symplectic Groups

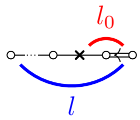
Let W be a finite dimensional (of dimension $2l$) symplectic vector space over k of q elements, q odd, let j be a non-degenerate bilinear skew-symmetric form on W, $j(w_1, w_2) = \langle w_1, w_2\rangle$, and $i = \frac{j}{2}$.

The **Heisenberg group** $H(W; j)$ is defined as $W \times k$ with multiplication $(w_1, z_1) \cdot (w_2, z_2) = (w_1 + w_2, z_1 + z_2 + \frac{\langle w_1, w_2\rangle}{2})$. Expression in matrices form in given by $\begin{pmatrix} 1 & i(w) & z \\ & 1 & w \\ & & 1 \end{pmatrix}$, and the sequence is $0 \longrightarrow k \xrightarrow{e} H(W; j) \xrightarrow{e'} W \longrightarrow 0$, where $e : z \mapsto (0, z), e' : (w, z) \mapsto w$. Similar to the $H(V)$ case, we have $H(W; j)$ is a two-step nilpotent group with center $Z = e(k)$. And maximal commutative subgroups of $H(W)$ is the pullback of $W_\pm \times k$, where $W_\pm$ are the maximal isotropic spaces of W (isotropic means $W \subseteq W^\perp$, $W^\perp$ is the orthogonal space of W according to j).

Let $Sp(W; j)$ be the symplectic group associated with space W, where $g \in Sp(W)$ acting on $H(W; j)$ as $(w, z) \mapsto$

(gw, z). Following the notion in Gerardin's paper, we use $SH(W;j)$ to denote the semidirect product of $Sp(W;j)$ and $H(W;j)$(where Sp is the group of automorphisms on H).

Proposition 4.1.3. *Let $H(W')$ denote the pullback of subspace $W' \subseteq W$, and $P(W')$(resp. $P^0(W')$) the subgroup of $Sp(W)$ that leaves W' (resp. pointwise) invariant. For W_+ the isotropic subspace of $W, W_0 = W_+^{\perp}/W_+$ and W_- the isotropic complement of $W_+^{\perp}$, we have*

(i) $P(W_+)$ is a maximal parabolic subgroup of $Sp(W)$, it has the form $\begin{pmatrix} GL(W_+) & * & * \\ & Sp(W_0) & * \\ & & GL(W_-) \end{pmatrix}$, and it has the Dynkin diagram of type $C_l(l = \frac{1}{2}\dim W, l_0 = \frac{1}{2}\dim W_0)$, which is 

.

(ii) $P(W_+)$ is the semidirect product from the decomposition $1 \longrightarrow P^0(W_+) \longrightarrow P(W_+) \longrightarrow GL(W_+) \longrightarrow 1$.
(iii) There is a unique nontrivial one-dimensional representation on $P(W_+)$ expressed as $\chi_{W_+}(g) = (\det_{W_+} g)^{\frac{q-1}{2}}$.

Proof. [Gér77], P63. ☐

Now take the Witt decomposition of W into $W = W_+ \oplus W_-$, where $W_\pm$ are lagrangians, for $G = P(W_+) \cap P(W_-)$, by Weil representation ω_W^G we mean the Weil representation of $G(W_+)$ we get in the last section, notice that W_- is the dual of W_+ and satisfies $\langle x, y\rangle + \langle y, x\rangle = 0, x \in W_+, y \in W_-$. Further, ω_W is also the representation of $G(W_-)$ by Fourier transform. Next, we want to get the Weil representation of $Sp(W)$, which is fundamental for learning the Howe correspondence over finite fields. One main theorem in Gerardin's paper gives the explicit structure and calculations of this case.

Theorem 4.1.4. *Take the nontrivial character ζ of Z, the center of $H(W)$, we get the class of irreducible representations η_ζ^W of $H(W)$ given by ζ, then there is a unique class of representations $\omega_{\zeta,W}$ of $Sp(W)$, called **Weil representation** of symplectic group, such that(Assume $l \neq 1, \text{and odd}\, q \neq 3$).*
(i) In the realization space of η_ζ^W, there is a representation ρ of $SH(W)$ such that its restriction on $Sp(W)$ is in $\omega_{\zeta,W}$.
(ii) For any nontrivial totally isotropic subspace $W_+ \subseteq W$, then W has decomposition $W = W_+ \oplus W_0 \oplus W_-$, let η_0 be the representation of $SH(W_0, j_0$ as in (i), where the form j_0 of W_0 is induced by j of W. Let π_+ be the representation of $P(W_+)H(W)$ given by η_0 with two surjective maps $P(W_+) \twoheadrightarrow Sp(W_0, j_0)$ and $H(W_+^{\perp}) \twoheadrightarrow H(W_0)$.
Then in the realization space of π_+, there is a unique representation of $SH(W)$ that is in $\omega_{\zeta,W}$.
(iii) For the Witt decomposition $W = W_+ \oplus W_-$, the restriction of $\omega_{\zeta,W}$ to $G = P(W_+) \cap P(W_-)$ is $\chi_G \otimes \omega_W^G$, where χ_G is a nontrivial one dimensional real representation of G(and is unique), ω_W^G is the Weil representation of G introduced above.

Proof. **Step 1** First consider the maximal isotropic case $W = W_+ \oplus W_-$, as in the above proposition, there is a unique nontrivial character χ_{W_+} of $P(W_+)$, then π_+ of $P(W_+)H(W)$ is induced by $\chi_{W_+} \cdot 1 \cdot \zeta$, the representation of the semidirect product of $P(W_+)$ and the pullback of W_+ and Z. From Proposition 4.1.2 we know that π_+ on H is some η in η_ζ, and π_+ on $P(W_+)$ is π which extends η. From Ger lma 1.5, η has an extension ρ to $SH(W)$ Moreover, the realization space of the above π_+ is $\mathbb{C}[W_-]$, let $\psi(z) := \eta(e(z))$ be the character of k. The action of $H(W)$ is given by

$$\eta((x_0)(y_0)f_-)(y) = \psi(j(y, x_0))f_-(y + y_0)$$

The action of $P(W_+)$ is given by

$$(\pi(g)f_-)(y) = \chi_{W_+}(g)f_-(g^{-1}y) \quad \text{if s stabilizes } W_+ \text{and } W_-$$
$$(\pi(g)f_-)(y) = \psi\left(\frac{j(by, y)}{2}\right)f_-(y) \quad \text{if } g = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$$

For $f \in \mathbb{C}[W_-]$ and with the Haar measure $\int_{W_-} dy = q^{\frac{\dim W}{2}}$. The intertwining operator $\iota(g)$ between isotropic spaces $W_\pm$ is acting on $\mathbb{C}[W_-]$ by the Fourier transform.

$$(\iota(g)f)(y) = \mathcal{F}f_-\left(g^{-1}y\right), \quad \mathcal{F}f_- = \int_{W_-} f(y)\psi(j(y,x))dy$$

since $Sp(W)$ is generated by $P(W_+)$ (which is maximal) and symplectic transform $s = \begin{pmatrix} 0 & \gamma' \\ \gamma & 0 \end{pmatrix}$ which changes W_+ and W_-. Consider representation ρ of $SH(W)$ that extends η, its restriction σ on $Sp(W)$ is given by

$$\sigma(g) = \varphi(b)\chi(a)\pi(g), \qquad g = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}\begin{pmatrix} a & 0 \\ 0 & a' \end{pmatrix} \in P(W_+), a \in GL(W_+), a' \in GL(W_-)$$
$$\sigma(g) = \phi\left(\gamma^{-1}\right)\iota(g), \qquad g = \begin{pmatrix} 0 & \gamma' \\ \gamma & 0 \end{pmatrix}$$

Here χ is a character of $GL(W_+)$, φ is the character of set of i -symmetric isomorphism $b : W_- \longrightarrow W_+$, and ϕ is the function on the set of isomorphisms W_- to W_+. We determine them in the next.

Step2 φ is the character such that $\varphi\left(a_+ b a_-^{-1}\right) = \varphi(b)$, assume $l \neq 1$ and $q \neq 3$, then φ is constant thus should be 1 To determine ϕ, we use the relation from Weil:

$$g_b = \begin{pmatrix} 0 & b \\ -b^{-1} & 0 \end{pmatrix}\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & -b \\ b^{-1} & 0 \end{pmatrix}\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -b \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 & b \\ -b^{-1} & 0 \end{pmatrix}, \quad b : W_- \to W_+, b^{-1} : W_+ \to W_-$$

then σ acting with the left hand side is

$$\begin{aligned}(\sigma(g_b) f)(y) &= \phi(-b)\phi(b)\int_{W_-\times W_-} \psi\left(\frac{j(by',y')}{2}\right)\psi\left(\frac{j(by'',y'')}{2}\right)\psi\left(j(y',-by)\right)\psi\left(j(y'',by')\right) f(y'')\, dy'dy'' \\ &= \phi(-b)\phi(b)c_{W,\psi}(b)\psi\left(\frac{j(by,y)}{2}\right)^{-1}\int_{W_-} f(y'')\psi\left(j(y'',-by)\right)dy'' \text{ from Weil, and } c_{W,\psi} \text{ is a complex number}\end{aligned}$$

Comparing with the σ action with right hand side, which is $\phi(-b)\psi\left(\frac{j(by,y)}{2}\right)^{-1}\mathcal{F}f_-(-by)$, we get $\phi(b)c_{W,\psi}(b) = 1$. Here $c_{W,\psi}(b) = c(\psi)\Delta(b)^{\frac{q-1}{2}}$, where $c(\psi) = \int_k \psi\left(\frac{t^2}{2}\right)dt, c(\psi)^2 = (-1)^{\frac{1-q}{2}q}$ is the Gauss sum, and Δ is the discriminant.

Notice that $\begin{pmatrix} 0 & \gamma' \\ \gamma & 0 \end{pmatrix}\begin{pmatrix} a & 0 \\ 0 & a' \end{pmatrix} = \begin{pmatrix} 0 & \gamma' a' \\ \gamma a & 0 \end{pmatrix}$, then

$$\phi\left(\gamma^{-1}\right)\chi(a)\left(\det_{W_+} a\right)^{\frac{q-1}{2}} = \phi\left(a^{-1}\gamma^{-1}\right)$$

Without lose of generality, take a to be a split semisimple one, then there is $b : W_- \to W_+$ such that $\gamma^{-1} = ab$ together with the above formulas, we have $\chi(a)\left(\det_{W_+} a\right)^{\frac{q-1}{2}} = \left(\det_{W_+} a\right)^{\frac{q-1}{2}}$, it means $\chi = 1$, and $\phi\left(\gamma^{-1}\right) = c(\psi)^{-1}\left(\det_{W_+}\gamma\right)^{\frac{1-q}{2}}$. Thus the representation σ on $Sp(W)$ is determined:

$$(\sigma(g)f_-)(y) = (\pi(g)f_-)(y), \quad g = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}\begin{pmatrix} a & 0 \\ 0 & a' \end{pmatrix}$$
$$(\sigma(g)f_-)(y) = c(\psi)^{-1}\left(\det_{W_+}\gamma\right)^{\frac{1-q}{2}}\mathcal{F}f_-\left(\gamma^{-1}y\right), \quad g = \begin{pmatrix} 0 & \gamma' \\ \gamma & 0 \end{pmatrix}$$

Hence, for the maximal isotropic case, the unique extension of any representation π of $P(W_+)H(W)$ associated with character ψ is determined, which is $\rho = \sigma \cdot \eta$ of $Sp(W) \times H(W)$, and $\sigma \in \omega_{\zeta,W}$, the class of Weil representations, and the property (iii) is straightforward by considering σ on contragredient $a_{\pm}$.

Step 3 For the general case of Witt decomposition $W = W_+ \oplus W_0 \oplus W_-$, where $\dim W_0 = 2l_0$, and symplectic form j on W_0 is denoted by j_0. Consider $(W_0, j_0) = W_{0+} \oplus W_{0-}$, then the realization space of π on $P(W_+)$ will be $\mathbb{C}[W_+] \otimes V[\eta_0]$. Denote by σ_0 the Weil representation of $Sp(W_0)$, which is obtained as in **Step 2**, $\sigma_0 \cdot \eta_0$ the representation of $SH(W_0)$, and $V[\eta_0]$ the space of $\eta_0 \in \eta_{\psi_0, W_0}$ then π acting as

$$(\pi(g)f_-)(y) = \det_{W_+}(a_+) f_-\left(a_-^{-1}y\right), \quad g = \begin{pmatrix} a_+ & & \\ & 1 & \\ & & a_- \end{pmatrix}, a_{\pm} \in GL(W_{\pm})$$

$$(\pi(g)f_-)(y) = \sigma_0(s_0) f_-(y), \quad g = \begin{pmatrix} 1 & & \\ & s_0 & \\ & & 1 \end{pmatrix}, s_0 \in Sp(W_0)$$

$$(\pi(g)f_-)(y) = \eta_0 \begin{pmatrix} 1 & -i(b''y) & i(by,y) + i(b''y, b''y) \\ & 1 & -b''y \\ & & 1 \end{pmatrix} f_-(y), \quad g = \begin{pmatrix} 1 & b' & b \\ & 1 & b'' \\ & & 1 \end{pmatrix} \in \text{ unipotent radical of } P(W)$$

The unique extension of π to $Sp(W)$ will be obtained by considering π acting on the elements exchanges W_+ and W_- while leaving W_0 fixed. Let $i_{\pm}$ be the restriction of i to $W_+ \oplus W_-$, then those elements are expressed as $g = \begin{pmatrix} & & -\gamma \\ & 1 & \\ \gamma^{-1} & & \end{pmatrix}$, where $\gamma : W_- \to W_+$ is $i_{\pm}-$symmetric isomorphism. π action with g is the Fourier transform:

$$(\pi(g)f_-)(y) = c_{W+\oplus W_-,\psi}(\gamma)\mathcal{F}f_-(\gamma y), \quad \mathcal{F}f_-(x) = \int_{W_-} f_-(y)\psi(j(y,x))dy$$

Then π differs from the restriction of σ to $P(W_+)$ is a one-dimensional representation of $Sp(W_0)$, which is trivial $\square$

4.2 Uniform projection of Weil Representations

Take (D, ι) to be the finite field with involution ι, it is either **(i)** $(D,\iota) = (F, \mathrm{id})$ or **(ii)** $(D,\iota) = (E, \sigma)$, where σ is the nontrivial Galois involution.

Fix $\epsilon = \pm 1$, for $D-$ vector space V and V', with $\epsilon-$ Hermitian forms $\langle,\rangle_V$ and $\langle,\rangle_{V'}$. Let $G = U(V)$ and $G' = U(V')$ be the isometry groups (type I dual pair) under the nondegenerate Hermitian forms. We can define a nondegenrate $F-$ bilinear alternating form on $\mathbb{W} = V \otimes_D V'$ to be $\langle\langle,\rangle\rangle_{\mathbb{W}} : (x_1 \otimes y_1, x_2 \otimes y_2) \mapsto \mathrm{tr}_{D/F}(\langle x_1, x_2\rangle_V \langle y_1, y_2\rangle_{V'})$.

There is a natural map $G \times G' \longrightarrow GG' \longrightarrow \mathrm{Sp}(\mathbb{W})$. Take $\mathbb{T}$ and $\mathbb{T}'$ to be Witt towers of V and V'. And let $(G_k, G_l^{'})$ denote the corresponding dual pair, $\omega_{k,l}$ denote the restriction of the corresponding Weil representation to $G_k \times G_l^{'}$.

Definition 4.2.1. Suppose $G = G_m, G' = G_n^{'}$. For each $(k.l)$, we have $\omega_{k,l} = \oplus_{\pi,\pi'} m_{\pi,\pi'} \pi \otimes \pi'$, where $\pi \in \mathrm{Irr}(G_k)$, and $\pi' \in \mathrm{Irr}(G_l^{'})$ runs over the irreducible representations of the dual groups. We say π is in **theta correspondence** to $\pi^{'}$ if $\pi \otimes \pi^{'}$ occurs in the decomposition of $\omega_{k,l}$.

We introduce Srinivasan's work on formulating the uniform part of restriction of Weil representations on dual reductive paris(symplectic-even orthogonal dual pair).

Without causing ambiguity, for now we take Sp_{2n} (resp. $\mathrm{O}_n, \mathrm{GL}_n, \mathrm{U}_n$, etc) to denote the rational points of **Sp**(fixed by standard Frobenius map $F = \mathbb{F}_q$, and other nomal fonts are used to denote finiteness. We still use the **bold fonts** to denote the surronding algebraic reductive groups when there is a need.

Theorem 4.2.2. *(Srinivasan, 1979 [Sri79]) Assume that q is sufficiently large such that for torus $\boldsymbol{T}$ of $\mathbf{G}$ and $\mathbf{G}'$, the characters have at least two regular orbits under $\mathbf{W}(\boldsymbol{T})-$action. Let $\omega_{n,m}$ denote the restriction of Weil representation ω on $G \times G' = \mathrm{Sp}_{2n} \times SO^{\epsilon}_{2m}$, then we have for symplectic-even orthogonal pair*
Case(i) $n < m$

$$\omega^{\sharp}_{n,m} = \sum_{k=0}^{n} \sum_{(\boldsymbol{T})\subset \boldsymbol{Sp}_{2k}} \frac{1}{|W(T)|} \sum_{\theta\in\widehat{T}} \epsilon\epsilon' R^{\mathrm{Sp}_{2n}}_{\boldsymbol{T}\times \mathrm{Sp}_{2(n-k)},\theta\otimes 1} \otimes R^{\mathrm{SO}^{\epsilon}_{2m}}_{\boldsymbol{T}\times \mathrm{SO}^{\epsilon'}_{2(m-k)},\theta\otimes 1}$$

Case(ii) $n \geq m$

$$\omega^{\sharp}_{n,m} = \sum_{k=0}^{m-1} \sum_{(\boldsymbol{T})\subset \boldsymbol{Sp}_{2k}} \frac{1}{|W(T)|} \sum_{\theta\in\widehat{T}} \epsilon\epsilon' R^{\mathrm{Sp}_{2n}}_{\boldsymbol{T}\times \mathrm{Sp}_{2(n-k)},\theta\otimes 1} \otimes R^{\mathrm{SO}^{\epsilon}_{2m}}_{\boldsymbol{T}\times \mathrm{SO}^{\epsilon'}_{2(m-k)},\theta\otimes 1}$$
$$+ \sum_{(\boldsymbol{T})\subset \boldsymbol{SO}^{\epsilon}_{2m}} \frac{2}{|W(T)|} \epsilon(-1)^{n+m} \sum_{\theta\in\widehat{T}} R^{\mathrm{Sp}_{2n}}_{\boldsymbol{T}\times \mathrm{Sp}_{2(n-m)},\theta\otimes 1} \otimes R^{\mathrm{SO}^{\epsilon}_{2m}}_{\boldsymbol{T},\theta}$$

Remark. (i) Notice that in $\mathrm{SO}^{\epsilon}_{2m}$ if $T \subset \mathrm{SO}^{\epsilon''}_{2k}$, then the sign of the orthogonal groups satisfy $\epsilon\epsilon' = \epsilon''$.
(ii) In the formula we take the sum over the $\mathbf{G}^F$ conjugacy classes. We sometimes may also take the sum over tori labelled by the corresponding Weyl group for the convience. The equivalent expression should be (take case (i) for example)

$$\omega^{\sharp}_{n,m} = \sum_{k=0}^{n} \frac{1}{|W(\mathrm{Sp}_{2k})|} \frac{1}{\left|W(\mathrm{Sp}_{2(n-k)})\right|} \frac{1}{\left|W(\mathrm{SO}_{2(m-k)})\right|} \sum_{v\in W(\mathrm{Sp}_{2k})} \sum_{\theta\in\widehat{\mathbf{T}_v}} \epsilon\epsilon' \sum_{w\in W(\mathrm{Sp}_{2(n-k)})} \sum_{w'\in \mathrm{SO}_{2(m-k)}} R^{\mathrm{Sp}_{2n}}_{\mathbf{T}_v\times\mathbf{T}_w,\theta\otimes 1} \otimes R^{\mathrm{SO}^{\epsilon}_{2m}}_{\mathbf{T}_v\times\mathbf{T}_{w'},\theta\otimes 1}$$

the substitution of $\mathbf{L} = \mathrm{Sp}$(resp. $\mathbf{L}' = \mathrm{SO}$) in the formula is due to

$$1_{\mathbf{L}} = \frac{1}{|\mathbf{W}(\mathbf{L})^F|} \sum_{v\in \mathbf{W_L}} R^{\mathbf{L}}_{\mathbf{T}_v,1} \quad \text{and} \quad R^{\mathbf{G}}_{\mathbf{T}\times\mathbf{L}} \circ R^{\mathbf{T}\times\mathbf{L}}_{\mathbf{T}\times\mathbf{T}_v} = R^{\mathbf{G}}_{\mathbf{T}\times\mathbf{T}_v} (\text{ transitivity of functor by 3.1.1})$$

(iii) For the pair $(\mathrm{Sp}_{2n}, \mathrm{O}^{\epsilon}_{2m})$, the uniform projection of Weil representation should be (see [AMR96] Proposition 2.1)

$$\omega^{\sharp}_{n,m} = \sum_{k=0}^{n} \frac{1}{|W(\mathrm{Sp}_{2k})|} \frac{1}{\left|W(\mathrm{Sp}_{2(n-k)})\right|} \frac{1}{\left|W(\mathrm{O}_{2(m-k)})\right|} \sum_{v\in W(\mathrm{Sp}_{2k})} \sum_{\theta\in\widehat{\mathbf{T}_v}} \epsilon\epsilon' \sum_{w\in W(\mathrm{Sp}_{2(n-k)})} \sum_{w'\in \mathrm{O}_{2(m-k)}} R^{\mathrm{Sp}_{2n}}_{\mathbf{T}_v\times\mathbf{T}_w,\theta\otimes 1} \otimes R^{\mathrm{O}^{\epsilon}_{2m}}_{\mathbf{T}_v\times\mathbf{T}_{w'},\theta\otimes 1}$$

(iv) We won't show Srinivasan's proof of the uniform projection of Weil representation, she provides also the cases of linear and the unitary dual pairs, which are actually easier to demonstrate. And the trace theorem in [Gér77] provides the foundation for the calculations in her proof.
(v) It is shown in [Pan19] that the theorem remains true without the condition on sufficient large q.

Based on Srinivasan's work, Adams-Moy stated the correspondence of unipotent representations, the proof is straightforward.

Theorem 4.2.3. *(Adams-Moy, 1993 [AM93]) Under the same assumption as Theorem 4.2.2, and if the representation $\pi \otimes \pi'$ of dual pair $(G, G') = (\mathrm{Sp}_{2n}, \mathrm{SO}^{\epsilon}_{2m})$ occurs in the Weil representation, then π is unipotent if and only if π' is unipotent.*

Proof. The proof use the notion of the geometric conjugation and Lusztig series, which represents the irreducible components of Deligne-Lusztig characters $R^{\mathbf{G}}_{\mathbf{T},\theta}$. For the dual pair in the theorem, we consider the inclusion $G \times G' \to G \cdot G' \to \mathrm{Sp}_{4mn}$. Denote $G \cdot G'$ by H, and the corresponding $R^{\mathbf{G}}_{\mathbf{T},\theta} \otimes R^{\mathbf{G}'}_{\mathbf{T}',\theta'}$ by $R^{H}_{\mathbf{T}\cdot\mathbf{T}',\theta\otimes\theta'}$. It is known that irreducible characters of H facts into disjoint series $\coprod_s \mathcal{E}(H, (s))$, where s running over the conjugacy classes of semisimple elements in the dual group H^*.
Then we consider $s = s_{G^*} s_{G'^*}$, since irreducible constituents of $R := R^{H}_{\mathbf{T}\cdot\mathbf{T}',\theta\otimes\theta'}$ are in the same Lusztig series, thus we consider the corresponding semisimple element as $s = s(R)$. The uniform projection $\mathcal{E}(H, (s))^{\sharp}$ should be spanned

by those geometric conjugacy class of $(\mathbf{T}\cdot\mathbf{T}', \theta\otimes\theta')$. If $\tau = \pi\otimes\pi'$ occurs as component of $\omega_{n,m}$, with either (but not both) π or π' unipotent, then the associating $s(\tau) = s_{G^*}s_{G'^*}$ should has exactly one of s_{G^*}, $s_{G'^*}$ being 1. We take the comparison between $\tau^\sharp$ in $\omega^\sharp_{n,m}$ and expression in Theorem 4.2.2, then a contradiction is obtained. □

In the same spirit as Adams and Moy, for π an irreducible representation of G_n with its character χ located in some $\mathcal{E}(G_n,(s))$, one may think about in which Lusztig series $\Theta_{G'_m}(\chi)$ is located. In [AMR96], injections between dual groups of dual pairs are considered. We specifically choose the dual pairs to be of the type $(\mathrm{Sp}_{2n}, \mathrm{O}^\epsilon_{2m})$. It is shown in the section 3.3, $(\mathbf{Sp}^*_{2n}, \mathbf{O}^{\epsilon^*}_{2m}) = (\mathbf{SO}_{2n+1}, \mathbf{O}^\epsilon_{2m})$. Then define $\mathcal{I}_{n,m} := \begin{cases} G^*_n \to G'^*_m & n<m, \\ G^*_m \to G'^*_n & n\geq m. \end{cases}$, to be more precisely, if we write in matrix, it is

Case(i) $n<m$

$$\mathcal{I}: \mathrm{SO}_{2m+1} \hookrightarrow \mathrm{O}^\epsilon_{2m}, \quad \begin{pmatrix} a & 0 & b \\ 0 & 1 & 0 \\ c & 0 & d \end{pmatrix} \mapsto \begin{pmatrix} a & 0 & b \\ 0 & \mathrm{Id}_{2(m-n)} & 0 \\ c & 0 & d \end{pmatrix}$$

Case(ii) $n\geq m$

$$\mathcal{I}: \mathrm{O}^\epsilon_{2m} \hookrightarrow \mathrm{SO}_{2m+1}, \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \begin{pmatrix} a & 0 & b \\ 0 & \mathrm{Id}_{2(n-m)+1} & 0 \\ c & 0 & d \end{pmatrix}$$

Proposition 4.2.4. *Let χ (resp. χ') be the irreducible character of Sp_{2n} (resp. O^ϵ_{2m}), then if $\chi\otimes\chi'$ occurs in $\omega_{n,m}$, the geometric conjugacy classes in which χ and χ' located are correspond by $\mathcal{I}_{n,m}$.*

Proof. Let (s) denote the conjugacy associated to χ (resp. (s') to χ'), we suppose $n<m$ if $(s')\neq\mathcal{I}(s)$, then we know from Theorem 4.2.2, $\left\langle \omega^\sharp_{n,m}, R^{G_n\cdot G'_m}_{(\mathbf{T}\cdot\mathbf{T}')^*, s\cdot s'} \right\rangle = 0 = \left\langle \omega_{n,m}, R^{G_n\cdot G'_m}_{(\mathbf{T}\cdot\mathbf{T}')^*, s\cdot s'} \right\rangle$, then $\omega_{n,m}$ projection on $\mathbb{C}[\mathcal{E}(G_n,(s))]\otimes\mathbb{C}[\mathcal{E}(G'_m,(s')]$ is zero, a contradiction. □

It is know that each conjugacy class of semisimple elements in G_n uniquely corresponding to a diagonal element $\mathrm{diag}(\lambda_1,...,\lambda_n)$ (consider the eigenvalue), and the injection $\mathcal{I}_{n,m}$, take the case $n<m$ as example, actually corresponding to the immersion $\mathrm{diag}(\lambda_1,...,\lambda_n) \hookrightarrow \mathrm{diag}(\lambda_1,...,\lambda_n,1,...,1)$. We have the partition of ω, following the notation in [AMR96].

Definition 4.2.5. Let $\mathbf{T}_{n,0}$ be the set of elements in diagonal maximal torus without being 1, if $n<m, s_n\in\mathbf{T}_{n,0}$, denote s'_n the image of s_n under $\mathcal{I}_{n,m}$ and vice versa, then according to Proposition 4.2.4, we have partition

$$\omega_{n,m} = \bigoplus_{l=0}^{\min(n,m)} \bigoplus_{s_l\in\mathbf{T}_{l,0}} \omega_{n,m,s_l}$$

where ω_{n,m,s_l} is the projection of $\omega_{n,m}$ onto $\mathbb{C}[\mathcal{E}(G_n,(s_l,1_{n-l}))]\otimes\mathbb{C}[\mathcal{E}(G'_m,(s'_l,1_{m-l})]$. And denote by $\omega_{n,m,1}$ the unipotent part of $\omega_{n,m}$ (it does not contradict to the choice of s_l, since it occurs only when $l=0$), denote by $\omega^\sharp_{n,m,s}$ the projection of uniform $\omega^\sharp_{n,m}$ onto $\mathbb{C}[\mathcal{E}(\mathrm{Sp}_{2n},(s_n)]\otimes(\mathbb{C}[\mathcal{E}(\mathrm{O}^+_{2m},(s_m)]\oplus\mathbb{C}[\mathcal{E}(\mathrm{O}^-_{2m}(s_m)])$ for the symplectic-orthogonal case.

Let the diagonal maximal torus $\mathbf{T}_n$ be the fixed maxima torus in G_n, for $w\in W_n$, there is an isomorphism $\mathbf{T}^{wF}_n \cong \mathbf{T}^F_w$. Write the pair $(wF,\tilde{s})$ for the geometric conjugacy class corresponding to Lusztig series $\mathcal{E}(G_n,(s))$.

Proposition 4.2.6. *Suppose $s\in\boldsymbol{T}_{l,0}$, then we have*

$$\omega^\sharp_{n,m,s} = \sum_{k=l}^{\min(n,m)} \frac{1}{|W(s_k)|}\frac{1}{|W(G_{n-k})|}\frac{1}{|W(G'_{m-k})|}\sum_{v\in Z(s_k)}\epsilon\epsilon'\sum_{w\in W(G_{n-k})}\sum_{w'\in W(G'_{m-k})} R^{G_n}_{\boldsymbol{T}^{vF}\times\boldsymbol{T}^{wF},\widetilde{s_k}\otimes 1}\otimes R^{G'^\epsilon_m}_{\boldsymbol{T}^{vF}\times\boldsymbol{T}^{w'F},\widetilde{s_k}\otimes 1}$$

Where $W(s_k)\subseteq W(G_k)$ is the subset that fixes s_k, and $Z(s_k)\subseteq W(G_k)$ is of those w that wF fix s_k.

Proof. The proof is just a rewrite of Remark (ii) of 4.2.2, notice the Lusztig series give the correspondence between $\theta \in \widehat{\mathbf{T}_v}$ and $s \in (\mathbf{T}_v^*)^{F*}$, thus the equality between $\sum_{v\in W(G_k)}\sum_{\theta\in\widehat{\mathbf{T}_v}}$ and $\sum_{v\in W(G_k)}\sum_{s\in(\mathbf{T}_v^*)^{F*}}$ holds, the later one in our setting is just $\sum_{s\in\mathbf{T}_k}\sum_{w\in Z(s)}$, we then get $\omega_{n,m}^{\sharp}$ is

$$\sum_{k=0}^{\min(n,m)}\frac{1}{|W(G_k)|}\frac{1}{|W(G_{n-k})|}\frac{1}{|W(G'_{m-k})|}\sum_{s\in\mathbf{T}_k}\sum_{v\in Z(s)}\epsilon\epsilon'\sum_{w\in W(G_{n-k})}\sum_{w'\in W(G'_{m-k})}R^{G_n}_{\mathbf{T}^{vF}\times\mathbf{T}^{wF},\tilde{s}\otimes 1}\otimes R^{G'^{\epsilon}_m}_{\mathbf{T}^{vF}\times\mathbf{T}^{w'F},\tilde{s}\otimes 1}$$

$$=\sum_{k=0}^{\min(n,m)}\frac{1}{|W(G_{n-k})|}\frac{1}{|W(G'_{m-k})|}\sum_{s\in(\mathbf{T}_k)}\frac{1}{|W(s)|}\sum_{v\in Z(s)}\epsilon\epsilon'\sum_{w\in W(G_{n-k})}\sum_{w'\in W(G'_{m-k})}R^{G_n}_{\mathbf{T}^{vF}\times\mathbf{T}^{wF},\tilde{s}\otimes 1}\otimes R^{G'^{\epsilon}_m}_{\mathbf{T}^{vF}\times\mathbf{T}^{w'F},\tilde{s}\otimes 1}$$

By Proposition 4.2.4, for a fixed $s\in\mathbf{T}_{l,0}$, since the class of tori $(\mathbf{T}_k)$ in each dimension $l\le k\le\min(n,m)$ containing s is uniquely determined, then the same as the projection of $\omega_{n,m}^{\sharp}$ on each layer. □

Before going further into more results in [AMR96], we consider the uniform projection of representation $R^{\mathbf{G}^F}$ of $\mathbf{G}^F\times\mathbf{G}^F$ realized on $\bar{\mathbb{Q}}_l[\mathbf{G}^F]$, given by $R^{\mathbf{G}^F}(g_1,g_2)f(g)=f(g_1gg_2^{-1})$, for $f\in\bar{\mathbb{Q}}_l[\mathbf{G}^F]$, and $g_1,g_2,g\in\mathbf{G}^F$.

Proposition 4.2.7. $(R^{\mathbf{G}^F})^{\sharp}=\frac{1}{|\mathbf{W(G)}|}\sum_{w\in\mathbf{W(G)}}\sum_{\theta\in\mathrm{Irr}(\boldsymbol{T}_w^F)}R^{\boldsymbol{G}}_{\boldsymbol{T}_w,\theta}\otimes R^{\boldsymbol{G}}_{\boldsymbol{T}_w,\theta}$.

Proof. The proof based on computations with the uniform functor obtained in Theorem 3.2.19 and the Mackey formula 3.1.2. Notice for class functions f_1,f_2 on $\mathbf{G}^F$, we have $\langle f_1,f_2\rangle=\left\langle R^{\mathbf{G}^F},f_1\otimes f_2\right\rangle$. Then we compute $\left\langle f_1^{\sharp},f_2^{\sharp}\right\rangle=\left\langle R^{\mathbf{G}^F},f_1^{\sharp}\otimes f_2^{\sharp}\right\rangle=\left\langle(R^{\mathbf{G}^F})^{\sharp},f_1\otimes f_2\right\rangle$, which also implies

$$\begin{aligned}\left\langle f_1^{\sharp},f_2^{\sharp}\right\rangle&=\left\langle\sum_{(\mathbf{T})}\frac{1}{|\mathbf{W(T)}^F|}R^{\mathbf{G}}_{\mathbf{T}}\circ{}^*R^{\mathbf{G}}_{\mathbf{T}}(f_1),\sum_{(\mathbf{T}')}\frac{1}{|\mathbf{W(T')}^F|}R^{\mathbf{G}}_{\mathbf{T}'}\circ{}^*R^{\mathbf{G}}_{\mathbf{T}'}(f_2)\right\rangle\text{(use 3.2.19)}\\&=\sum_{(\mathbf{T})}\frac{1}{|\mathbf{W(T)}^F|^2}\left\langle{}^*R^{\mathbf{G}}_{\mathbf{T}}(f_1),{}^*R^{\mathbf{G}}_{\mathbf{T}}\circ R^{\mathbf{G}}_{\mathbf{T}}\circ{}^*R^{\mathbf{G}}_{\mathbf{T}}(f_2)\right\rangle=\sum_{(\mathbf{T})}\frac{1}{|\mathbf{W(T)}^F|}\left\langle{}^*R^{\mathbf{G}}_{\mathbf{T}}(f_1),{}^*R^{\mathbf{G}}_{\mathbf{T}}(f_2)\right\rangle\text{(Mackey)}\\&=\left\langle\sum_{(\mathbf{T})}\frac{1}{|\mathbf{W(T)}^F|}\sum_{\theta\in\mathrm{Irr}(\mathbf{T}^F)}\theta\otimes\theta,{}^*R^{\mathbf{G}}_{\mathbf{T}}(f_1)\otimes{}^*R^{\mathbf{G}}_{\mathbf{T}}(f_2)\right\rangle=\left\langle\sum_{(\mathbf{T})}\frac{1}{|\mathbf{W(T)}^F|}\sum_{\theta\in\mathrm{Irr}(\mathbf{T}^F)}R^{\mathbf{G}}_{\mathbf{T},\theta}\otimes R^{\mathbf{G}}_{\mathbf{T},\theta},f_1\otimes f_2\right\rangle\end{aligned}$$

We see the expression in the Proposition. □

Remark. Denote by $(R^{\mathbf{G}^F})_1^{\sharp}$ the unipotent part of $(R^{\mathbf{G}^F})^{\sharp}$, then $(R^{\mathbf{G}^F})_1^{\sharp}=\frac{1}{|\mathbf{W(G)}|}\sum_{w\in\mathbf{W(G)}}R^{\mathbf{G}}_{\mathbf{T}_w,1}\otimes R^{\mathbf{G}}_{\mathbf{T}_w,1}$.

In [AMR96], a description of $\omega_{n,m,s}^{\sharp}$ is given under Lusztig correspondence 3.3.7. Let $s\in\mathbf{T}_{l,0}$, denote by $v_\lambda(s)$ the number of λ occurs in eigenvalue of s, let $\mathbf{G}_\lambda(s)$ be the reductive group of semisimple rank $v_\lambda(s)$. then $C_{\mathbf{G}}(s)=\prod_\lambda\mathbf{G}_\lambda(s)$.

Proposition 4.2.8. *For the symplectic-orthogonal case, let* $\mathbf{G}_\sharp(s):=\prod_{\lambda\neq\pm}\mathbf{G}_\lambda(s)$*, then* $\omega_{n,m,s}^{\sharp}$ *is the image of*

$$(R^{\mathbf{G}_\sharp(s)^F})_1\otimes(R^{O^+_{2v_{-1}}(s)^F}\oplus R^{O^+_{2v_{-1}}(s)^F})_1^{\sharp}\otimes\omega_{n-l,m-l,1}^{\sharp}$$

under the Lusztig correspondence

$$\mathbb{C}\mathcal{E}(\mathrm{Sp}_{2n}\times\mathrm{O}^+_{2m},(s_n)\times(s_m))\oplus\mathbb{C}\mathcal{E}(\mathrm{Sp}_{2n}\times\mathrm{O}^-_{2m},(s_n)\times(s_m))\xrightarrow{\simeq}\mathbb{C}\mathcal{E}(C_{\mathrm{Sp}_{2n}}(s_n)\times C_{\mathrm{O}^+_{2m}}(s_m),1)\oplus\mathbb{C}\mathcal{E}(C_{\mathrm{Sp}_{2n}}(s_n)\times C_{\mathrm{O}^-_{2m}}(s_m),1)$$

Definition 4.2.9. For a connected $\mathbf{G}$, fix a maximal torus $\mathbf{T}$ with $W=\mathbf{W(T)}^F$, we define the character associated to $\chi\in\mathrm{Irr}(W)$ to be $R_\chi:=\frac{1}{|W|}\sum_{w\in W}\chi(w)R_w$, called the **almost character**. Here we abbreviate $R^{\mathbf{G}}_{\mathbf{T}_w,1}$ as R_w.

Remark. (i) From definition, we have $R_w=\sum_{\chi\in\mathrm{Irr}(W)}\chi(w)R_\chi$, then we know R_χ forms an orthonormal basis of $\mathcal{V}(\mathbf{G}^F)^{\sharp}\cap\mathcal{V}(\mathbf{G}^F)_1$. For $\mathbf{G}^F$ of type **A**, the almost characters are irreducible characters, and the construction of almost

characters gives a bijection from $\mathrm{Irr}(W)$ to $\mathcal{E}(\mathbf{G}^F,(1))$, notice it may not be true for other cases. Let γ_w^W be the class function on W which takes value 1 at w conjugacy class in W and 0 otherwise, then $\chi = \frac{1}{|W_n|}\sum_{w\in W_n}\chi(w)\gamma_w$ and $R_w = R_{\gamma_w}$.
(ii) If we consider O_n^ϵ, the non-connected case, it will be more complex. Recall the Weyl group of SO_{2n+1} and Sp_{2n} is $W_n = S_n \ltimes \{\pm 1\}^n$, i.e. the permutation group on the set $\{1,...,n,n^*,...,1^*\}$. It has the generator set $\{s_1,...,s_{n-1},t_n\}$, where $s_i = (i,i+1)$ and $t_n = (n,n^*)$, the homomorphism $W_n \to \{\pm 1\}$ is $\varphi: s_i \mapsto 1, t_n \mapsto -1$. Let $W_n^+ := \mathrm{Ker}(\phi), W_n^- := W_n/W_n^+$, then $|W_n^+| = |W_n^-|$ and W_n^+ has generators $\{s_1,...,s_{n-1},t_ns_{n-1}t_n\}$. $W_n^\pm$ are as Weyl groups of $\mathrm{SO}_{2n}^\pm$, then define

$$R_\chi^{\mathrm{O}_{2n}^\epsilon} = \frac{1}{|W_n^\epsilon|}\sum_{w\in W_n^\epsilon}\chi(w)R_{\mathbf{T}_w,1}^{\mathrm{O}_{2n}^\epsilon} = \frac{1}{|W_n^\epsilon|}\sum_{w\in W_n^\epsilon}\chi(w)\,\mathrm{Ind}_{\mathrm{SO}_{2n}^\epsilon}^{\mathrm{O}_{2n}^\epsilon}(R_{\mathbf{T}_w,1}^{\mathrm{SO}_{2n}^\epsilon})$$

and for $w \in W_n$, let $\mathbf{T}_w$ denote the corresponding torus in $\mathrm{SO}_{2n}^{\phi(w)}$, then

$$R_\chi^{\mathrm{O}_{2n}^+\oplus \mathrm{O}_{2n}^-} = \frac{1}{|W_n|}\sum_{w\in W_n}\chi(w)R_{\mathbf{T}_w,1}^{\mathrm{O}_{2n}^{\phi(w)}} = \frac{1}{2}(R_\chi^{\mathrm{O}_{2n}^+} + R_\chi^{\mathrm{O}_{2n}^-}) \quad \text{abbreviate as} \quad R_\chi^{\mathrm{O}_{2n}^+\oplus \mathrm{O}_{2n}^-} = \frac{1}{|W_n|}\sum_{w\in W_n}\chi(w)R_w$$

(ii) For the case of $\mathbf{GL}_n$ (**type A**), there is a bijection between $\widehat{S_n}$ and $\mathcal{E}(GL_n,(1))$. And thus almost characters are all irreducible unipotent characters. However, it is not ture in general for other types.

We have a connection between Weyl groups and unipotent, uniform projection of $\omega_{n,m}$, through the almost characters.

Proposition 4.2.10. *For the symplectic-even orthogonal dual pair, we have $\omega_{n,m,1}^\sharp$ being*

$$\sum_{k=0}^{\min(n,m)}\sum_{\chi\in \mathrm{Irr}(W_k)} R^{\mathrm{Sp}_{2n}}_{\mathrm{Ind}_{W_k\times W_{n-k}}^{W_n}(\chi\otimes 1)} R^{\mathrm{O}_{2m}^+\oplus \mathrm{O}_{2m}^-}_{\mathrm{Ind}_{W_k\times W_{m-k}}^{W_m}(\varphi\chi\otimes 1)}$$

Consider O_{2m}^ϵ separately, it becomes $\omega^\sharp_{\mathrm{Sp}_{2n},\mathrm{O}_{2m}^\epsilon} = \frac{1}{2}\sum_{k=0}^{\min(n,m)}\sum_{\chi\in\mathrm{Irr}(W_k)} R^{\mathrm{Sp}_{2n}}_{\mathrm{Ind}_{W_k\times W_{n-k}}^{W_n}(\chi\otimes 1)} R^{\mathrm{O}_{2m}^\epsilon}_{\mathrm{Ind}_{W_k\times W_{m-k}}^{W_m}(\varphi\chi\otimes 1)}$.

Proof. In Remark (i) of Definition 4.2.9, we have $\chi = \frac{1}{|W_k|}\sum_{w\in W_k}\chi(w)\gamma_w$ and $1_{W_{n-k}} = \frac{1}{|W_{n-k}|}\sum_{v\in W_k}\gamma_v$, then

$$\chi\otimes 1 = \frac{1}{|W_k|\,|W_{n-k}|}\sum_{w\in W_k}\chi(w)\sum_{v\in W_{n-k}}\gamma_{wv}^{W_k\times W_{n-k}}, \quad \mathrm{Ind}_{W_k\times W_{n-k}}^{W_n}(\chi\otimes 1) = \frac{1}{|W_k|\,|W_{n-k}|}\sum_{w\in W_k}\chi(w)\sum_{v\in W_{n-k}}\gamma_{wv}^{W_n}$$

Here we use the fact that $\mathrm{Ind}_{W_k\times W_{n-k}}^{W_n}(\gamma_{wv}^{W_k\times W_{n-k}}) = \gamma_{wv}^{W_n}$. And thus we can compute the irreducible character

$$R^{G_n}_{\mathrm{Ind}_{W_k\times W_{n-k}}^{W_n}(\chi\otimes 1)} = \frac{1}{|W_k|\,|W_{n-k}|}\sum_{w\in W_k}\chi(w)\sum_{v\in W_{n-k}}R_{wv}$$
$$R^{G'_m}_{\mathrm{Ind}_{W_k\times W'_{m-k}}^{W_m}(\varphi\chi\otimes 1)} = \frac{1}{|W_k|\,|W'_{m-k}|}\sum_{w\in W_k}\varphi\chi(w)\sum_{v'\in W'_{m-k}}R_{wv'}$$

Compare with the expression in Proposition 4.2.6 we get the conclusion. ∎

Remark. (i) We shall compute some examples in the next section with formula in the above Proposition.
(ii) In [AMR96], they get an explicit expression of unipotent theta correspondence by considering the bipartition of $\mathfrak{S}_n$.

4.3 More on Cuspidal Representations and Hecke Algebras

Take $\mathbb{T}$ and $\mathbb{T}'$ to be Witt towers of V and V'. And let $(G_k, G_l^{'})$ denote the corresponding dual pair, $\omega_{k,l}$ denote the restriction of the corresponding Weil representation to $G_k \times G_l^{'}$. Suppose $G = G_n, G' = G_m^{'}$. For each $(k.l)$, we have $\omega_{k,l} = \oplus_{\pi,\pi'} m_{\pi,\pi'} \pi \otimes \pi'$, where $\pi \in \mathrm{Irr}(G_k)$, and $\pi' \in \mathrm{Irr}(G_l^{'})$ runs over the irreducible representations of the dual groups. We say π corresponds to $\pi^{'}$ if $\pi \otimes \pi^{'}$ occurs in the decomposition of $\omega_{k,l}$. A result of [AM93] shows

Theorem 4.3.1. *If π is a cuspidal representation of G, and π correspond to π' for $G \times G^{'}$. If π does not occur in any correspondence for all $G \times G_l^{'}$, $\forall l < m$, then $\pi^{'}$ is also cuspidal.*

Corollary 4.3.2. *Together with 4.2.3, we have an immediate corollary of Theorem 4.3.1. In the dual pair of symplectic-special orthogonal case, if π is a unipotent cuspidal representation of G, and π does not occur in the theta correspondence for $(G, G_l')\forall l < m$, then π' is also a unipotent cuspidal representation if G'.*

Proof of the theorem. Take a Witt decomposition of $V' = V^{'+} \oplus V_0 \oplus V^{'-}$, assume $\dim V^{'+} = \dim V^{'+} = d < n$. Then is a maximal parabolic of $G^{'}$ assciates to this decomposition, $P' = M'N'$, where $M' = \mathrm{GL}(V^{'+}) \times G_{n-j}^{'}$ is the Levi, and N' the unipotent radical of M'. Let $(\omega_{m,n})_{N'}$ denote the Jacquet module.
If π' is not cuspidal, then there is some decomposition of V', which corresponding to a maximal parabolic $P' = M'N'$ such that there exists representation $\sigma' \otimes \tau'$ of $M' = \mathrm{GL}(V^{'+}) \times G_{n-j}^{'}$ such that $\pi' \subseteq \mathrm{R}_{P'}^{G'}(\sigma' \otimes \tau' \otimes 1)$. Then we have

$$\begin{aligned} 0 &\neq \langle \omega_{m,n}, \pi \otimes \pi' \rangle_{G\times G'} \\ &\leq \left\langle \omega_{m,n}, \pi \otimes \mathrm{R}_{P'}^{G'}(\sigma' \otimes \tau' \otimes 1) \right\rangle_{G\times G'} && \text{Here 1 indicates trivial extension on } N' \\ &= \left\langle (\omega_{m,n})_{N'}, \pi \otimes \mathrm{R}_{P'}^{G'}(\sigma' \otimes \tau') \right\rangle_{G\times \mathrm{GL}(V'^{+})\times G_{n-j}^{'}} && \text{Consider the Jacquet module} \end{aligned}$$

Denote by $R(M', \omega_{m,n})_{\text{cusp}}$ the cuspidal part of $(\omega_{m,n})_{N'}$ to G. We have the following lemma

Lemma 4.3.3. *$R(M', \omega_{m,n})_{cusp} \simeq \omega_{m,n-j}$, where j is the dimension of $V^{'+}$.*

Proof of the lemma. Let $\mathbb{W} = \mathbb{W}_+ \oplus \mathbb{W}_0 \oplus \mathbb{W}_- = (V \otimes V^{'+}) \oplus (V \otimes V_0) \oplus (V \otimes V^{'-})$. Consider the mixed model.

$$\omega_{m,n} = \mathbb{C}[\mathrm{Hom}(V^{'+}, V)] \otimes \omega_{m.n-j} = \left\{ f : \mathrm{Hom}(V^{'+}, V) \longrightarrow \omega_{m,n-j} \right\}$$

Here $\mathrm{Hom}(V^{'+}, V) = V \otimes V^{'+}$ is identified with V^j Now $N' = Z(N')N_1^{'}$, where $\forall z \in Z(N') \subseteq \mathrm{Hom}(V^{'+}.V^{'-})$, here z could be regarded as a bilinear form on $V \otimes V^{'+}$ by $z \mapsto \beta_z : (x,y) \mapsto \langle\langle zx, y \rangle\rangle$, then the action of z is given by $\omega_{m,n}(z)f(x) = \chi(\mathrm{tr}(\frac{\langle\langle zx,x\rangle\rangle}{2}))f(x)$. thus $\omega_{m,n}(z)f = f$ iff ${}^t xx = 0$, or equivalently $\langle x_i, x_j \rangle = 0$. Let $\mathcal{N} := \{x | \langle x_i, x_j \rangle = 0, x_i \in V\}$, then

$$(\omega_{m,n})_{Z(N')} = \mathbb{C}[\mathcal{N}] \otimes \omega_{m,n-j} = \mathbb{C}[\oplus_{i=1}^{j} \mathcal{N}_i] \otimes \omega_{m,n-j} = \bigoplus_{i=0}^{\min\{j,m\}} \mathbb{C}[\mathcal{N}_i] \otimes \omega_{m,n-j} \quad \text{where } \mathcal{N}_i \text{ denote the rank i piece of } \mathcal{N}$$

Easy to see the restriction of $(\omega_{m,n})_{Z(N')}$ on G has cuspidal summand unless $i = 0$, in which case the cuspidal part of $(\omega_{m,n})_{Z(N')}|_G$ is $\omega_{m,n-j}$. Additionally, $N_1^{'}$ acts transitively on $\omega_{m,n-j}$. Hence $R(M', \omega_{m,n})_{\text{cusp}} \simeq \omega_{m,n-j}$, the lemma is proved. ☐

back to our proof of the theorem,

$$\begin{aligned} 0 \neq \left\langle (\omega_{m,n})_{n'}, \pi \otimes \mathrm{ind}_{p'}^{g'}(\sigma' \otimes \tau') \right\rangle_{g\times \mathrm{gl}(v'^{+})\times g_{n-j}^{'}} &\leq \left\langle (\omega_{m,n})_{n'}|_{g\times g_{n-j}^{j}}, \pi \otimes \tau^{'} \right\rangle_{g\times g_{n-j}^{'}} \\ = \left\langle r(m', \omega_{m,n})_{\text{cusp}}, \pi \otimes \tau^{'} \right\rangle_{g\times g_{n-j}^{'}} &= \left\langle \omega_{m,n-j}, \pi \otimes \tau^{'} \right\rangle_{g\times g_{n-j}^{'}} = 0 \qquad \text{by assumption} \end{aligned}$$

hence our theorem is proved by showing this contradiction. □

The following result of cuspidal representations is due to Lusztig [Lus77]

Lemma 4.3.4. *For groups of type* $\boldsymbol{Sp}_{2\mathrm{n}}$, $\mathrm{Sp}_{2n}(q)$ *has a unique unipotent cuspidal representation if and only if* $n = k^2 + k$. *For groups of type* $\boldsymbol{SO}^{\epsilon}_{2\mathrm{n}}$, $\mathrm{SO}^{\epsilon}_{2n}(q)$ *has a unique unipotent cuspidal representation if and only if* $n = k^2$ *and* $\epsilon = \mathrm{sgn}(-1)^k$. *For groups of type* $\boldsymbol{SO}_{2\mathrm{n}+1}$, $\mathrm{SO}_{2n+1}(q)$ *has a unique unipotent cuspidal representation if and only if* $n = k^2 + k$.

Since unipotent representation of $\mathrm{O}^{\epsilon}_{2n}$ is obtained from $\mathrm{R}^{O^{\epsilon}_{2n}}_{SO^{\epsilon}_{2n}}\pi$, where π is unipotent. Then the induce representation of $\mathrm{O}^{\epsilon}_{2n}$ by unipotent cuspidal π_{cu} is the direct sum $\pi^{I}_{cu} \oplus \pi^{II}_{cu}$ and this occurs only when $n = k^2$ and $\epsilon = \mathrm{sgn}(-1)^k$. Both π^{I}_{cu} and π^{II}_{cu} are irreducible unipotent cuspidal representations of $\mathrm{O}^{\epsilon}_{2n}$ and they differ by tensoring sgn character of $\mathrm{O}^{\epsilon}_{2k^2}$ (since $\left\langle \rho_1^{\sharp}, \rho_2^{\sharp} \right\rangle_{\mathrm{O}^{\epsilon}} = 1$ implies $\rho_1 = \rho_2$ or $\rho_1 = \rho_2 \cdot \mathrm{sgn}$ from Proposition 3.2.22). We have the following in [AM93]

Theorem 4.3.5. *For the dual pair* $(\mathrm{Sp}_{2n}, \mathrm{O}^{\epsilon}_{2m})$ *theta correspondence for unipotent cuspidal representation is given by*
(i) $\left(Sp(2k(k+1)), O^{\epsilon}\left(2k^2\right)\right)\left(\epsilon = \mathrm{sgn}\left((-1)^k\right)\right), \quad \pi_{cu} \to \pi^{II}_{cu}$
(ii) $\left(\mathrm{Sp}(2k(k+1)), O^{\epsilon}\left(2(k+1)^2\right)\right)\left(\epsilon = \mathrm{sgn}\left((-1)^{k+1}\right)\right), \quad \pi_{cu} \to \pi^{I}_{cu}$

4.3.1 Induced Hecke Rings from Cuspidal Representations

We would like to study on the generalized Hecke algebra associated to cuspidal representations, especially its structure theory, by following [HL80], and Theorem 3.1.7 is part of this section. Complicated demonstrations and computations are omitted, we extract skeleton of Howlett and Lehrer's paper.

Denote by $\mathbf{P}$ the parabolic and $\mathbf{P} = \mathbf{LU}$ its Levi decomposition, and D the representation of $L = \mathbf{L}^F$ which lift to $P = \mathbf{P}^F$ module D^* through the projection $P \to L$ with kernel U. Let $W(D) := \left\{w \in G = \mathbf{G}^F | {}^w L = L, {}^w D = D\right\}$. For convenience, we choose P to be standard $P_J = L_J U_J$. The complex representation space is V, and thus $D : L_J \to GL(V)$.

Denote by $\mathrm{ind}(w) := |U_w^-|$ and $U_{w,J}^{\pm} := U_{wJ} \cap U_{w^{-1}}^{\pm}$, let $\hat{\Omega} := \{a \in \Sigma | \exists w \in W, w(J \cup \{a\}) \subseteq \Pi\}$ and for $a \in \hat{\Omega}$, denote $v(a, J) = w_{I,0} w_{J,0}$, where $I = J \cup \{a\}$ and $w_{I,0}$ denote the longest element in corresponding W_I (the same as $w_{J,0}$). Let $\mathcal{H}_D = \mathrm{End}(\mathrm{R}_{P_J}^G(D^*))$, we first determine the basis of $\mathcal{H}_D$ by constructing.

For any $w \in W$ with $wJ \subseteq \Pi$, we can give a representation wD of $L_{wJ} = {}^w L_J$ by

$$wD(x)v := D({}^{w^{-1}}x)v, \quad \forall x \in L_{wJ}, v \in V.$$

If $w \in W(D)$, easy to know $wD \cong D$, thus by **Schur's Lemma**, we have $wD(x) = \tilde{D}(w^{-1})w(x)\tilde{D}(w)$ for some linear operator $\tilde{D}(w)$ on V. For $w_1, w_2 \in W(D)$, there is a cocycle condition $\tilde{D}(w_1 w_2) = \lambda(w_1, w_2)\tilde{D}(w_1)\tilde{D}(w_2)$, $\lambda : W(D) \times W(D) \to \mathbb{C}$ with $\lambda(u, v)\lambda(uv, w) = \lambda(u, vw)\lambda(v, w)$. By direct computation with λ, we have $\lambda(w^{-1}, w) = \lambda(w, 1) = \lambda(1, w) = 1$. And thus $\lambda(u, v)\lambda(uv, v^{-1}) = 1, \lambda(v^{-1}, u) = (\lambda(u, v))^{-1}$.

We conclude λ is a root of unity by finiteness of $W(D)$ and above. Now consider $\tilde{L}_J$ generated by L_J and $W(D)$, we can lift $\tilde{D}$ to representation of $\tilde{L}_J$ realized in V by setting $\tilde{D}(xw) := D(x)\tilde{D}(w), \forall x \in L_J, w \in W(D)$. And the similar cocycle relation holds, $\tilde{D}(\tilde{x}\tilde{y}) = \lambda(w, v)\tilde{D}(\tilde{x})\tilde{D}(\tilde{y}), \forall \tilde{x} \in L_J \cdot w, \tilde{y} \in L_J \cdot v$.

Definition 4.3.6. For $w \in W(D)$, let B_w be the operator on $\mathrm{R}_{P_J}^G(D^*) = \{f : G \to V | f(pg) = D^*(p)f(g), \forall p \in P_J, g \in G\}$ given by

$$(B_w f)(x) := \frac{1}{|U_J|}\tilde{D}(w)\sum_{y \in U_J} f(w^{-1}yx), \quad \forall x \in G$$

We see $f(w^{-1}ypg) = D^*(w^{-1}yp)f(g) = D^*({}^{w^{-1}}yp)D^*(w^{-1})f(g) = D^*(yp)D^*(w^{-1}y)f(g) = D^*(p)f(g)$ by definition. And thus $B_w f \in \mathrm{R}_{P_J}^G(D^*)$, B_w is an operator on the space. Since $U_J = U \cap U_J = (U_w^+ \times U_w^-) \cap U_J$, for $\forall y \in U_J, y = y^+ y^-$, $y^+ \in U_w^+ \cap U_J, y^- \in U_w^- \cap U_J$, and since $D(w_J u_J w_J^{-1}) = 1$, we have $f(w^{-1}yx) = f(w^{-1}ww_J u w_J^{-1} w^{-1} ww^{-1} y^- x) = f(w^{-1}yx)$, and thus

$$(B_w f)(x) := \frac{1}{|U_J|}\tilde{D}(w)\sum_{y \in U_J} f(w^{-1}yx) = \frac{1}{|U_w^-|}\tilde{D}(w)\sum_{y \in U_w^-} f(w^{-1}yx), \quad \forall x \in G$$

The following lemma is provided by Springer,

Lemma 4.3.7. *$\{B_w | w \in W(D)\}$ forms a basis of $\mathcal{H}_D = \mathrm{End}_G(\mathrm{R}_{P_J}^G(D^*))$.*

Definition 4.3.8. For other $w \in W$ with $wJ \subseteq \Pi$, we can also define a map $B_{D,w} : \mathrm{R}_{P_J}^G(D^*) \to \mathrm{R}_{P_{wJ}}^G((wD)^*)$ by

$$(B_{D,w} f)(x) := \frac{1}{|U_{wJ}|}\sum_{y \in U_{wJ}} f(w^{-1}yx) = \frac{1}{\left|U_{w,J}^-\right|}\sum_{y \in U_{w,J}^-} f(w^{-1}yx) \quad \forall x \in G$$

It is obvious to see that $B_w = \tilde{D}(w)B_{D,w}$, and we can verify $B_{D,w}$ is a $G-$equivariant map that has image in $\mathrm{R}_{P_{wJ}}^G((wD)^*)$.

Proposition 4.3.9. *If $w_1, w_2 \in W$ with $w_2 J \subseteq \Pi, w_1 w_2 J \subseteq \Pi$, $l(w_1 w_2) = l(w_1) + l(w_2)$, then $B_{D,w_1 w_2} = B_{w_2 D, w_1} B_{D, w_2}$.*

Proposition 4.3.10. *Let $a \in \Pi \backslash J$, and denote $v = v(a, J), I = J \cup \{a\}$, then we have the following relation on operators*

$$B_{vD,v^{-1}} B_{D,v} = (\operatorname{ind} v)^{-1} \operatorname{id} + \beta B_v$$

Here $\beta \in \mathbb{C}$ is 0 when $v \notin W(D)$, and $\operatorname{ind} v \beta$ is an algebraic integer when $v \in W(D)$.

Proof. First we see that $B = B_{vD,v^{-1}} B_{D,v} : \mathrm{R}_{P_J}^G D \to \mathrm{R}_{P_{vJ}}^G vD \to \mathrm{R}_{P_J}^G D$ by definition, then it is an operator on $\mathrm{R}_{P_J}^G D$. And B is the linear combination of $\{B_w | w \in W_I \cap W(D)\}$. By our setting, $W_I \cap W(D) \subseteq \{1, v\}$, thus $B = \alpha \operatorname{id} + \beta B_v, \alpha, \beta \in \mathbb{C}$, if $v \notin W(D)$, $W_I \cap W(D) = 1$, then $\beta = 0$ in this case. Compute whenever v is in $W(D)$

$$(Bf)(x) = \frac{1}{\left|U_{v^{-1},vJ}^-\right|} \frac{1}{\left|U_{v,J}^-\right|} \sum_{y \in U_{v^{-1},vJ}^-, z \in U_{v,J}^-} f(v^{-1} zvyx) = (\alpha \operatorname{id} + \beta B_v(f))(x) = \alpha f(x) + \beta \tilde{D}(v) \frac{1}{\left|U_{v,J}^-\right|} \sum_{y \in U_{v,J}^-} f(v^{-1} yx)$$

fix some $v_0 \in V$, associate with $f_{v_0}(x) := \begin{cases} v_0, x = 1 \\ 0, x \neq P_J \end{cases}$. It is clear $f_{v_0} \in \mathrm{R}_{P_J}^G(D^*)$ and $(Bf_{v_0})(1) = (\operatorname{ind} v)^{-1} v_0 = \alpha v_0 + \beta \tilde{D}(v) \frac{1}{|U_{v,J}^{-1}|} \sum_{y \in U_{v,J}^{-1}} f_{v_0}(v^{-1} y) = \alpha v_0$. Thus $\alpha = (\operatorname{ind} v)^{-1}$.
For the proof of $(\operatorname{ind} v)\beta$ is an algebraic integer, see [HL80]. □

As a corollary of the above Proposition, we have the relation $B_v^2 = (\operatorname{ind} v)^{-1} \operatorname{id} + \beta B_v$.

Lemma 4.3.11. *Let $w \in W$ with $wJ \subseteq \Pi$ and for $a \in \pi \backslash wJ$, let $v = v(a, wJ)$. Assume $w^{-1} a \in \Sigma^+$, then*

$$B_{wD,v} B_{D,w} = (\operatorname{ind} v \operatorname{ind} w)^{-\frac{1}{2}} (\operatorname{ind} vw)^{\frac{1}{2}} B_{D,vw}$$

Set $\Omega = \left\{a \in \hat{\Omega} | v(a, J)^2 = 1\right\}$, denote by the subset $\Gamma \subseteq \Omega$ with $v(a, J) \in W(D), \forall a \in \Gamma$, $wa \in \Gamma, \forall w \in W(D), a \in \Gamma$. $\Gamma^+ := \Gamma \cap \Sigma^+$. Recall for $w \in W$, $N(w)$ denotes the set $\{a \in \Sigma^+ | wa \in \Sigma^-\}$, define $\Delta = \{a \in \Sigma^+ | N(v(a, J)) \cap \Sigma = \{a\}\}$. Let $R(D) \subseteq W(D)$ generate by $v(a, J), a \in \Gamma$, and $C(D)$ is the set with $W(D) = R(D)C(D)$.

Theorem 4.3.12. *For $w \in W(D), t \in C(D), a \in \Delta, v = v(a, J)$, we have*
(i) $B_w B_t = (\operatorname{ind} w \operatorname{ind} t)^{-\frac{1}{2}} \lambda(w, t) (\operatorname{ind}(wt))^{\frac{1}{2}} B_{wt}$, $\quad B_t B_w = (\operatorname{ind} t \operatorname{ind} w)^{-\frac{1}{2}} \lambda(t, w) (\operatorname{ind}(tw))^{\frac{1}{2}} B_{tw}$.
(ii) $B_v^2 = (\operatorname{ind} v)^{-1} \operatorname{id} + \varepsilon_a (p_a \operatorname{ind} v)^{\frac{1}{2}} B_v$, where p_a is some integral power of characteristic p, $\varepsilon_a \in \mathbb{C}$ both of them are determined by a, see [HL80] for detail.

Corollary 4.3.13. *For $a \in \Delta$, $v = v(a, J)$, define $T_v := \varepsilon_a (p_a \operatorname{ind} v)^{\frac{1}{2}} B_v$. For $w \in w(D)$, let $p_w = \prod p_a$ take multiplication over $N(w) \cap \Sigma$. As a corollary of Theorem 4.3.12, we have $T_v^2 = p_a \operatorname{id} + (p_a - 1) T_v$. And in fact $T_w = \varepsilon_w (p_w \operatorname{ind} w)^{\frac{1}{2}} B_w$, where ε_w is a root of unity. $\{T_w | w \in W(D)\}$ is a linear basis for $\operatorname{End}_G(\mathrm{R}_{P_J}^G(D^*)) = \mathcal{H}_D$.*

Theorem 4.3.14. *Define $\mu(v, w) = \varepsilon_v \varepsilon_w \varepsilon_{vw}^{-1} \lambda(v, w)$ for $v, w \in W(D)$. Then for $w \in W(D), x \in C(D), a \in \Delta, v = v(a, J)$*

$$(i) T_w T_x = \mu(w, x) T_{wx}, \qquad (ii) T_x T_w = \mu(x, w) T_{xw}$$

$$(iii) T_v T_w = \begin{cases} T_{vw} \text{ if } w^{-1} a \in \Gamma^+ \\ p_a T_{vw} + (p_a - 1) T_w \quad \text{if } w^{-1} a \in \Gamma^- \end{cases} \quad (iv) T_w T_v = \begin{cases} T_{wv} \text{ if } wa \in \Gamma^+ \\ p_a T_{wv} + (p_a - 1) T_w \text{ if } wa \in \Gamma^- \end{cases}$$

We get a structure of $\mathcal{H}_D$ similar to the Hecke algebra in the normal setting.

For $T \in \operatorname{Hom}_P(\sigma, \operatorname{Ind}_P^G \sigma) : v \in (\sigma, V) \mapsto f_v : G \mapsto \mathrm{GL}(V)$, and $\operatorname{Supp}(f_v) \subseteq P, f_v(p) = \sigma(p)(v)$. Then $\mathcal{H}_\sigma$ acts as $\mathcal{H}_\sigma T(v) := \mathcal{H}_\sigma f_v$. More explicitly, if consider the basis B_w, it will be $(B_w f_v)(x) = \left|\operatorname{ind}_w^-\right|^{-1} \sigma(w) \sum_{y \in U_w^{-1}} f_v(w^{-1} yx)$. The ring homomorphism from $\mathcal{H}_\sigma$ to $\mathcal{H}$ is given by $(\Delta : G \to V) \mapsto \phi(g \mapsto [G : P] \dim(\sigma) \operatorname{tr} \Delta(g))$. Now assume σ extends to $W(\sigma)$, then for $\Phi \in \mathcal{H}_\sigma$ with support PwP. for some $w \in N(\sigma)$, by definition, we have $\sigma^\vee(wlw^{-1})\Phi(w) =$

$\Phi(wl) = \Phi(w)\sigma^\vee(l), \forall l \in L$, thus $\Phi(w) \in \mathrm{Hom}_L(\sigma^\vee, (\sigma^\vee)^w)$. Easy to see $\sigma(w^{-1})^\vee \in \mathrm{Hom}_L(\sigma^\vee, (\sigma^\vee)^w)$, then there is Φ_w such that $\mathrm{Supp}\,\Phi_w = Pw^{-1}P$ and $\Phi_w(w) = \sigma(w^{-1})^\vee$. Let ϕ_w be the image of Φ_w in $\mathcal{H}$ under the algebra homomorphism, we have

$$\begin{aligned}\phi_w f_v(w) &= \frac{1}{|G|}\sum_{g\in G}\phi_w(g)f_v(wg) = \frac{[G:P]\dim\sigma}{|G|}\sum_{g\in G}\mathrm{tr}(\Phi_w(g))f_v(wg)\\ &= \frac{[G:P]\dim\sigma}{|G|}\sum_{g\in U^-_{w^{-1}}wP=PwP}\mathrm{tr}(\Phi_w(uw^{-1}p))f_v(wuw^{-1}p) \qquad (\mathrm{Supp}\,\Phi_w = PwP)\\ &= \frac{[G:P]\dim\sigma}{|G|}\sum_{p\in P}\mathrm{tr}(\Phi_w(w^{-1}p))f_v(p) \qquad (\mathrm{Supp}\,f_v = P, \text{and } wU^-_{w^{-1}}w^{-1}\cap P = \{1\})\\ &= \sigma(w)v\end{aligned}$$

Comparing to $(B_w f_v)(w) = \frac{1}{|U^-_w|}\sigma(w)f_v(1) = \frac{1}{|U^-_w|}\sigma(w)v$ (Notice that the support of $B_w f_v$ is PwP and $wU^-_{w^{-1}}w^{-1}\cap P = \{1\}$). One can conclude that B_w corresponds to ϕ_w by multiple $\frac{1}{|U^-_w|}$ under such circumstance. Now for some representation π of G, with $v \in \pi[\sigma]$, the Hecke module by ϕ_w is $\phi_w\cdot v = \frac{1}{|G|}\sum_{g\in G}\phi_w(g)\pi(g)v = \sum_{u\in U^-_{w^{-1}}}\pi(uw^{-1})\sigma(w)v$. Hence we define for general $B_w := \frac{1}{|U^-_w|}\sum_{u\in U^-_{w^{-1}}}\pi(uw^{-1})\sigma(w)$.

We now compute examples.

Example 4.3.15. Let (G, G') be the dual pair $(\mathrm{Sp}_2 = \mathrm{Sp}(V), \mathrm{O}_2^+ = \mathrm{O}(V'))$. Fix the character $\psi : \mathbb{F}_q^\times \to \mathbb{C}$.
Take the decomposition of V into maximal isotropic subspaces, then $G \times G' \to G \cdot G' \to \mathrm{Sp}(\mathbb{W})$, where $\mathbb{W} = \mathbb{W}_- \oplus \mathbb{W}_+ = \mathrm{Hom}(V^+, V') \oplus \mathrm{Hom}(V^-, V')$ and $\mathrm{Hom}(V^+, V') \cong \mathrm{Hom}(V', V^-), x \mapsto x^*$.
Recall the construction of Weil representation, $\omega_{\mathrm{Sp}(\mathbb{W})}$ is realized on the space $\mathbb{C}[\mathbb{W}_-]$. To study the theta correspondence is thus to consider $\omega = \omega_{n,m} = \omega_{\mathrm{Sp}(\mathbb{W})}|_{G\times G'}$.
In [AMR96], the theta correspondence between unipotent characters of dual pair $(\mathrm{GL}_n\,\mathrm{GL}_m)$ can be write down explicitly by considering some combinatorics regard of the bipartition of $\mathscr{R}(\mathfrak{S})$, the Hope algebra of symmetric group (Weyl groups of type **A**). Inspired by this, we want to consider Hecke algebras (which are isomorphic to $\mathbb{C}[W]$), and their modules.
We compute $\omega^{B\times B'}$. Notice first that the Borel $\mathbf{B}_{2m}$ of $\mathbf{SO}_{2m}$ is $\mathbf{SO}_{2m}\cap\mathbf{T}_{2m}$, hence we regard $\mathbf{B}'$ as some product of $\mathbb{G}_m \cong k^\times$. And $\omega^B = \omega^{TN} = (\omega_N)^T$, where $N = \left\{\begin{pmatrix}1 & z\\ & 1\end{pmatrix}\right\}$. For $f \in \mathbb{C}[\mathbb{W}_-]$ We have the action of N on $\mathbb{C}[\mathbb{W}_-]$ being

$$\omega\left(\begin{pmatrix}1 & z\\ 0 & 1\end{pmatrix}f\right)(y) = \psi\left(\frac{\langle\langle zy, y\rangle\rangle}{2}\right)f(y) = \psi\left(\frac{(zy)^*y}{2}\right)f(y) = \psi\left(\frac{zyy^*}{2}\right)f(y), \quad \forall y \in \mathrm{Hom}(V^+, V')$$

here $z \in \mathrm{Hom}(V^-, V^+)$ could be regarded as a bilinear form on $\mathbb{W}_- \times \mathbb{W}_- \cong (V'\otimes V^+)\times(V'\otimes V^+)$ by $z \mapsto \beta_z : (x, y) \mapsto \langle\langle zx, y\rangle\rangle = (zx)^*y$. In the same way as proof of Theorem 4.3.1, we have

$$\omega^{B\times B'} \cong \omega^B \cong \omega_N \cong \mathbb{C}[\mathcal{N}] \subseteq \mathbb{C}[\mathbb{W}_-], \quad \mathcal{N} = \left\{T \in V'\otimes V^+ \cong V'^{\dim V^+} \,|\, \langle T_i, T_j\rangle_{V'} = 0, \forall i, j\right\}$$

In our case, $\dim V^+ = 1$, $\mathcal{N} = \{v \in V' \,|\, \langle v, v\rangle = 0\}$. Hence $\mathcal{N}^\times = \mathcal{N}\setminus\{0\}$ is spanned by v_+, v_-, with corresponding characteristic function $f_\pm \in \mathbb{C}[\mathcal{N}]$ supported on $\mathbb{F}_q[v_\pm]$ respectively.
Consider the canonical Hecke algebra $\mathcal{H}_G$, which is spanned by $\{\phi_1 = 1_B, \phi_s = 1_{BsB}\}$, since $\mathrm{Sp}_2 \cong \mathrm{SL}_2$, $s =$

$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, then $\omega^{B\times B'}$ as $\mathcal{H}_G$ module is

$$\begin{aligned}
\phi_s f_+(y) &= \frac{1}{|B|}\sum_{g\in BsB}\phi_s(g)\omega(g)f_+(y) = \frac{1}{|B|}\sum_{u\in U_{s^{-1}}b\in B}\omega(bsb)f_+(y) = \sum_{u\in U_{s^{-1}=s}}\omega(us)f_+(y) \quad f_+ \text{ is fixed by } B\\
&= \sum_{u\in U_s}\omega(u)c_\psi\mathcal{F}f_+(y) = \sum_{u\in U_s}\omega(u)\int_x f_+(x)\psi(\langle\langle x,y\rangle\rangle)\,dx = \sum_{u\in U_s}\omega(u)q^{\frac{-\dim V'}{2}}\sum_{x\in\mathbb{F}_q[v_+]}f_+(x)\psi(\langle\langle x,y\rangle\rangle)\\
&= \begin{cases}\sum_u\omega(u)qq^{-1}, y\in\mathbb{F}_q^\times[v_+]\\ 0,\ \text{others}\end{cases} \quad (\text{ Notice } \sum_{a\in\mathbb{F}_q}\psi(a) = \begin{cases} q, \psi \text{ trivial}\\ 0,\ \text{others}\end{cases}, \text{ it is from first orthogonality of characters })\\
&= \sum_{u\in U_s}\omega(u)f_+(y) = \sum_{z\in\mathbb{F}_q}\psi(\frac{1}{2}y^*yz)f_+(y) = qf_+(y)
\end{aligned}$$

And we should notice under the canonical quadratic relations of generators of Hecke ring, we have the equality $\phi_s\circ\phi_s f_+ = q^2 f_+ = (q\phi_1 + (q-1)\phi_s)f_+ = (\phi_s*\phi_s)f_+$. And similar to our computation above, we have $\phi_s f_- = qf_-$. Then we can compute $\phi_s f_0$ by quadratic relation

$$\begin{aligned}
&(q\phi_1 + (q-1)\phi_s)(f_+ + f_- + f_0) = (\phi_s*\phi_s)(f_+ + f_- + f_0) = (\phi_s\circ\phi_s)(f_+ + f_- + f_0)\\
&q^2(f_+ + f_-) + qf_0 + (q-1)\phi_s f_0 = q^2(f_+ + f_-) + \phi_s^2 f_0 \quad \Longrightarrow \phi_s f_0 = f_+ + f_- - f_0.
\end{aligned}$$

Now Hecke algebra $\mathcal{H}_{G'}$ is spanned by $\{\phi_1 = 1_{B'}, \phi_{s'} = 1_{Bs'B}\}$, where $s' = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, then

$$\phi_{s'}f_+(y) = f_+(s'y) = f_-(y), \quad \phi_{s'}f_-(y) = f_-(s'y) = f_+(y).$$

The action of s' is by translation, then exchanges $v_\pm$. Thus $\phi_{s'}f_+$ is supported on $\operatorname{span} v_-$, $\phi_{s'}f_+ = f_-, \phi_{s'}f_- = f_+$. And the $\mathcal{H}_G'$ is thus regular representation.
Hence if we consider the submodule $\omega_\pm := \mathbb{C}[\mathcal{N}^\times] \subseteq \omega^{B\times B'}$ of $\mathcal{H}_G\times\mathcal{H}_{G'}$, it is isomorphic to $1_{\mathcal{H}_G}\otimes \operatorname{reg}_{\mathcal{H}_{G'}}$.
And since $\phi_s f_0 = f_+ + f_- - f_0$, $\phi_{s'}f_0 = f_0$, then $\omega/\omega_\pm$ is $\operatorname{sgn}_{\mathcal{H}_G}\otimes 1_{\mathcal{H}_{G'}}$.
Compute the character constructed in Definition 4.2.9, let $\mathbf{T}_0$ denote the rational torus and $\mathbf{T}_1$ conjugate by s, we have

$$R^{\mathrm{Sp}_2}_{\mathrm{sgn}} = \frac{1}{|W(\mathrm{Sp}_2)|}\sum_{w\in W(\mathrm{Sp}_2)}\operatorname{sgn}(w)R^{Sp_2}_{\mathbf{T}_w,1} = \frac{1}{2}(R^{Sp_2}_{\mathbf{T}_0,1} - R^{Sp_2}_{\mathbf{T}_1,1}).$$

Since $\mathrm{Sp}_2\cong\mathrm{SL}_2$, and it is well known $R^{Sp_2}_{\mathbf{T}_0,1} = 1+\mathrm{St}$, $R^{Sp_2}_{\mathbf{T}_1,1} = 1-\mathrm{St}$, then $R^{\mathrm{Sp}_2}_{\mathrm{sgn}} = \mathrm{St}$ is the Steinberg character.
And notice the character $\operatorname{reg}_{\mathrm{O}_2^+} = 1_{\mathrm{O}_2^+} + \operatorname{sgn}_{\mathrm{O}_2^+}$. We can read off from $\omega_\pm$ and $\omega/\omega_\pm$ the correspondence $1\longleftrightarrow\operatorname{reg}$, $\operatorname{sgn}\longleftrightarrow 1$ respectively, which implies the theta correspondence between characters. It is the same result as in [Pan19].

$$1_{\mathrm{Sp}_2}\longleftrightarrow\operatorname{reg}_{\mathrm{O}_2^+} = 1_{\mathrm{O}_2^+} + \operatorname{sgn}_{\mathrm{O}_2^+} \quad \text{and} \quad \mathrm{St} = R^{\mathrm{Sp}_2}_{\mathrm{sgn}}\longleftrightarrow 1_{\mathrm{O}_2^+}.$$

Notice also the Proposition 4.2.10 of [AMR96] holds, we can compute

$$[1_{\mathrm{Sp}_2}\otimes(1_{\mathrm{O}_2^+} + \operatorname{sgn}_{\mathrm{O}_2^+}) + \mathrm{St}\otimes 1_{\mathrm{O}_2^+}]^\sharp = 1_{\mathrm{Sp}_2}\otimes\left(1_{\mathrm{O}_2^+} + \operatorname{sgn}_{\mathrm{O}_2^+}\right) + \mathrm{St}\otimes\frac{1}{2}\left(1_{\mathrm{O}_2^+} + \operatorname{sgn}_{\mathrm{O}_2^+}\right) = \omega^\sharp_{\mathrm{Sp}_2,\mathrm{O}_2^+,1}$$

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